

Theory of Metastable States in Many-Body Quantum Systems

Chao Yin^{1,*}, Federica M. Surace^{2,†} and Andrew Lucas^{1,‡}

¹*Department of Physics and Center for Theory of Quantum Matter, University of Colorado, Boulder, Colorado 80309, USA*

²*Department of Physics and Institute for Quantum Information and Matter, California Institute of Technology, Pasadena, California 91125, USA*



(Received 20 August 2024; revised 11 January 2025; accepted 6 February 2025; published 19 March 2025)

We present a mathematical theory of metastable pure states in closed many-body quantum systems with finite-dimensional Hilbert space. Given a Hamiltonian, a pure state is defined to be metastable when all sufficiently local operators either stabilize the state or raise its average energy. We prove that short-range-entangled metastable states are necessarily eigenstates (scars) of a perturbatively close Hamiltonian. Given any metastable eigenstate of a Hamiltonian, in the presence of perturbations, we prove the presence of prethermal behavior: Local correlation functions decay at a rate bounded by a timescale nonperturbatively long in the inverse metastability radius, rather than Fermi’s golden rule. Inspired by this general theory, we prove that the lifetime of the false vacuum in certain d -dimensional quantum models grows at least as fast as $\exp(\epsilon^{-d})$, where $\epsilon \rightarrow 0$ is the relative energy density of the false vacuum; this lower bound matches, for the first time, explicit calculations using quantum field theory. We identify metastable states at finite energy density in the PXP model, along with exponentially many metastable states in “helical” spin chains and the two-dimensional Ising model. Our inherently quantum formalism reveals precise connections between many problems, including prethermalization, robust quantum scars, and quantum nucleation theory, and applies to systems without known semiclassical and/or field-theoretic limits.

DOI: [10.1103/PhysRevX.15.011064](https://doi.org/10.1103/PhysRevX.15.011064)

Subject Areas: Condensed Matter Physics,
Particles and Fields, Statistical Physics

I. INTRODUCTION

The notion of metastability has a long and rich history in physics. Perhaps the most famous illustration of metastability is the abrupt freezing of supercooled water at temperature $T < T_c$ (the freezing temperature), which occurs only after the water is shaken to induce ice crystal nucleation. This phenomenon is not only easy to demonstrate experimentally, but is also well understood through the framework of nucleation theory in classical statistical physics [1]. The classical theory of metastability can be formulated more precisely using the framework of Markov chains [2], where one shows that the “mixing time needed to escape” the metastable part of state space is nonperturbatively large in the strength of a small parameter, such as a small free-energy density difference between the two phases.

Metastability is also an extremely important phenomenon in *quantum systems*. Radioactive decay is a property of metastable atomic isotopes decaying to a lower energy state by emitting energetic particles [3]. Such metastability amounts to a *few-body* problem, which has been carefully analyzed for many decades and which (in its simplest realization) is textbook quantum mechanics. There are also intrinsically many-body metastable quantum states. For example, evidence of metastable unconventional superconducting states or other types of “hidden” quantum states have been observed in various materials after the excitation with strong laser pulses [4–6]. Metastability is also crucial in fields beyond condensed matter physics. In cosmology, a famous field-theoretic calculation proposed that the Universe may be in a metastable state—a *false vacuum* [7,8]. Extrapolations based on current estimates of the parameters of the standard model suggests that the electroweak vacuum might be metastable [9–11].

The simplest manifestation of this false vacuum arises in a quantum ferromagnet, with $\mathbb{Z}_2$ symmetry breaking, placed in a small longitudinal field of strength ϵ (which picks out one of the two degenerate ground states as lower energy). One expects that “the lifetime” of the false vacuum—the original ground state which is now antialigned with the longitudinal field—is nonperturbatively large: $t \sim \exp[C\epsilon^{-d}]$, for some constant C [12,13].

*Contact author: chao.yin@colorado.edu

†Contact author: fsurace@caltech.edu

‡Contact author: andrew.j.lucas@colorado.edu

Unlike in the classical models, we *do not* know of a mathematically precise theory of metastability in many-body quantum systems. This does not mean that useful calculations have not been done—for example, very famous calculations have estimated the lifetime of false vacua in quantum field theory [7]. But these calculations are *not well posed mathematically*—as physicists, we know “what” is being calculated but perhaps have only a hand-wavy argument for “why” the calculation works in the first place. Careful attempts to even give a “physicist’s derivation”—within a nonrigorous path integral formalism—of quantum metastability are quite recent [14].

In this paper, we aim at formulating a rigorous theory of metastability for quantum many-body systems. We essentially define a metastable pure state $|\psi\rangle$ in a many-body quantum system as one in which $\mathcal{O}|\psi\rangle - \langle\psi|\mathcal{O}|\psi\rangle|\psi\rangle$ has higher energy than $|\psi\rangle$, for arbitrary operators $\mathcal{O}$ that act on R or fewer degrees of freedom, for some “large” number R . This definition is (hopefully) intuitive and appears closely related to the definition used in recent work [15]. Given this simple definition, we are able to prove that given a spatially local Hamiltonian in a finite number of spatial dimensions, metastable states have nonperturbatively long lifetimes scaling at least as fast as e^{R^α} for any $\alpha \in (0, \frac{1}{2})$, as measured by local correlation functions. As in a recent paper by two of us [16], we are able to give a precise definition for the lifetime of a metastable state and explain why our definition is relevant to experimental and measurable physics.

The remainder of this introduction serves primarily to give a slightly more quantitative historical background to previous theories of metastability, as well as to outline the remainder of the paper.

A. Few-body metastability

To contrast the hardness of defining a theory of many-body quantum metastability, let us first review the much simpler few-body problem.

We first discuss classical physics (with stochastic fluctuations). Consider—just for pedagogical purposes—a one-dimensional particle moving along the line and undergoing overdamped dynamics. Suppose that the system is in thermal equilibrium, such that the (unnormalized) probability density function in equilibrium is

$$P_{\text{stationary}}(x) = \exp[-U(x)/T], \quad (1)$$

where $U(x)$ is the potential energy of the particle. Under Markovian dynamics with Gaussian noise, the dynamics of the probability density $P(x, t)$ can be described by the Fokker-Planck equation [17, 18]:

$$\partial_t P = \partial_x(\gamma(x)(T\partial_x + U(x))P). \quad (2)$$

One can show explicitly that if $U(x)$ has a deep local minimum, as depicted in Fig. 1(c), then so long as $\gamma(x)$

varies sufficiently slowly, the particle is trapped in this metastable region for a time [19]

$$t_c \sim \exp[c_c \Delta U/T], \quad (3)$$

where ΔU is the energy barrier and c_c is an $\mathcal{O}(1)$ constant. This is the textbook Arrhenius law.

In quantum mechanics, *tunneling* plays the role of stochastic fluctuations in a closed system (which does not decohere with its environment). Given a single quantum particle of mass m obeying the nonrelativistic Schrödinger equation, we can estimate the rate at which it tunnels out of the barrier using the WKB approximation:

$$t_q \sim \exp[c_q \sqrt{m\Delta U} R_q / \hbar], \quad (4)$$

where c_q is an $\mathcal{O}(1)$ constant and R_q is the length scale of the barrier. The physical interpretation given for t_q is that it is the time needed for the particle to escape the metastable region R :

$$\int_R dx |\psi(x, t)|^2 \sim e^{-t/t_q}. \quad (5)$$

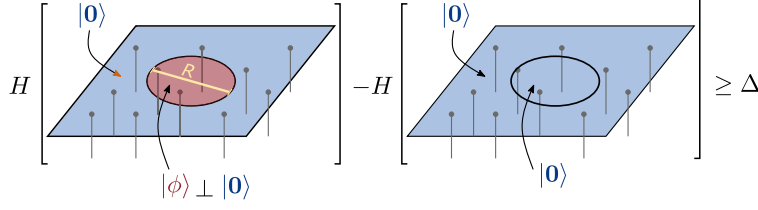
There are two related ways [20] that such a calculation is usually done in the literature. The first is to solve the Schrödinger equation subject to *outgoing* (radiation) boundary conditions on the wave function [3, 22]. This leads to a complex eigenvalue

$$E = E_0 - i \frac{\hbar}{2t_q}, \quad (6)$$

which is aligned with the interpretation (5) that t_q is the decay time of the metastable state. Such a complex eigenvalue is problematic: It renders the quantum dynamics of the system nonunitary, in direct contradiction to the starting assumption that the quantum system was isolated. Related to the nonunitarity, the resulting eigenfunctions are not normalizable—they exponentially diverge at large distances. Despite the lack of mathematical rigor, the predictions of this approach are physically sound and have been compared to experiments, e.g., on the decay of radioactive nuclei [3]. The second approach, which is more common in modern calculations, is to perform a path integral calculation in imaginary time [14, 23]; the resummation of instanton contributions corresponding to particles escaping the metastable state also leads to Eq. (6) and the same physical interpretation for t_q .

We remark that the prediction (4) makes sense only in a *semiclassical* limit. If the metastable region R does not admit a large number of bound states [assuming that the potential $U(x)$ continued to increase outside R], then the approximation that $E \ll \Delta U$ —required to justify the WKB prediction (4)—fails.

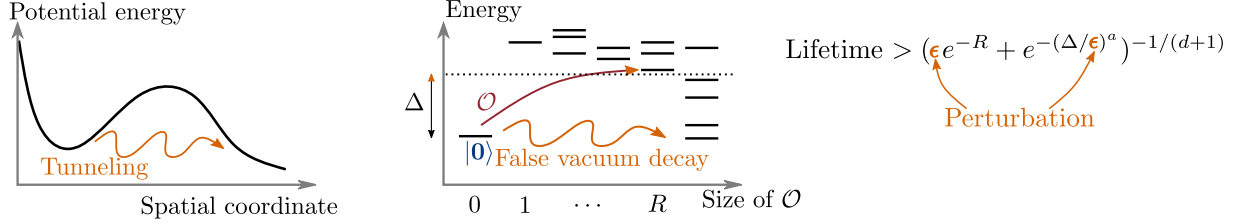
(a) Metastability as a local gap condition



(b) Metastable systems as perturbations of stable systems



(c) Nonperturbatively slow decay of metastable states



(d) Applications

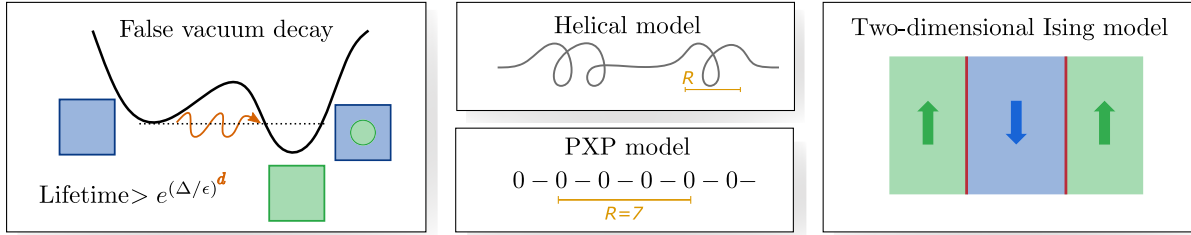


FIG. 1. A summary of our results. (a) Metastable states have a “local gap”—all sufficiently small operators acting on a set of diameter $\leq R$ raise the average energy by $\geq \Delta$. While our exposition mostly focuses on the all-zero metastable state (or, more generally, short-range-entangled states) as depicted here, the metastability concept generalizes to any state. (b) Theorem 5 proves that any short-range-entangled metastable state of Hamiltonian H is an *eigenstate* of a “nearby” Hamiltonian $H_0 = H - V$, where (heuristically) V is perturbatively small in R^{-1} . (c) Theorem 9 proves that, given the decomposition $H = H_0 + V$, all metastable states exhibit prethermal slow dynamics: Local correlation functions decay on the nonperturbatively slow timescale depicted. (d) Our metastability theory leads us to new insight into a broad range of many-body models, giving us rigorous guarantees on the lifetime of false vacua, as well as uncovering slow thermalization in helical spin chains. We also provide a new perspective on slow thermalization in the PXP model and the two-dimensional Ising model.

B. False vacuum: A many-body metastable state

Let us now turn to the many-body theory of metastability. Again, we begin with the classical setting and focus on an illustrative example: the d -dimensional Ising model with $d > 1$ in a longitudinal field, with Hamiltonian

$$H = -\Delta \sum_{i \sim j} Z_i Z_j - \epsilon \sum_i Z_i, \quad (7)$$

where $i \sim j$ represents nearest-neighbor pairs in d spatial dimensions and each $Z_i = \pm 1$. Suppose we start in the state $Z_i(t=0) = -1$ and consider temporal dynamics generated by a Gibbs sampler [24], which tends toward a very low temperature stationary state $P(Z) \propto \exp[-\beta H(Z)]$. At low enough temperature, the system is overwhelmingly likely to be found in a high magnetization $\langle Z_i \rangle \approx +1$ state, yet

nevertheless one can prove that the time it takes to reach a state in which the average magnetization is positive will grow as

$$t'_c \gtrsim \exp[\beta \Delta \cdot (\Delta/\epsilon)^{d-1}], \quad (8)$$

where β is the inverse temperature. This notion of hitting time can be bounded using the classical theory of Markov chains [2], and so we have a precise theory of classical metastability, following Ref. [1].

The problem above can be made quantum by simply considering Z to be a Pauli operator acting on a qubit (or spin-1/2) on each site of a lattice and considering Hamiltonian

$$H = -\Delta \sum_{i \sim j} Z_i Z_j - g \sum_i X_i - \epsilon \sum_i Z_i. \quad (9)$$

This time, nearly all calculations are done by the heuristic path integral calculations [7,25] which neatly generalize the few-body case mentioned previously. In a path integral approach, one can identify the false vacuum as a local minimum of the classical effective potential. In analogy, one could, for example, consider the (classical) product state with $Z_i = -1$ as the false vacuum state for the Hamiltonian (9).

However, the use of the path integral does not have a rigorous foundation, and perhaps more importantly, when Feynman diagrammatic perturbation theory fails at strong coupling, very basic questions remain mysterious. (i) Is there a meaningful notion of a minimum of a semiclassical effective potential, when our framework for strongly coupled (e.g., conformal) quantum field theory is not organized around a Lagrangian but instead around operator content? In contrast, we expect that metastability is a property of quantum systems that does not rely on any semiclassical approximation: For example, we expect that metastability should, in general, arise close to any first-order quantum phase transition, even when semiclassical approximations fail. (ii) What is the false vacuum state which has a long lifetime? In the Ising example above, the product state with $Z_i = -1$ at time $t = 0$ will rapidly decohere because of the nonzero transverse field g , which does not need to be small. (iii) What do we mean by the lifetime of the false vacuum? Taken literally, in perturbative quantum field theory, the probability of remaining trapped in the false vacuum after time t is estimated by path integral to be

$$P_{\text{trapped}} = \exp[-C\Lambda t], \quad (10)$$

where Λ denotes the spatial volume of the entire system. We might be tempted to define the characteristic timescale of the decay as $(C\Lambda)^{-1}$, but this quantity vanishes in the thermodynamic limit. This is the effect of the orthogonality catastrophe [26]: The false vacuum state $|\psi\rangle$ is not a “long-lived state” as a wave function; it is a rapidly decohering superposition of many-body eigenstates, so $\langle\psi(t)|\psi\rangle \rightarrow 0$ for any finite t , in the thermodynamic limit. So, we instead interpret the smallness of prefactor C as controlling the lifetime t_q of the false vacuum:

$$C^{-1} \sim t_q \gtrsim \exp[(\Delta/\epsilon)^d]. \quad (11)$$

Intuitively, this is reasonable: Experiments in physics measure local observables, so we should define the lifetime of the state to be the decay time of local correlation functions in the false vacuum. But the decay time of these correlation functions is not directly what the path integral computed. Recent works studied the decay of the false

vacuum using methods for real-time evolution [13,27–35] (for example, as the evolution after a quantum quench) rather than Euclidean path integrals. The connection of these results with traditional calculations of the decay rates is often unclear. Clear answers to these questions are now pertinent, as ultracold atom experiments and other quantum simulation platforms are able to probe false vacuum decay directly [36–38].

Two of us have recently suggested some precise answers to these questions [16], in quantum many-body lattice models. We found an explicit (but not elegant) expression for a particular many-body state, which we conjectured represented the correct nonperturbative definition of the false vacuum. In this state, we proved that local correlation functions decay on a nonperturbatively slow timescale t'_q . The rigorous bound found in Ref. [16] showed that the decay time obeys

$$t'_q \geq \exp[(\Delta/\epsilon)^{1/(2d-1)}], \quad (12)$$

which is nonperturbatively long in ϵ . Still, the path integral calculation [7] suggests a much longer timescale (11) in $d > 1$, suggesting that the actual physics that protects the false vacuum (in local correlation functions) is not properly captured by the framework of Ref. [16] in $d > 1$. This observation led us to conjecture that a better understanding of what it means for a quantum system to be metastable is necessary, before we can attempt to prove Eq. (11) quantum mechanically.

C. Slow thermalization, robustness of scars, Hilbert space shattering, and beyond

The fate of the quantum false vacuum has been studied for half a century. The recent literature has also seen extensive work on slow (or absent) thermalization, which turns out to also be deeply related to the rigorous theory of quantum metastability.

A metastable state like the one of the quantum Ising model (9) is a high-energy state, in the sense that it has an extensive energy proportional to $\epsilon\Lambda$ (a finite energy density) above the ground state. The evolution of a state with finite energy density is generally understood within the framework of the eigenstate thermalization hypothesis (ETH) [39–42], which is the quantum manifestation of the assumptions of statistical mechanics that all correlation functions will eventually appear thermal. In this context, such metastable state is expected to thermalize. Metastability can, thus, be understood as an example of slow thermalization [16,42–52] (sometimes described as prethermalization, i.e., the thermalization to a different Hamiltonian for transient, but long times).

A prominent example of slow (or absent) thermalization at high energies is “scarring” in many-body quantum systems, observed in a Rydberg atom experiment [53] and investigated in a plethora of recent theoretical

works [54–63]. Scars are eigenstates of the many-body Hamiltonian that violate the ETH. They represent a very small number of states, compared to the exponential number of thermal states that satisfy the ETH. Nevertheless, they can have significant practical consequences: A simple initial state $|\psi\rangle$ can have a large overlap with scar eigenstates, and, hence, the time-evolved state $e^{-iHt}|\psi\rangle$ will exhibit nonthermal local correlation functions for very long times.

However, the existence of scar states does not truly settle the question of why such slow dynamics ought to be visible experimentally—scar states are typically fine-tuned and are believed to decay with Fermi’s golden rule (or with a parametrically smaller, but still perturbative rate) [64,65] in the presence of generic perturbations [42]. A metastable state is, in general, not an exact scar (it is not an energy eigenstate), but for sufficiently small system sizes it will have large overlap with a single eigenstate, which may appear as an approximate scar in the many-body spectrum. (This can be understood from the long lifetime, which implies a small energy variance.) In contrast to fine-tuned scar states, a false vacuum is expected to be robust, having a nonperturbative decay rate rather than the perturbative Fermi’s golden rule rate.

If we wish to prepare a scarlike state in the lab, either as a check on the quantum coherence of a simulator or for metrological purposes [66], it is desirable to find models where slow thermalization is robust at higher orders in perturbation theory. One setting where such models can be found is in systems with Hilbert space shattering (also called fragmentation), whereby kinematic (or other) constraints forbid matrix elements between certain “Krylov sectors” in Hilbert space [61,67–70], which cannot be associated with any local symmetry. Again, in most models, this Hilbert space shattering is expected to be fragile to generic perturbations, in the sense that if H_0 is shattered, $H_0 + V$ will not be for a generic small V ; however, see Refs. [71,72] for some recent progress toward more stable shattered models. In the general case, the timescale over which one predicts “shattering physics” can be detected experimentally is still often quite large: If $V \sim \epsilon$ and H_0 has a spectrum with gap Δ , the timescale over which the shattered dynamics would be detectable in experiment scales as

$$t_p \gtrsim \exp[\Delta/\epsilon]. \quad (13)$$

The bound in Eq. (13) can be derived rigorously in more general settings [44] (that do not depend in shattering). Consider a Hamiltonian $H = H_0 + V$, where H_0 has an integer spectrum and is a sum of commuting terms and V represents a small perturbation. An example of physical interest for such a model is the Fermi-Hubbard model at large repulsive interaction strength, $H_0 = \sum_i \Delta c_{i\uparrow}^\dagger c_{i\uparrow} c_{i\downarrow}^\dagger c_{i\downarrow}$ corresponds to the number of doubly occupied sites, while

$V = \epsilon \sum_{i \sim j, \sigma} c_{i\sigma}^\dagger c_{j\sigma}$ corresponds to the hopping of single fermions from one site to the next. In this model, Eq. (13) manifests in the approximate conservation of the number of doubly occupied sites for $t \lesssim t_p$: Although a doubly occupied site is highly energetic, one must go to very high order Δ/ϵ in perturbation theory to find an energetically resonant many-body state with only singly occupied sites. Mathematically, the methods used to obtain the rigorous bound (12) are similar to those used to prove Eq. (13).

The intuition that prethermalization arises due to the lack of many-body resonances is quite similar to our previous argument for metastability in classical and quantum Ising models [16]. To obtain a genuine quantum theory of metastability, however, we must relax (at least, *a priori*) many of the assumptions listed above. It is not clear that a metastable Hamiltonian H is close to a solvable Hamiltonian H_0 in the sense of $H = H_0 + V$ nor that a Hamiltonian H_0 even exists for which the metastable $|\psi\rangle$ is an exact eigenstate. The purpose of this paper is to develop a theory of quantum metastability, starting from rather minimal assumptions. We present a precise, and intrinsically quantum, theory of metastable states and show that they indeed exhibit the expected phenomenology of slow (pre)thermal dynamics.

D. Summary of our results

We now briefly summarize the key ideas in this paper, which are illustrated in Fig. 1.

Our first step is to give a precise definition of a metastable state. In Sec. II, we define a pure state $|\psi\rangle$ and a Hamiltonian H to be (Δ, R) metastable if (loosely) all operators acting on $\leq R$ bits increase $\langle H \rangle$ by a finite amount $\geq \Delta$ [Fig. 1(a)]. This definition follows closely our classical intuition about metastable states as being *local* minima in the energy landscape, and a similar intuition was also used to define a “quantum local minimum” in Ref. [15]. By analogy with the tunneling picture in quantum mechanics, the local energy gap Δ and the size R represent the height and the width of the barrier, respectively. A difference with respect to other possible definitions (such as the one used in Ref. [15]) is that metastability is here a robust property with respect to local perturbations of a state: Rather than single isolated states, our definition defines “metastable basins” of states. A metastable state can be pictured as a state that is *sufficiently close* to a local minimum in the energy landscape (with this clarification in mind, in the following, we simply refer to metastable states as a *local minima*).

We then show that, with this definition, a metastable state is stable with respect to a close Hamiltonian. Theorem 5 proves that if a short-range-entangled state $|\psi\rangle$ is metastable, then the Hamiltonian H can always be written as $H = H_0 + V$, where $|\psi\rangle$ is an exact eigenstate of H_0 and V is a small perturbation (possibly on each lattice site) whose magnitude decays with R [Fig. 1(b)]. Rather surprisingly,

then, this demonstrates that a broad range of metastable states do indeed imply the existence of “scars” in a perturbatively close Hamiltonian.

The implications of Theorem 5 are profound. In Sec. III, we prove in Theorem 9 that all such metastable states exhibit prethermal behavior, whereby local correlation functions decay over a nonperturbatively long timescale $\exp[R^\alpha]$ for some $\alpha > 0$ [Fig. 1(c)]. This demonstrates that our quantum definition of metastability is a good one: It generally implies that experimental observables are well approximated (over very long times) by their value in $|\psi\rangle$, even if $|\psi\rangle$ is not close to a single eigenstate of the Hamiltonian H .

The remainder of the paper explores a number of applications and extensions of our formal theory of metastability [Fig. 1(d)]. Theorem 10 in Sec. IV proves that the lifetime of the false vacuum in Ising-like models is lower bounded by Eq. (11), which provides a mathematically well-posed confirmation of a 50-year-old prediction of quantum field theory [7]. Section V then describes a broad range of explicit microscopic models with metastable states. We find a metastable pair of product states in the one-dimensional PXP model [55], a popular model of arrested thermalization and quantum scarring, which provably exhibit extraordinarily slow thermalization, despite being at finite energy *density* above the ground state. We numerically study a family of “helical” spin chains for which there are an exponential number of metastable states and observe numerically that these finite energy density states are extremely long lived and generate many-body entanglement extremely slowly. Lastly, we also demonstrate that two-dimensional Ising models exhibit metastable states, which provides a connection to recent discussions of prethermal Hilbert space shattering [73,74] in this setting.

Theorems 5, 9, and 10 all have rather technically involved proofs, which are contained in appendixes.

II. DEFINING METASTABLE STATES

In this section, we carefully introduce both the class of models which we study as well as our definition of quantum metastable pure states.

A. Formal preliminaries: Notation, graphs, and local Hamiltonians

We consider a system of N qudits. The local Hilbert space dimensions of the qudits need not be equal, but so long as they are uniformly upper bounded by q , we might as well take each to be q . A local basis for a single qudit is $\{|0\rangle, |1\rangle, \dots, |q-1\rangle\}$. In the special case of $q = 2$ (qubits or spin- $\frac{1}{2}$ degrees of freedom), we define the Pauli operators by $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$, $X|0\rangle = |1\rangle$, and $X|1\rangle = |0\rangle$.

We place each qudit on one vertex of a graph with vertex set $\Lambda = \{i : i = 1, 2, \dots, N\}$. This graph is naturally imbued with a Manhattan distance function $d(i, j)$ between any

pairs of sites i, j , which counts the length of the shortest path between i and j in the graph. [75] The distance function induces a natural definition for distance between a site and a subset $S \subset \Lambda$: $d(i, S) = \min_{j \in S} d(i, j)$. The distance between two sets $d(S, S') = \min_{i \in S, j \in S'} d(i, j)$. The diameter of a subset $\text{diam} S := \max_{i, j \in S} d(i, j)$. We assume the graph is d -dimensional, in the sense that there exists a finite constant c_d such that for any connected subset $S \subset \Lambda$,

$$\begin{aligned} |\partial S| &\leq c_d \cdot \max(1, \text{diam} S)^{d-1} \quad \text{and} \\ |S| &\leq c_d \cdot \max(1, \text{diam} S)^d, \end{aligned} \quad (14)$$

where $\partial S := \{i \in S : d(i, S^c) = 1\}$ is the boundary of S . In the equations above, $|S|$ is the number of sites contained in S , and S^c is the complement of set S in Λ , or elements in Λ not contained in S . For the sake of efficiency, we say that a constant depends on d if it depends on d along with the constant c_d in Eq. (14). Note that the graph does not need to be a (subset of a) regular lattice; nevertheless, we sometimes refer to the graph as the “lattice.”

Let us now briefly remind the reader of some additional common mathematical notations that prove useful for us. The subtraction of two sets is defined as $S \setminus S' := \{i \in S : i \notin S'\}$. For example, $S^c := \Lambda \setminus S$ is the complement of S . A subset S is connected, if $\forall i, j \in S$, there exists a path in S that connects the two sites; i.e., there exists a set of edges $(i, i_1), (i_1, i_2), \dots, (i_p, j)$ of the graph where $i_1, \dots, i_p \in S$. We define the ball $B(S, r) := \{i \in \Lambda : d(i, S) \leq r\}$ which contains all sites with distance $\leq r$ to a given set S . We use $|\mathbf{0}\rangle_S$ to denote the all-zero state in qudit set S , with $\langle \mathbf{0} |_S$ being its bra and $|\mathbf{0}\rangle := |\mathbf{0}\rangle_\Lambda$ for the whole system. We use shorthand notation $|\phi\rangle_S \langle \psi|_S$ for operator $|\phi\rangle_S \langle \psi|_S$ supported in S . The floor function $\lfloor a \rfloor$ is defined as the largest integer no larger than a . We use big-O notations, where $f = O(g)$ means there exists a constant c such that $f \leq cg$. $f = \Omega(g)$ similarly means $f \geq cg$. $f = \Theta(g)$ means both $f = O(g)$ and $f = \Omega(g)$.

To these N qudits, we associate a Hamiltonian

$$H = \sum_S H_S. \quad (15)$$

The sum S runs over subsets of Λ . We demand that all subsets S appearing in the sum are connected (unless explicitly stated, this is an implicit assumption for the remainder of the paper) and that each local term H_S is supported in set S , i.e., $H_S = I_{S^c} \otimes \mathcal{O}_S$, where I_{S^c} is the all-identity operator on S^c and $\mathcal{O}_S$ acts nontrivially in S . The decomposition (15) is not unique: e.g., $Z_1 + Z_2$ (Pauli- Z operators on qubits $i = 1, 2$) can be either two terms supported in $S = \{1\}, \{2\}$ or a single term supported in $S = \{1, 2\}$. It will suffice for us to find one valid decomposition to analyze; all things considered, we wish to associate terms in H with the smallest S possible [76].

Given decomposition (15), we assume the Hamiltonian is sufficiently local in the d -dimensional lattice:

$$\max_{i \in \Lambda} \sum_{S \ni i} e^{2\mu \text{diam} S} \|H_S\| \leq h, \quad (16)$$

for some constants μ, h . Equation (16) implies the local terms $\|H_S\| \leq h e^{-2\mu \text{diam} S}$ decay no slower than an exponential of the support diameter. Clearly, any finite-range Hamiltonian can be normalized to satisfy Eq. (16).

As Eq. (16) effectively bounds the “local norm” of H , such a local H generates dynamics with a “linear light-cone structure,” as guaranteed by the following Lieb-Robinson bound (LRB) [78].

Theorem 1 (Lieb-Robinson theorem). For any H satisfying Eq. (16), there exists a constant u determined by d, h, μ such that

$$\begin{aligned} & \| [e^{itH} \mathcal{O}_S e^{-itH}, \mathcal{O}'_{S'}] \| \\ & \leq 2 \| \mathcal{O}_S \| \| \mathcal{O}'_{S'} \| \min \left[1, c_{\text{LR}} \min(|\partial S|, |\partial S'|) e^{\mu(u|t| - d(S, S'))} \right], \end{aligned} \quad (17)$$

for any pair of local operators $\mathcal{O}_S, \mathcal{O}'_{S'}$. The constant c_{LR} is determined by d and is independent of H .

The proof is found in Theorems 3.7 and 3.11 of a recent review [79]. This locality bound is a crucial ingredient for our results. We say H has a (μ, u) -LRB if Eq. (17) holds.

Lastly, we sometimes also need a slightly different local norm than Eq. (16). For any operator $\mathcal{O} = \sum_S \mathcal{O}_S$ decomposed like Eq. (15), we define a local norm parametrized by κ and $\alpha \in (0, 1)$:

$$\| \mathcal{O} \|_{\kappa} := \max_{i \in \Lambda} \sum_{S \ni i} e^{\kappa(\text{diam} S)^{\alpha}} \| \mathcal{O}_S \|, \quad (18)$$

where we do not show the α dependence explicitly. Throughout the paper, α can be viewed as a fixed number in $(0, 1)$, and our results are asymptotically tightest in the limit $\alpha \rightarrow 1$, where operators have an almost exponential decay with the support diameter.

B. Metastability is an energy barrier to local excitations

Given the formal setup, we are now ready to introduce what a metastable state is. Intuitively, metastability should be associated to “local minima in the energy landscape.” In classical statistical physics, such a statement need not be in quotes: The Hamiltonian of each state in state space is a well-defined function. In quantum many-body systems, however, a Hamiltonian H has complex and entangled *eigenstates* which are not calculable in any thermodynamic limit. To give a precise definition for metastability, therefore, we need to be careful about what “local minimum” refers to. Crucially, we do *not* want our definition of

metastability to relate in any way to exact eigenstates of a Hamiltonian; in contrast, checking that a state is metastable should be computationally straightforward.

To overcome the issue raised above, we say that state $|\psi_0\rangle$ is a local minimum when any sufficiently local Hermitian operator $\mathcal{O} = \mathcal{O}_S$ acting on $|\psi_0\rangle$ either stabilizes the state $\mathcal{O}|\psi_0\rangle = \langle \mathcal{O} \rangle_{\psi_0} |\psi_0\rangle$, [80] or raises the energy by at least a constant Δ . More precisely,

$$\begin{aligned} & H[(\mathcal{O} - \langle \mathcal{O} \rangle_{\psi_0})|\psi_0\rangle] - H[|\psi_0\rangle] \geq \Delta, \quad \text{for any} \\ & \mathcal{O} = \mathcal{O}^\dagger = \mathcal{O}_S \text{ acting on connected } S \text{ with } \text{diam} S \leq R, \end{aligned} \quad (19)$$

where R is an $O(1)$ —but, ideally, large—user-defined constant. Here,

$$\langle \mathcal{O} \rangle_{\psi_0} := \frac{\langle \psi_0 | \mathcal{O} | \psi_0 \rangle}{\langle \psi_0 | \psi_0 \rangle} \quad (20)$$

is the component that stabilizes the state, and

$$H[|\psi\rangle] := \begin{cases} \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle}, & \text{if } |\psi\rangle \neq 0 \\ +\infty, & \text{if } |\psi\rangle = 0 \end{cases} \quad (21)$$

is the average energy of an unnormalized state $|\psi\rangle$. Here, the $+\infty$ energy for a vanishing state is chosen such that Eq. (19) is satisfied if $\mathcal{O}$ stabilizes $|\psi_0\rangle$. This is intuitive because the state is “stable” with respect to its stabilizers. We arrive at the following.

Definition 2. We say the pair $(H, |\psi_0\rangle)$ is (Δ, R) metastable, if Eq. (19) holds with $R \geq 1$.

A state which is metastable under H is not metastable under $-H$ by this definition. However, as it is straightforward to simply reverse the sign of H for the remainder of the paper, we remark that any state which is a local maximum will exhibit all of the same phenomena that metastable states have.

We can relate this definition to the quantum Ising models reviewed in the introduction.

- (i) If R is chosen as the system size $\text{diam} \Lambda$, this definition is equivalent to $|\psi_0\rangle$ having large overlap [see Eq. (24)] with a nondegenerate gapped ground state, because $\mathcal{O}$ can be any operator in Eq. (19); such a state would be metastable for any gapped system.
- (ii) For $H = H_{\text{TFIM}}$ being the ferromagnetic transverse-field Ising Hamiltonian—i.e., the first two terms in Eq. (9)—and $|\psi_{\pm}\rangle$ being its two ground states with negative and positive polarization [81] for any finite R each of $|\psi_{\pm}\rangle$ is $(\Delta_{\text{TFIM}}(1-c), R)$ metastable for some small constant c if the volume Λ is sufficiently large. The reason is that any local operator $\mathcal{O}$ (that acts on a finite number of sites) has a matrix element $\langle \psi_+ | \mathcal{O} | \psi_- \rangle = O(\exp(-\Lambda))$ in the limit of large Λ ,

so the state $(\mathcal{O} - \langle \mathcal{O} \rangle_{\psi_+})|\psi_+\rangle$ (and similarly for $|\psi_-\rangle$) will be mostly supported on excited states, and c can be taken as small as $O(\exp(-\Lambda))$.

- (iii) For H being the longitudinal-field Ising Hamiltonian (7), the all-zero state $|\mathbf{0}\rangle$ is the unique ground state. On the other hand, $(H, |\mathbf{1}\rangle)$ is $(\Delta/2, c\Delta/\epsilon)$ metastable with a constant c , where $|\mathbf{1}\rangle$ is the all-one state. In this case, one still has a parametrically large $R \sim 1/\epsilon$ at small ϵ , because one has to flip (roughly speaking) a radius- R ball of qubits to get a state with similar energy, so that the energy gain in the bulk $\sim \epsilon R^d$ compensates the energy loss at the boundary $\sim \Delta R^{d-1}$. (iii) When adding a transverse field so that H is given by Eq. (9), one expects $(H, |\psi_0\rangle_{\text{TFIM}})$ is still $(\Delta_{\text{TFIM}}/2, c\Delta/\epsilon)$ metastable for some constant c . Here, $|\psi_0\rangle_{\text{TFIM}}$ is one of the ferromagnetic ground states of H without the longitudinal field defined above. The physical argument (although not rigorous) is that the excitations of H_{TFIM} are smoothly connected to those of the solvable point $H_{\text{Ising}} = -\Delta \sum_{i \sim j} Z_i Z_j$, because they are in the same ferromagnetic phase. Therefore, the metastability radius R of H should be similar to that of Eq. (7). Of course, given a suitable state, our definition can be explicitly checked.

Our first goal is to prove that, for any system satisfying the metastability condition Definition 2 with a large R and a short-range-entangled state $|\psi_0\rangle$ (to be rigorously defined shortly), $|\psi_0\rangle$ exhibits the *dynamical* phenomena associated with metastability, namely, when evolved by H , up to a long timescale t_* that roughly scales exponentially with R , that there exists a “dressed” version of $|\psi_0\rangle$, denoted as $|\tilde{\psi}_0\rangle$, for which local correlation functions of $e^{-itH}|\tilde{\psi}_0\rangle$ remain nearly the same as the initial $|\tilde{\psi}_0\rangle$ for $|t| < t_*$. [82] Hence, we need to show that Definition 2 implies $|\psi_0\rangle$ is close to a state $|\tilde{\psi}_0\rangle$. This will not depend on whether $|\psi_0\rangle$ is the *global* minimum: As in the introduction, we see that even if $|\psi_0\rangle$ is a highly excited (superposition of exponentially many eigenstates of H), the fact that it is a local minimum strongly suppresses the decay of local correlation functions, as any “path” in the Hilbert space connecting $|\psi_0\rangle$ to any such $|\phi\rangle$ requires “tunneling through” a large energy barrier. Crucially, in the many-body setting, when the Hamiltonian only contains few-body operators, we are able to bound this “tunneling” through radius R , since the state is evolved only by local operators and, thus, any perturbation of size R must grow in an “off-shell” way through many orders in perturbation theory.

As long as $\mathcal{O}|\psi_0\rangle$ (and their inner products) can be computed easily for a given $\mathcal{O}$ (which is the case for short-range-entangled $|\psi_0\rangle$, which we mostly focus on), Definition 2 can be verified numerically with complexity that grows exponentially with R^d , but at most linearly with system size N ; with translation invariance, the metastability

criterion can be checked in time independent of N . Our results, therefore, provide a rigorous way of using moderate-size classical computation to predict long-time dynamics of very large quantum many-body systems, as we do in Sec. V.

We expect that Definition 2 does not capture all kinds of quantum many-body metastable phenomena. The most clear shortcoming of our definition is that Eq. (19) requires the energy to raise by a finite amount when acting with any local operator, but the more physical condition—which we indeed make heavy use of in Sec. IV—is that the *minimum barrier height* is finite. The barrier height may be largest when acting with operators of size R , not size 1. We expect this “shortcoming” of our definition to be mainly technical in nature; as our theory already requires somewhat involved proofs, we elect to focus on understanding the implications of the simpler Definition 2 for this work. It is also likely, of course, that the dynamical implications of such a more relaxed notion of metastability will be weaker than those proved in this paper. We nevertheless see that our definition of quantum metastability encapsulates a large number of examples of interest across multiple subfields of physics.

Another shortcoming of our definition is that it does not account for the metastability of mixed states, such as the thermal ensemble of supercooled water which is metastable to the solid phase. Such a model should, in principle, have a quantum mechanical formulation, although we expect that this type of metastability is largely captured by classical models. The metastability of mixed states was recently discussed from a quantum computer science perspective [15].

C. Metastability as a local gap

This paper mostly focuses on $|\psi_0\rangle$ which are short-range entangled (SRE): $|\psi_0\rangle = U_0|\mathbf{0}\rangle$, where $|\mathbf{0}\rangle$ is the all-0 state, and unitary U_0 represents finite-time evolution generated by a quasilocal Hamiltonian [quasilocal in the sense that its local strength is bounded similarly as Eq. (16)]. It is reasonable to focus on such states for two reasons. First, this SRE condition is satisfied by the canonical models with false vacua described above. As we will see shortly, the physical meaning of the metastability condition also becomes more clear for SRE states, so it will be easy for us to cook up new models of metastability in Sec. V. Second, long-range entanglement can introduce a number of complications: For example, the long-range-entangled state can be *completely stable* (not metastable) in the thermodynamic limit, e.g., in theories with topological order, or highly sensitive to perturbations (and, thus, not even metastable). This sensitivity is well illustrated by the Greenberger-Horne-Zeilinger (GHZ) state $|\psi_0\rangle = (|\mathbf{0}\rangle + |\mathbf{1}\rangle)/\sqrt{2}$, which is readily dephased by a small Z field on one site to $|\mathbf{0}\rangle - |\mathbf{1}\rangle$, which seems to violate our intuition that metastable states ought to be locally robust. Insofar as we care about only local correlation functions, the GHZ state should instead be viewed as a classical

mixture of two SRE states $|\mathbf{0}\rangle, |\mathbf{1}\rangle$, and both of these states may be metastable. However, we show that some of our results about the phenomenology of metastable states do generalize to long-range-entangled metastable states, in Appendix B.

The quasilocal unitary U_0 can be viewed as a change of local basis or frame. From now on, we just work in the frame dressed by U_0 , so that

$$|\psi_0\rangle = |\mathbf{0}\rangle, \quad (22)$$

and H is actually $U_0^\dagger H_{\text{original}} U_0$, where H_{original} is the Hamiltonian in the original frame. If H_{original} is quasilocal in the sense of obeying Eq. (16), H remains quasilocal due to Theorem 1; the only price to pay is to modify constants such as μ, h, Δ , and R by $O(1)$ factors that depend on the “depth” of U_0 . Therefore, we still assume Eq. (16) in the dressed frame without loss of generality. Because of the same argument, the results obtained in the dressed frame can be automatically translated back to the original frame by just adjusting some constants. As an example of this local-frame change, consider the metastable system $(H_{\text{original}}, |\psi_0\rangle_{\text{TFIM}})$ with H_{original} being Eq. (9). Here, $|\psi_0\rangle_{\text{TFIM}}$ is in the same ferromagnetic phase as $|\mathbf{1}\rangle$, so U_0 is just the finite-time (quasi)adiabatic unitary evolution connecting them, [83] followed by a trivial relabeling $0 \leftrightarrow 1$ of the local states. $H = U_0^\dagger H_{\text{original}} U_0$, on the other hand, is still quasilocal containing the dressed Z field $-\epsilon \sum_i U_0^\dagger Z_i U_0$. The Z field after dressing can have terms like X_i , which no longer preserve the $|\mathbf{0}\rangle$ state (here we have relabeled $0 \leftrightarrow 1$) and causes this false vacuum to decay.

After restricting to the all-zero state (or SRE states accompanied by dressing), the meaning of the metastability condition (19) becomes more clear by the following proposition.

Proposition 3. The pair $(H, |\mathbf{0}\rangle)$ is (Δ, R) metastable, if and only if

$$H[|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}] - H[|\mathbf{0}\rangle] \geq \Delta, \quad \text{for any } |\phi\rangle_S \perp |\mathbf{0}\rangle_S \quad (23)$$

with connected S and $\text{diam} S \leq R$.

Proof. For any $|\phi\rangle_S \perp |\mathbf{0}\rangle_S$, one can define $\mathcal{O} = |\phi\rangle_S \langle \mathbf{0}| + \text{H.c.}$ so that Eq. (19) implies Eq. (23). On the other hand, any state $(\mathcal{O} - \langle \mathcal{O} \rangle_0) |\mathbf{0}\rangle$ is proportional to the form $|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}$ for some $|\phi\rangle_S \perp |\mathbf{0}\rangle_S$, so Eq. (23) also implies Eq. (19). ■

Proposition 3 implies that, when fixing the quantum state (i.e., boundary conditions) in S^c , a state is metastable if and only if it has sufficient overlap with the ground state. If the gap between the fixed-boundary-condition Hamiltonian’s ground state and first excited state is $\tilde{\Delta}$, then the state can [85] be (Δ, R) metastable so long as the overlap between $|\mathbf{0}\rangle$ and the true ground state $|\text{gs}\rangle$ in region S obeys

$$|\langle \mathbf{0} | \text{gs} \rangle_S|^2 \geq \frac{1}{2} \left(1 + \frac{\Delta}{\tilde{\Delta}} \right). \quad (24)$$

We can, alternatively, bound the local gap Δ :

$$\Delta \leq H[|1\rangle_i \otimes |\mathbf{0}\rangle_{i^c}] - H[|\mathbf{0}\rangle] \leq 2 \sum_{S \ni i} \|H_S\| \leq 2h. \quad (25)$$

Flipping one spin increases the energy by at most an $O(1)$ amount. The key point is that acting a local operator $\mathcal{O}_S$ on $|\mathbf{0}\rangle$ does not change the expectation value of $H_{S'}$ for any S' that does not intersect with S . These ideas play an important role in our proofs.

D. Metastable systems as perturbations of stable systems

Our definition of metastability does not depend in any way on the form of H . However, our motivating example—the false vacuum problem—does have special structure in the Hamiltonian H , which can be expressed as

$$H = H_0 + V, \quad (26)$$

where H_0 is the “ideal” part of H for which $|\psi_0\rangle$ is an exact eigenstate and V is a generic small perturbation such that $|\psi_0\rangle$ is not an eigenstate of the full H . For example, for the metastable system $(H, |\psi_0\rangle_{\text{TFIM}})$ with H being Eq. (9), it can be separated to $H_0 = H_{\text{TFIM}}$ and $V = -\epsilon \sum_i Z_i$, where $|\psi_0\rangle_{\text{TFIM}}$ is one gapped ground state of H_0 and V has small $\sim \epsilon$ local strength. In this case, the gap of H_0 guarantees that under a generic perturbation V , $e^{-itH} U |\psi_0\rangle_{\text{TFIM}}$, where U is an appropriate small local rotation (see Sec. III), is effectively constrained in the two-dimensional ground state subspace of H_0 (probed by local correlation functions) up to a long timescale $t_* \sim e^{(\Delta/\epsilon)^a}$. Since $|\psi_0\rangle_{\text{TFIM}}$ cannot be mapped to the other ground state by local operators, this gap condition already proves the slow decay of the false vacuum, albeit with a loose exponent $a \approx 1/(2d-1)$ [16].

Generalizing the *global* gap condition to *local* minimum condition (19), one expects that perturbing a stable system $(H_0, |\psi_0\rangle)$ where $|\psi_0\rangle$ is an eigenstate and local minimum of H_0 generically leads to metastability. Indeed, it is straightforward to show that, for SRE states, the metastability condition is robust under arbitrary small perturbation to the Hamiltonian.

Proposition 4. If $(H_0, |\mathbf{0}\rangle)$ is (Δ, R) metastable and

$$\max_{i \in \Lambda} \sum_{S \ni i} \|V_S\| \leq \epsilon, \quad (27)$$

then $(H, |\mathbf{0}\rangle)$ is $(\Delta/2, R')$ metastable, where H is given by Eq. (26) and $R' = \min(R, (\Delta/2c_d \epsilon)^{1/d})$.

Proof. We use the simpler condition (23). For any $|\phi\rangle_S \perp |\mathbf{0}\rangle_S$ with connected S and $\text{diam} S \leq R'$,

$$\begin{aligned}
& H[|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}] - H[|\mathbf{0}\rangle] \\
& = H_0[|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}] - H_0[|\mathbf{0}\rangle] + V[|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}] - V[|\mathbf{0}\rangle] \\
& \geq \Delta - \epsilon|S| \geq \Delta - \epsilon c_d(R')^d \geq \Delta/2,
\end{aligned} \tag{28}$$

where the term $\epsilon|S|$ comes from a straightforward generalization of Eq. (25). [86] ■

Note that Proposition 4 does not assume $|\mathbf{0}\rangle$ is an eigenstate of H_0 . In many examples, it will still be useful to start with an eigenstate $|\psi_0\rangle$ of H_0 . In this setting, if $|\psi_0\rangle$ is a gapped ground state of H_0 (which, in a nontrivial model, is metastable with thermodynamically large R), Proposition 4 shows that perturbing this stable system leads to metastability. Intriguingly, the converse is almost also true: Any SRE metastable system $(H, |\mathbf{0}\rangle)$ has a decomposition (26) with H_0 stabilizing the state (which is still metastable with respect to H_0) and small local norm for V ; the price to pay is that the metastability range R may shrink for $(H_0, |\mathbf{0}\rangle)$ to $R_0 \sim \sqrt{R}$.

Theorem 5. Suppose H is quasilocal by Eq. (16) and $(H, |\mathbf{0}\rangle)$ is (Δ, R) metastable. Suppose $R \geq c_R$ with c_R determined by $d, h/\Delta, \mu$, and the system is translation invariant on a d -dimensional square lattice. Then, for any constants $\alpha, \delta \in (0, 1)$, H can be decomposed as Eq. (26), where $(H_0, |\mathbf{0}\rangle)$ is $(\Delta/2, R_0)$ metastable with $R_0 = c'_\delta R^{1/2-\delta/d}$ and

$$H_0|\mathbf{0}\rangle = E_0|\mathbf{0}\rangle, \tag{29}$$

for some number E_0 , while the local norms are bounded by

$$\max_{i \in \Lambda} \sum_{S \ni i} e^{2\mu \text{diam} S} \|H_{0,S}\| \leq c_H h, \tag{30a}$$

$$\|V\|_{\kappa_1} \leq c_\delta h R^{\delta-d/2}, \tag{30b}$$

where $\kappa_1 = \mu$ and constants c_H , c_δ , and c'_δ are determined by d , α , h/Δ , μ , and δ .

The proof is given in Appendix A and proceeds by explicitly constructing the Hamiltonians H_0 and V by their local decompositions

$$H_0 = \sum_{S \subset \Lambda} H_{0,S}, \quad V = \sum_{S \subset \Lambda} V_S. \tag{31}$$

There, we also show (i) the assumption of translation invariance is not fundamental, but it does simplify the statement of the theorem; (ii) an explicit example which (at least naively) saturates the scaling $R_0 \sim \sqrt{R}$, $\|V\|_{\kappa_1} \sim R^{-d/2}$ at $\delta \rightarrow 0$.

Because of this reduction, starting from the next section we take the perturbation scenario (26) as the starting point and aim to show long lifetime of the metastable state. The local norm $\|V\|_{\kappa_1}$ is defined in Eq. (18) and requires the interaction strength of V to decay only stretched

exponentially with the support diameter of its local terms. However, for latter technical reasons, we do require that H_0 decays exponentially in Eq. (30a) and has an exponential LRB (17) as a consequence.

Before proceeding, we remark that the robustness Proposition 4 also implies robustness of metastability when dressing the state locally.

Proposition 6. Let $U = \mathcal{T} e^{\int_0^1 ds A(s)}$ ($\mathcal{T}$ means time ordering) be any quasilocal unitary where the anti-Hermitian $A(s)$ has local norm bounded by a sufficiently small $\epsilon > 0$:

$$\max_s \max_{i \in \Lambda} \sum_{S \ni i} e^{2\mu \text{diam} S} \|A_S(s)\| \leq \frac{\epsilon}{c_- h} (2\mu)^{4d-2} < (2\mu)^{6d+2}, \tag{32}$$

with constant $\mu \leq 1/(2e)$ and c_- determined by d . If $(H, |\mathbf{0}\rangle)$ is (Δ, R) metastable with H satisfying Eq. (16), then $(U^\dagger H U, |\mathbf{0}\rangle)$ is $(\Delta/2, R')$ metastable with R' the same as in Proposition 4.

Proof. Using the κ -norm notation in Eq. (18) with $\alpha = 0.5$, we have, e.g., $\|H\|_{2\mu} \leq h$ from Eq. (16) and $\text{diam} S \geq (\text{diam} S)^\alpha$. We then invoke Proposition 12 in Appendix B, [87] where the condition $\max_s \|A(s)\|_{2\mu} \leq (2\mu)^{6d+2}$ holds from Eq. (32), so that

$$\|U^\dagger H U - H\|_0 \leq c_- (2\mu)^{-2(2d-1)} \|H\|_{2\mu} \max_s \|A(s)\|_{2\mu} \leq \epsilon. \tag{33}$$

Since $U^\dagger H U = H + (U^\dagger H U - H)$, this reduces to the setting in Proposition 4, and metastability of $(U^\dagger H U, |\mathbf{0}\rangle)$ follows. ■

Alternatively, we can study the metastability of $(H, U|\mathbf{0}\rangle)$. Ignoring the dependences on constants d and μ , Proposition 6 implies that given one metastable SRE state $|\psi_0\rangle$, any state of the form $U|\psi_0\rangle$ is also metastable, as long as the quasilocal unitary U has local strength $\epsilon \lesssim h, \Delta$.

III. NONPERTURBATIVELY SLOW DECAY OF METASTABLE STATES

The most important consequence of metastable states is the nonperturbatively slow decay of local correlation functions. In this section, we outline the proof of this, and related, statements.

A. Long lifetime from local diagonalizability

We prove the long “lifetime” of $|\psi_0\rangle$ under H by showing that H can be almost “locally diagonalized” to stabilize $|\psi_0\rangle$. Here, “local” is crucial: One does not want to fully diagonalize H using a global unitary, because no true eigenstates of H will be close to $|\psi_0\rangle$ due to the orthogonality catastrophe. Instead, one asks if any quasilocal unitary U approximately diagonalizes H up to a small but still

extensive error term [16]. Formally, we define local diagonalizability by the following structure for H_0, V .

Definition 7. Suppose H_0 and V are two Hamiltonians and $|\psi_0\rangle$ is a quantum state. We say the triple $(H_0, V, |\psi_0\rangle)$ is $(v_*, \mathbb{d}_*, a_*, \kappa_*)$ locally diagonalizable, if the following statements hold.

There exists a quasilocal unitary $U = \mathcal{T}e^{\int_0^1 ds A(s)}$ (recall that $\mathcal{T}$ means time ordering) with anti-Hermitian $A(s)$ of local norm

$$\|A(s)\|_{\kappa_*} \leq a_*, \quad (34)$$

such that the rotated Hamiltonian (26)

$$H_* + V_* := U^\dagger (H_0 + V) U \quad (35)$$

$(H_*, V_*$ both Hermitian) satisfies

$$H_* |\psi_0\rangle = E_* |\psi_0\rangle, \quad (36)$$

for some number E_* , and

$$\|H_* + V_* - H_0\|_{\kappa_*} \leq \mathbb{d}_*, \quad (37a)$$

$$\|V_*\|_0 \leq v_*. \quad (37b)$$

Here, U is a quasilocal unitary, because it is generated by finite-time evolution of some time-dependent quasilocal Hamiltonian $iA(s)$. The parameter a_* describes the local strength of U that brings H to an almost diagonalized form (here diagonalization is only with respect to the particular state $|\psi_0\rangle$). After the rotation, $\mathbb{d}_*$ quantifies how far the new Hamiltonian is with respect to the unperturbed H_0 . v_* is the most important parameter that quantifies the local strength of the remaining undiagonalized part V_* .

This local diagonalizability implies long lifetime $t_* \sim v_*^{-1/(d+1)}$ of the system, among other things.

Proposition 8 (Corollary 4 in Ref. [16]). If the triple $(H_0, V, |\psi_0\rangle)$ is $(v_*, \mathbb{d}_*, a_*, \kappa_*)$ locally diagonalizable with corresponding U, H_* defined above, and $\|H\|_{\kappa_1} \leq h$ for Eq. (26), there exist constants c_U, c_{oper} determined by κ_1, h such that the following statements hold.

- (1) *Locality of U .*—Given local operator $\mathcal{O} = \mathcal{O}_S$ supported in connected set S , $U^\dagger \mathcal{O} U$ is quasilocal in the sense that

$$U^\dagger \mathcal{O} U = \mathcal{O} + \sum_{r=0}^{\infty} \mathcal{O}_r, \quad (38)$$

where $\mathcal{O}_r$ is supported in $B(S, r)$, but not in $B(S, r-1)$, and $\mathcal{O}_r$ decays rapidly with r :

$$\|\mathcal{O}_r\| \leq c_U a_* |S| \|\mathcal{O}\| e^{-\kappa_* r^\alpha}. \quad (39)$$

- (2) *Local operator dynamics is approximately generated by H_* .*—Given local operator $\mathcal{O} = \mathcal{O}_S$,

$$\begin{aligned} & \|e^{itH} \mathcal{O} e^{-itH} - U e^{itH_*} U^\dagger \mathcal{O} U e^{-itH_*} U^\dagger\| \\ & \leq c_{\text{oper}} v_* |S| (|t| + 1)^{d+1} \|\mathcal{O}\|. \end{aligned} \quad (40)$$

As a result, for $|t| \ll t_* \sim v_*^{-1/(d+1)}$, one can effectively evolve a local operator by first going to the dressed frame $U^\dagger \mathcal{O} U$, evolving by H_* , and finally transforming to the original frame.

- (3) *The dressed $|\psi_0\rangle$ is locally preserved for $|t| \ll t_* \sim v_*^{-1/(d+1)}$.*—Define the dressed metastable state

$$|\tilde{\psi}_0\rangle = U |\psi_0\rangle, \quad (41)$$

along with its time-evolved reduced density matrix on any set S :

$$\rho_S(t) := \text{Tr}_{S^c} \left(e^{-itH} |\tilde{\psi}_0\rangle \langle \tilde{\psi}_0| e^{itH} \right). \quad (42)$$

$\rho_S(t)$ is close to the initial value in trace norm

$$\|\rho_S(t) - \rho_S(0)\|_1 \leq c_{\text{oper}} v_* |S| (|t| + 1)^{d+1}. \quad (43)$$

The proof follows verbatim from that of Corollary 4 in Ref. [16,88] where Eq. (43) follows from Eq. (40) by a state tomography argument, together with the fact that H_* preserves $|\psi_0\rangle$. Equation (43) is what we mean by the long lifetime of $|\psi_0\rangle$: If v_* is nonperturbatively small, there exists a locally dressed version (41) of $|\psi_0\rangle$, such that all local correlation functions in the dressed $|\tilde{\psi}_0\rangle$ are preserved for a nonperturbatively long time. This is because any local correlation function at time t is of the form $\text{Tr}(\mathcal{O}_S \rho_S(t)) \approx \text{Tr}(\mathcal{O}_S \rho_S(0))$ for some small set S . Although the rigorous lifetime bound holds only for the dressed state, physical arguments suggest that the undressed $|\psi_0\rangle$, as well as other locally rotated states, should share similar lifetimes [16]. One also does not need to prepare the exact N -qudit state (41) to see this effect (which is practically impossible at large N in an experiment), because we are probing local correlation functions: The result (43) holds for any initial state whose reduced density matrix in the backward light-cone region of S approximates that of $|\tilde{\psi}_0\rangle$. The size of the allowed region S does not depend on the total number of degrees of freedom N , although it does scale with v_* .

B. Metastability implies local diagonalizability

Our goal is now to show that metastability leads to local diagonalizability with nonperturbatively small v_* , which automatically implies that metastable states are long-lived in the sense of Proposition 8. This is established for general metastable systems with SRE metastable states by the

following key theorem, utilizing the perturbation decomposition $H = H_0 + V$ guaranteed by Theorem 5.

Theorem 9. Suppose $H_0 = \sum_S H_{0,S}$ is a sum of operators with finite norm:

$$\max_i \sum_{S \ni i} \|H_{0,S}\| \leq h_0, \quad (44)$$

H_0 has (μ, u) -LRB (17), and $(H_0, |\psi_0\rangle)$ is (Δ, R) metastable where H_0 stabilizes product state $|\psi_0\rangle$ as in Eq. (29). For any $\alpha \in (0, 1)$ and

$$\kappa_1 \leq \min\left(\frac{\mu}{5}, \frac{1}{e}\right), \quad (45)$$

there exist constants c_R, c_*, c_V, c_A , and c_D determined by $\alpha, d, \mu, \Delta/h_0, \Delta/u$, and κ_1 (where c_R does not depend on α, κ_1), such that if $R \geq c_R$, for any perturbation V with small local norm

$$\epsilon := \|V\|_{\kappa_1} \leq c_V \Delta, \quad (46)$$

$(H_0, V, |\psi_0\rangle)$ is $(v_*, c_D \epsilon, c_A \epsilon, \kappa_1/2)$ locally diagonalizable, where v_* is small so that $|\psi_0\rangle$ has lifetime

$$t_* = \begin{cases} \epsilon^{-\frac{2(d-\alpha)}{(2d-1)(d+1)}} \exp\left[\Omega\left(\frac{R^\alpha}{\ln R}\right)\right], & \frac{\epsilon}{\Delta} = O\left(R^{-\frac{\alpha(2d-1)}{2\alpha-1}}\right) \\ \exp\left[\Omega\left(\left(\frac{\Delta}{\epsilon}\right)^{\frac{2\alpha-1}{2d-1}}\right)\right], & \frac{\epsilon}{\Delta} = O\left(R^{\delta - \frac{\alpha(2d-1)}{2\alpha-1}}\right) \end{cases} \quad (47)$$

in the sense of Proposition 8. Here, Eq. (47) keeps track of only the two large parameters Δ/ϵ and $R \gg 1$, and $\delta > 0$ is any constant.

The condition (45) is always achievable by freely reducing κ_1 because $\|V\|_\kappa \leq \|V\|_{\kappa'}$ for any $\kappa < \kappa'$. Intuitively, at large R the second line of Eq. (47) recovers the nonperturbative prethermalization lifetime (12) for perturbed gapped systems [16]; while for extremely small perturbation ϵ/Δ , the perturbation theory works at very high order k_* and involves large operators that feel the presence of the metastability length scale R . In the latter case, R determines an effective cutoff order that leads to $t_* \sim e^R$.

We prove Theorem 9 in Appendix B and sketch the idea here for $|\psi_0\rangle = |\mathbf{0}\rangle$ [see a cartoon illustration in Fig. 2(a)]. In the limit $R \rightarrow \infty$, the problem reduces to perturbing a gapped H_0 , where the gapped ground state $|\mathbf{0}\rangle$ needs to gain energy $\geq \Delta$ from perturbation V in order to become resonant with excited states of H_0 . Since the local strength of V is $\epsilon \ll \Delta$, the resonant decay process happens only at high enough perturbation order $k \gtrsim (\Delta/\epsilon)^a$ with $a = 1$ (naively). This high-order perturbation theory is usually formalized by a sequence of Schrieffer-Wolff transformations (SWT).

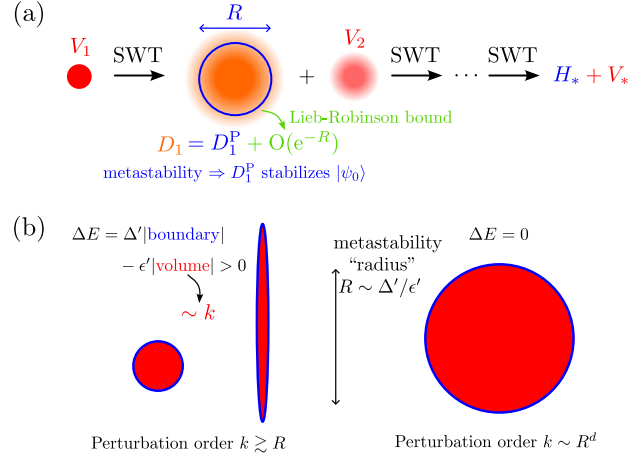


FIG. 2. Proof ideas for (a) Theorems 9 and (b) 10. (a) We perform Schrieffer-Wolff transformations to locally diagonalize the Hamiltonian as if H_0 was gapped, because metastability implies local operators feel an effective gap. (b) By keeping track of the operator volume instead of diameter, resonance happens at a much higher order $k \sim R^d$ instead of $k \sim R$.

As an example of how and why to use SWTs, consider the Fermi-Hubbard model $H_0 + V$, where $H_0 = \Delta \sum_i n_{i\uparrow} n_{i\downarrow}$ is on-site repulsion and $V = \epsilon \sum_{i \sim j, \sigma} c_{i,\sigma}^\dagger c_{j,\sigma} + \text{H.c.}$ is nearest-neighbor hopping with $\sigma = \uparrow, \downarrow$. At half filling and strong repulsion, H_0 prefers one fermion per site with a dangling spin-1/2, and the perturbation V induces an effective antiferromagnetic Heisenberg interaction acting on the spin degrees of freedom. This superexchange process comes from a SWT

$$e^{-A_1}(H_0 + V_1)e^{A_1} = H_0 + D_1 + V_2, \quad (48)$$

where A_1 is an anti-Hermitian operator that generates the SWT unitary e^{A_1} and the subscripts indicate the order of ϵ : $V_1 := V \propto \epsilon^1$, for example. To determine A_1, D_1 , and V_2 , one expands the left-hand side $e^{-A_1}(H_0 + V_1)e^{A_1} = H_0 + V_1 + [H_0 + V_1, A_1] + \dots$ and matches with the right-hand side order by order. The zeroth order $H_0 = H_0$ is trivial. The first order is

$$V_1 + [H_0, A_1] = D_1, \quad (49)$$

while the higher-order terms are all included in $V_2 = [V_1, A_1] + \frac{1}{2}[[H_0, A_1], A_1] + O(\epsilon^3)$. To solve Eq. (49), one demands that D_1 is block-diagonal $P^> D_1 P^< = 0$ with respect to the two subspaces separated by gap (projectors $P^<$ and $P^>$ project onto states below and above said gap, respectively), so that D_1 is the effective Hamiltonian inside the subspace, up to $O(\epsilon^2)$ corrections. Since V_1 always maps out of the subspace, $D_1 = 0$ in this example, and Eq. (49) is solved by

$$A_1 = \frac{\epsilon}{\Delta} \sum_{i \sim j, \sigma} P_{ij}^> (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{H.c.}) P_{ij}^< - \text{H.c.}, \quad (50)$$

where $P_{ij}^>$ ($P_{ij}^<$) denotes the subspace on i, j where one site is occupied by two fermions and the other site is empty (each site is occupied by one fermion). One can verify that the resulting $V_2 = \frac{1}{2}[V_1, A_1] + O(\epsilon^3)$, when restricted to the gapped subspace $P^<$, is dominated by the superexchange interaction of the order of ϵ^2 , serving as an effective Hamiltonian in the frame “rotated” by unitary e^{A_1} . Furthermore, since the SWT generator A_1 is local, even if V_2 contains higher orders that may not preserve subspace $P^<$, it is still quasilocal from evolution by A_1 . One can further perform a second-order SWT e^{A_2} to rotate $H_0 + D_1 + V_2$ viewing V_2 as perturbation, so that the remaining perturbation becomes $V_3 \propto \epsilon^3$, and so on.

In the above example, H_0 is commuting and has integer spectrum. Such cases are closely related to Floquet prethermalization [44–47], and the perturbation theory is proven to be convergent $\|V_k\|_\kappa \lesssim \epsilon^k$ up to the order of $k \sim \Delta/\epsilon$ where resonant decay process kick in. For general gapped H_0 like the transverse-field Ising model [the first two terms in Eq. (9)], however, technicality arises already at the first step of solving Eq. (49): The solutions for A_1 and D_1 are no longer strictly local:

$$A_1 = i \int_{-\infty}^{\infty} dt W(t) e^{itH_0} V_1 e^{-itH_0}, \quad (51a)$$

$$D_1 = \int_{-\infty}^{\infty} dt w(t) e^{itH_0} V_1 e^{-itH_0} \quad (51b)$$

involve long-time Heisenberg evolution of V_1 by H_0 , where the filter function $w(t)$ is (roughly speaking) the derivative of $W(t)$ so that Eq. (49) holds. More precisely, $W(t) = \pm 1/2$ is (only) discontinuous at $t = 0$, with $W(0^\pm) = \pm 1/2$, and decays to zero at $|t| \rightarrow \infty$, so that

$$\begin{aligned} [H_0, A_1] &= \int_{-\infty}^{\infty} dt W(t) \frac{d}{dt} (e^{itH_0} V_1 e^{-itH_0}) \\ &= - \int_{-\infty}^{\infty} dt \frac{dW(t)}{dt} e^{itH_0} V_1 e^{-itH_0} \\ &\quad - (W(0^+) - W(0^-)) V_1, \end{aligned} \quad (52)$$

from integration by parts, where the two contributions equal to D_1 and $-V_1$, respectively, by choosing $[dW(t)/dt] = -w(t)$ at $t \neq 0$ [see Eq. (B12) for the explicit formula]. A good filter function $w(t)$ exists such that (i) $w(t)$ has a compact Fourier transform

$$\hat{w}(E) = 0 \quad \forall |E| > \Delta/2, \quad (53)$$

so that D_1 (and A_1) is still block diagonal: For any pair of eigenstates $|E^{>,<}\rangle$ of H_0 in the two subspaces,

$$\begin{aligned} \langle E^{>} | D_1 | E^{<} \rangle &= \int_{-\infty}^{\infty} dt w(t) e^{it(E^{>} - E^{<})} \langle E^{>} | V_1 | E^{<} \rangle \\ &= \hat{w}(E^{>} - E^{<}) \langle E^{>} | V_1 | E^{<} \rangle = 0; \end{aligned} \quad (54)$$

(ii) $w(t)$ decays nearly exponentially: $w(t) \sim e^{-|t|/\ln^2|t|}$ at large $|t|$, so A_1 and D_1 are still “sufficiently local” in the sense that their κ -norm (18) is bounded, because $e^{itH_0} V_1 e^{-itH_0}$ contains terms of size $\lesssim ut$ from the Lieb-Robinson bound. Although the evolution generated by A_1 does not have the standard Lieb-Robinson bound (17) for strictly exponentially decaying interactions, one can nevertheless prove that [16] $V_2 = e^{-A_1} V_1 e^{A_1} - V_1 + \dots$ is also sufficiently local with a bounded $\|V_2\|_\kappa$ for a suitably chosen κ (recall that in the above commuting case, V_2 is quasilocal with strictly exponential tails). Based on these locality estimates, one can iterate the SWT procedure like the commuting case and show convergence $\|V_k\|_\kappa \lesssim \epsilon^k$ up to $k = k_* = (\Delta/\epsilon)^a$, where $a \approx 1/(2d-1)$ is no longer 1 due to the weaker locality bounds. Nevertheless, the remaining perturbation $V_* = V_{k_*} \sim \epsilon^{k_*}$ at this order is already small enough to yield the second line of Eq. (47).

We have briefly reviewed the proof for $R = \infty$, i.e., when H_0 is gapped [16]. Observe that SWT generated by Eq. (51) is defined even if H_0 has no gap: One just evolves V_1 by H_0 and integrates with the filter function $W(t)$. The corresponding D_1 in Eq. (51b) is still sufficiently local but does not have the block-diagonal structure anymore, because there may be eigenstates of H_0 with energy close to $|0\rangle$, between which D_1 can have nonzero matrix elements. Nevertheless, notice that *local terms* in D_1 with diameter $\leq R$ see an “effective gap” above $|0\rangle$ from the metastability condition, as shown in Fig. 2(a); hence, they nearly (instead of exactly) stabilize $|0\rangle$. More precisely, suppose (for now) $V_1 = V_{1,\{i\}}$ contains only one local term on site i , and suppose D_1 is *exactly* supported in the ball $B(i, R/2)$ centered at i . The metastability condition implies

$$H_0[(D_1 - \langle D_1 \rangle)|0\rangle] - H_0[|0\rangle] \geq \Delta. \quad (55)$$

However, this is impossible unless $(D_1 - \langle D_1 \rangle)|0\rangle = 0$, because $D_1|0\rangle$ has no support on eigenstates of energy $> \Delta/2$ due to Eq. (53). In reality, D_1 has tails $\lesssim \epsilon e^{-R}$ supported outside of $B(i, R/2)$. It turns out that the above argument is robust and yields $\|(D_1 - \langle D_1 \rangle)|0\rangle\| \lesssim \epsilon e^{-R}$, in general. Now let us consider extensive V_1 : The linearity of Eq. (51b) with respect to V_1 implies the existence of an extensive operator D_1^P that (i) approximates D_1 :

$$\|D_1 - D_1^P\|_\kappa \lesssim \epsilon e^{-R}, \quad (56)$$

and (ii) stabilizes $|0\rangle$ *exactly*. We repeat such arguments at higher orders.

To summarize, making R finite in the SWTs merely produces an extra error term $D_k - D_k^P \propto \epsilon^k e^{-R}$ that does not

stabilize $|\mathbf{0}\rangle$ at the order of k . Note that $D_1 - D_1^P$ cannot be rotated away by higher-order SWTs like typical terms in V_2 , because $D_1 - D_1^P$ is already dominated by large-size operators that do not see the effective gap. The dominant error for v_* is then just Eq. (56) at first order, which leads to

$$v_* \lesssim \epsilon e^{-R} + e^{-(\Delta/\epsilon)^a}, \quad (57)$$

where the second term, present for gapped H_0 , comes from the remaining perturbation at the terminating order k_* . The magnitude of v_* leads to Eq. (47), wherein we have further estimated which of the two terms in Eq. (57) dominates.

In Appendix B, we also prove that much of the critical phenomenology of metastability can be generalized to states with long-range entanglement, so long as the Hamiltonian has strictly few-body interactions, and $H = H_0 + V$ with the metastable state an eigenstate of H_0 . We do not have an analog of Theorem 5 for long-range-entangled metastable states, so this latter assumption becomes more nontrivial.

IV. FALSE VACUUM DECAY

Thus far, we have proved metastability implies non-perturbatively (in R or Δ/ϵ) slow relaxation of local correlation functions in complete generality. For the remainder of the paper, we discuss applications and extensions of this general framework for metastability. First, we revisit carefully the question of false vacuum decay in certain lattice models. Using the physical intuition from our theory of metastability, we prove a bound on the “lifetime” of the metastable state that matches the path integral calculation (11), far stronger in $d > 1$ spatial dimensions than any rigorous results (12) or (13) in the literature.

More precisely, we consider generic perturbations to the solvable Ising model in d dimension $H' = H'_0 + V'$, where

$$H'_0 = \frac{\Delta'}{2} \sum_{i \sim j} (1 - Z_i Z_j) \quad (58)$$

and

$$V' = \sum_{S: \text{diam}(S) \leq \ell_V} V'_S \quad (59)$$

has finite range ℓ_V . Here, we use primes because we construct a different Hamiltonian H_0 that corresponds to the previous metastable one. We prove the following.

Theorem 10. Consider H'_0 (58) that has two ground states $|\mathbf{0}\rangle, |\mathbf{1}\rangle$. Let ℓ_V be a finite number. There exist constants $c_V, c_A, c_D, c_*,$ and $\tilde{c}_*$ determined by d and ℓ_V such that the following holds: For any perturbation V' (59) with finite range ℓ_V and sufficiently small local norm

$$\epsilon' := \|V'\|_0 \leq c_V \Delta', \quad (60)$$

$(H'_0, V', |\psi_0\rangle)$ for either $|\psi_0\rangle \in \{|\mathbf{0}\rangle, |\mathbf{1}\rangle\}$ is $(\tilde{c}_* \epsilon' 2^{-k_*}, c_D \epsilon', c_A \epsilon', 1/6)$ locally diagonalizable (see Definition 7), where

$$k_* = \lfloor c_*(\Delta'/\epsilon')^d \rfloor. \quad (61)$$

Applying Proposition 8 then implies that the false vacuum that is either $|\mathbf{0}\rangle$ or $|\mathbf{1}\rangle$ relaxes slowly with lifetime

$$t_{\text{FVD}} \gtrsim \exp[(\Delta'/\epsilon')^d]. \quad (62)$$

In comparison, the previous best bound for this setting is the prethermalization bound $t_{\text{FVD}} \gtrsim \exp(\Delta'/\epsilon')$ (13), using the fact that H'_0 is commuting and has integer spectrum; this is much weaker than Eq. (62). Moreover, Eq. (62) matches the path integral calculation (11), putting the saddle-point approximation on a firmer ground.

Appendix C contains the proof of Theorem 10, where we, in fact, prove a stronger result: H'_0 can be any commuting and frustration-free qudit Hamiltonian, with a condition that its ground state degeneracy does not come from trivial effects (e.g., flipping a single qubit may not change the energy of a system consisting of noninteracting qubits; see Definition 19 for formal statements). In the main text, we focus on the simplest Ising example with $|\psi_0\rangle = |\mathbf{0}\rangle$ and sketch the proof idea.

Since H'_0 is gapped, one can perform a SWT (reviewed in the previous section) up to some finite order $k_d \sim 3d/2$, so that

$$U_{\text{finite}}^\dagger (H'_0 + V') U_{\text{finite}} = H_0 + V. \quad (63)$$

We now consider the Hamiltonian $H_0 + V$ in the frame rotated by the SWT unitary U_{finite} . Here, $H_0 = H'_0 + D'$, where D' is the block-diagonal effective Hamiltonian [i.e., D_1 in Eq. (49) for first-order SWT], and the remaining perturbation $V \sim \epsilon \propto (\epsilon')^{d+1}$. Once V is this small, the metastability radius should control the lifetime of the false vacuum [analogous to Eq. (47)]. The most naive way is to use Proposition 4 that implies $(H_0, |\mathbf{0}\rangle)$ is $(\Delta'/2, R)$ metastable with $R \sim (\Delta'/\epsilon')^{1/d}$, since the local norm of D' is $\sim \epsilon'$. However, this is too weak to yield Eq. (62). Physically, as sketched in Sec. II B, the metastability radius should scale as

$$R \sim \frac{\Delta'}{\epsilon'}, \quad (64)$$

because this is the shortest length scale of a true-vacuum bubble where the bulk energy gain $\sim \epsilon' R^d$ overcomes the boundary energy penalty $\sim \Delta' R^{d-1}$. It turns out that such a metastability radius is still not sufficient, because applying Theorem 9 directly would yield $t_{\text{FVD}} \gtrsim \exp(\Delta'/\epsilon')$.

Instead of Definitions 2 and 3, we need a slightly stronger notion of metastability that counts the *volume* of local operators: $(H_0, |\mathbf{0}\rangle)$ has metastability volume $\sim R^d$ with R being Eq. (64), in the sense that

$$H_0[|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}] - H_0[|\mathbf{0}\rangle] \geq \Delta, \quad \text{for any } |\phi\rangle_S \perp |\mathbf{0}\rangle_S \quad (65)$$

with connected S and $|S| \leq R^d$.

Equation (65) implies metastability radius (64) for Definition 2 due to Eq. (14) and is stronger than that because Eq. (65) also allows for S with $\text{diam} S \gg R$, like a long-stripe region. This is the key to obtain the tight exponent in Eq. (62), illustrated in Fig. 2(b). Intuitively, Eq. (65) comes from the geometrical fact that there exists a constant c'_d such that

$$|\partial S| \geq c'_d |S|^{1-1/d}, \quad (66)$$

for any subset S . As a result, any true-vacuum bubble that flips spins in S with volume bounded in Eq. (65) has larger boundary energy penalty than the bulk energy gain. A technical complication we overcome to show this result is that H_0 can map between computational basis states, so long as they do not (locally) look like $|\mathbf{0}\rangle$ or $|\mathbf{1}\rangle$; therefore, Eq. (65) does not directly follow from the form of H'_0 .

We then follow the idea of the previous section to show that, when doing SWT for $H_0 + V$, the obtained effective Hamiltonian [e.g., D_1 (51b) with $V_1 = V$] stabilizes $|\mathbf{0}\rangle$ approximately with local error $v_* \sim \exp(-R^d)$. Given Eq. (65), it suffices to show that local terms in, e.g., D_1 that act on a region of volume $\gtrsim R^d$ are highly suppressed $\sim v_*$. However, this is hard to prove using Eq. (51b), because $e^{itH_0} V e^{-itH_0}$ grows to diameter $\gtrsim R$ and, thus, volume $\gtrsim R^d$ at time $t \gtrsim R$ from the constant Lieb-Robinson velocity of H_0 , but at that timescale, $w(t) \sim e^{-t}$ is suppressed only by $\sim e^{-R} \gg v_*$.

This technical issue is the reason we restrict the original unperturbed part H'_0 to be commuting. For this special case, although H_0 is not commuting anymore, it is close to a commuting H'_0 , and we have more control over the operator locality than Eq. (51b). A similar technique has been used in Ref. [89] to prove stability of topological order given a commuting H_0 : We can write $H_0 = H'_0 + \epsilon' D'$ and solve the SWT equation (49) order by order in ϵ' . The zeroth order is just SWT for the commuting H'_0 , so that the corresponding zeroth-order solutions for A_1 and D_1 are strictly local for a local V_1 , as in the Hubbard model example; similarly, each order higher just grows the support of operators by a strictly finite amount. It turns out that this method avoids the dangerous operator growth that would arise in Eq. (51b) and leads to the desired bound v_* for local terms with large volume $\gtrsim R^d$.

To summarize the idea of Theorem 10, one first performs a finite round of SWTs to suppress the remaining

perturbation V to sufficiently high but constant (in ϵ') order (63), where the effective Hamiltonian H_0 is metastable with volume $\sim R^d$. One then performs SWTs for this new perturbation problem $H_0 + V$ up to high order (61), following the spirit of Theorem 9, whereas the metastability volume (instead of diameter) together with the solvability of H'_0 lead to a much stronger bound (62) than the more general theory of Theorem 9. In practice, for the second stage of perturbing $H_0 + V$, we do not follow directly the proof structure of Ref. [89], which has two layers of perturbation series: The outer layer performs SWTs, and each SWT step is embedded by an inner-layer perturbation series for solving Eq. (49). We find it more convenient to reorganize the orders of ϵ' to have just a single layer of perturbation series, where we always perform SWT using H'_0 instead of H_0 . This proof structure looks more like the original prethermalization proof [44]; see Appendix C for details.

V. CONSTRUCTING METASTABLE STATES

Next, we show that our metastability formalism can also be applied to systems and states which are not a symmetry-breaking false vacuum. Indeed, our theory of metastability leads to a mechanism to systematically generate “prethermal scars” and explain their long lifetime.

A. One-dimensional (generalized) helical models

We first turn our attention to one-dimensional models, where we can present a simple and explicit construction for finding *exponentially many finite energy-density* metastable states in system size. Consider the following model acting on a one-dimensional chain of qudits with local Hilbert space dimension q , which we dub the “unperturbed helical model”:

$$H_0 = \sum_i (1 - P_i) K_i (1 - P_i), \quad (67)$$

where P_i and K_i act on three sites $\{i-1, i, i+1\}$, K_i is an arbitrary operator with all positive eigenvalues: $K_i \geq \Delta > 0$, and P_i is a projector Hamiltonian defined as

$$P_i = |000\rangle\langle 000|_i + |001\rangle\langle 001|_i + |(q-1)00\rangle\langle (q-1)00|_i + \sum_{k=0}^{q-2} |k(k+1)(k+2)\rangle\langle k(k+1)(k+2)|_i, \quad (68)$$

where we define $|q\rangle = |0\rangle$. Here, the $|abc\rangle_i$ is shorthand for the state $|a_{i-1} b_i c_{i+1}\rangle$.

Observe that H_0 has exponentially many ground states of the form

$$|\text{gs}\rangle = |0_1 0_2 \cdots 0_a 1_{a+1} \cdots (q-1)_{a+q-1} 0_a 0_{a+1} \cdots 0_b 1_{b+1} \cdots\rangle. \quad (69)$$

It is easy to see that in a chain of length L there are at least $2^{\lfloor L/(q+1) \rfloor}$ such ground states—in each block of length $q+1$ we pick between motif $01 \cdots (q-1)0$ or $00 \cdots 00$. By construction, any state which is not one of these helical ground states has energy at least Δ , since it must be composed of orthogonal states to these ground states, all of which by construction must (in the computational basis) have at least one triplet of sites which is annihilated by P_i and which, thus, has energy of at least Δ .

To obtain metastable states, we simply need to modify H_0 by a weak perturbation, such as

$$H = H_0 - \mu \sum_{i=1}^N |0\rangle\langle 0|_i, \quad (70)$$

which picks out the helix-free ground state as unique. Any state with a finite helix density now has finite energy density, yet it is metastable because there is no way to locally flip the helix motif to 0's without an operator that acts on at least $q-1$ sites. At small μ , we thus deduce that this model is $(\Delta - (q-1)\mu, q-1)$ metastable, and any further perturbation to Eq. (70) leads to prethermalization proven in Theorem 9.

Notice that this is not the only valid construction. We can, in fact, look for much longer metastable strings. One explicit model has helix-antihelix motifs of the type $01 \cdots (q-1)(q-1) \cdots 10$. Let us consider, for example, the case $q=3$ and define the Hamiltonian [90]

$$H_0 = \sum_i \left(1 - \sum_{s \in B} P_i^{(s)} \right) + \sum_i (\mu_1 |1\rangle\langle 1|_i + \mu_2 |2\rangle\langle 2|_i), \quad (71)$$

where $P_i^{(s)} = |s\rangle\langle s|_{i-1,i,i+1}$ is the (three-site) projector on the s configuration and $B = \{000, 001, 012, 122, 221, 210, 100, 101, 020, 002, 200\}$. The set B is chosen such that for $\mu_1 = \mu_2 = 0$ there is an exponential number of degenerate ground states, consisting of blocks of motifs of the types 012210, 00200 interspersed with sequences of 0's. For $\mu_1, \mu_2 > 0$, the uniform 0 state is the only ground state, while long motifs of the type 012210 can be metastable depending on the parameters μ_1 and μ_2 . We study this model numerically for the case $\mu_1 = 0.2, \mu_2 = 0.12$: In this case, the repeated 012210 motif is (Δ_R, R) metastable for $R = 1, 2$ and

$$\Delta_1 = 1 - \mu_1 - \mu_2, \quad \Delta_2 = 1 - \mu_1 - 2\mu_2. \quad (72)$$

More generally, any state consisting of repeated motifs of the types 01221 and 012210 is similarly (Δ_R, R) metastable. Note that the number of these metastable states is exponentially large in the system size. We also observe that the ground state has a smaller local gap than these

metastable states, since applying a single flip from $\dots 00000 \dots$ to $\dots 00200 \dots$ costs only an energy $\Delta = \mu_2$. Similarly, a state with a certain number of 01221 motifs interspersed with long strings of zeros is therefore not (Δ_R, R) metastable (with the same $\Delta_R, R = 1, 2$ as the motifs-only states), because the 00200 flip is “easy.” However, we do expect that physically the helix-antihelix motifs are long lived, because they are locally protected by a large gap, and, therefore, the low-energy excitations represented by 00200 flips should not thermalize them.

We study the spectrum of the model for a chain of L qudits with periodic boundary conditions in the presence of a perturbation V : $H = H_0 + V$ with

$$V = \epsilon \sum_i (|0\rangle\langle 1|_i + |1\rangle\langle 2|_i + |2\rangle\langle 0|_i + \text{H.c.}). \quad (73)$$

The exact diagonalization results are shown in Fig. 3(a).

Even for relatively small system size ($L = 10$), we see that for $\epsilon = 0$ the state $|0122101221\rangle$ is in a rather dense region of the spectrum and is very close in energy with states that have no 012210 motifs. When the perturbation strength ϵ is nonzero, we still observe a distinct eigenstate in the spectrum that seems smoothly connected with the $\epsilon = 0$ eigenstate $|0122101221\rangle$. This persists to values of the perturbation strength that are much larger than the characteristic level spacing in the region of the spectrum around this eigenstate. These data show the effect of metastability on the robustness of energy eigenstates: Since $|0122101221\rangle$ is (Δ_R, R) metastable for $\epsilon = 0$, it (or, better, a state obtained from it through a quasilocal unitary transformation) has long lifetime and small energy variance. In other words, it is an approximate eigenstate and shows very little hybridization with the rest of the spectrum when perturbed. To test this long-lived robustness to perturbations, we also study the time evolution (quench) taking two different initial states: the metastable $|0122101221\rangle$ [Fig. 3(b)] and the state $|0000000012\rangle$ [Fig. 3(c)], which is not metastable (a single-site flip from 1 to 0 can lower its energy) but has a comparable energy to the $|0122101221\rangle$ state. We consider the expectation value of a local observable as a function of time after the quench for different values of the perturbation strength. For both initial states, we observe some oscillations, but the amplitudes are much smaller for the metastable state compared to the nonmetastable one. The small oscillations observed for the former can be understood as the effect of the quasilocal unitary transformation and should be accompanied by small spreading of correlations.

To verify this, we perform a numerical simulation of the quench for larger system sizes using the time-dependent variational principle algorithm for matrix product states [91–93]. We consider a system of $L = 30$ sites with open boundary conditions (including some boundary terms that mimic a prolonged chain with 0's for the external qudits)

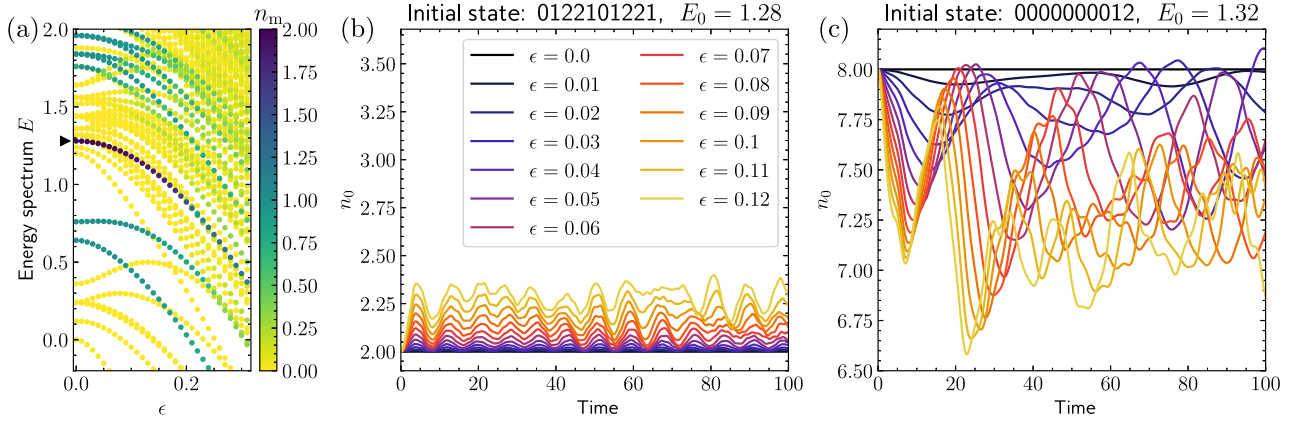


FIG. 3. (a) Energy spectrum vs perturbation strength ϵ in the zero momentum, inversion even sector. The color indicates, for each energy eigenstate, the expectation value of the number of motifs $n_m = \sum_i |012210\rangle \langle 012210|_{i,\dots,i+5}$. The arrow points at the metastable state $|0122101221\rangle$. (b) Expectation value of the number of 0's $n_0 = \sum_i |0\rangle \langle 0|_i$ as a function of time for different values of the perturbation strength ϵ . The initial state is the metastable state $|0122101221\rangle$. (c) The same as (b), but for the nonmetastable initial state $|0000000012\rangle$. All the results are obtained for $L = 10$, $\mu_1 = 0.2$, and $\mu_2 = 0.12$.

and study the time evolution starting from two initial states with the same energy $E = 3.2$: (i) a (Δ_R, R) -metastable state [with $R = 1, 2$ and Δ_R as in Eq. (72)], with “012210” motifs and (ii) a generic, nonmetastable state. For the timescale considered, the evolution from the metastable state shows no significant changes in the expectation values of local observables [Fig. 4(a)] and an almost undetectable

growth of the entanglement entropy [Fig. 4(c)]. The generic state, on the other hand, shows a much faster evolution of local observables [Fig. 4(b)] and faster growth of the entanglement entropy [Fig. 4(d)]. In Fig. 4, we also observe that, while the half-chain entanglement entropy of the generic state shows a saturation at long times due to the finite bond dimension χ utilized in the simulations, for

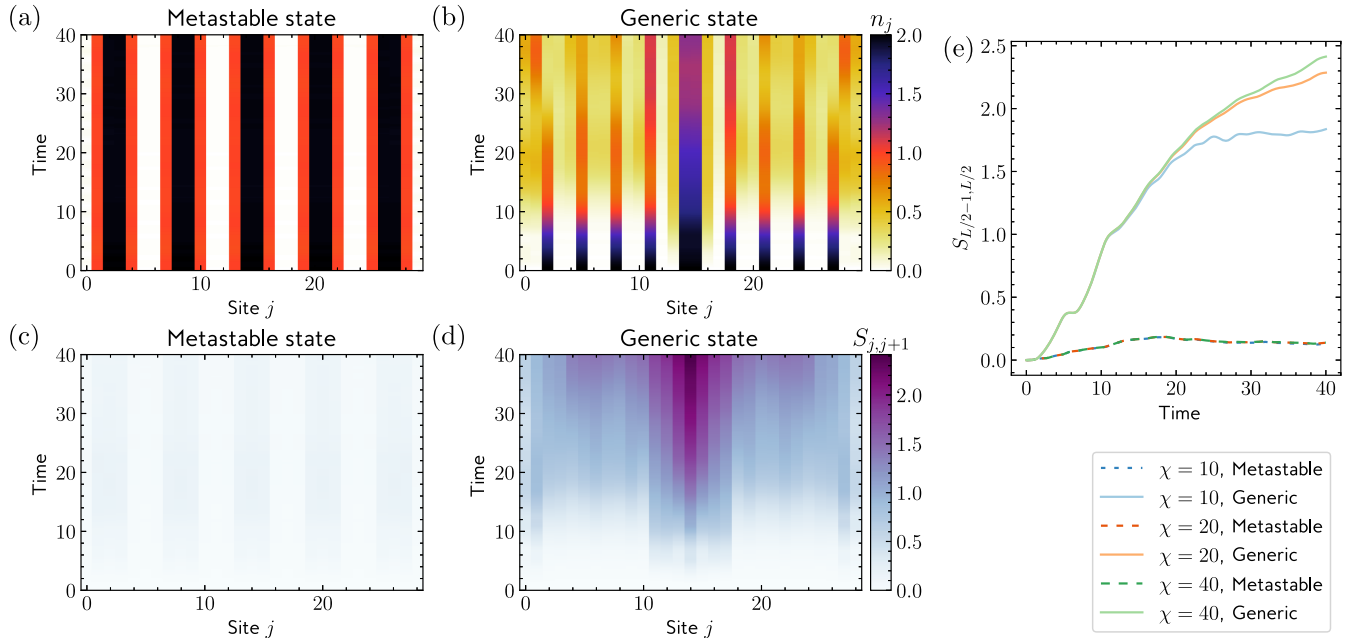


FIG. 4. Time evolution of the local observable $n_j = |1\rangle\langle 1|_j + 2|2\rangle\langle 2|_j$ for two initial states with the same energy $E = 3.2$: (a) a metastable state $|0122100122100\dots\rangle$ and (b) a nonmetastable state $|00200200\dots 00220020\dots\rangle$. Entanglement entropy $S_{j,j+1}$ at a cut between site j and $j+1$ in the evolution from the same (c) metastable and (d) nonmetastable states as in (a) and (b), respectively. All the results are obtained for $L = 30$, $\mu_1 = 0.2$, $\mu_2 = 0.12$, and maximal bond dimension $\chi = 40$. (e) Half-chain entanglement entropy for different values of the bond dimension, which demonstrates that the slow dynamics of the metastable state is not limited by the finite bond dimension.

comparable times the entanglement entropy of the metastable state remains small and independent on χ .

B. The PXP model

Next, we address the extent to which our theory of metastability can provide insight to the unexpectedly slow thermalization of the one-dimensional PXP model [55], whose Hamiltonian is

$$H = \sum_{i=1}^N P_{i-1} X_i P_{i+1}. \quad (74)$$

Here, each site of a one-dimensional lattice with periodic boundary conditions contains a qubit with Pauli matrices X , Y , Z , and $P_i = (1 + Z_i)/2$. H is able to excite a local $|0\rangle$ to $|1\rangle$ only if its two neighbors are all zero, and we focus on the invariant subspace $\mathcal{H}_{\text{PXP}}$ containing no neighboring excitations [11]. The PXP model has been realized experimentally in Rydberg atom arrays [53] and studied extensively as a model with approximate quantum many-body scars [55,58,62,63,94–103].

Most notably, starting from a charge density wave (CDW) state $|010101\dots\rangle$, the system exhibits long-lived oscillations probed by, e.g., the CDW order parameter, even if the dynamics is not perturbatively close to any known form of solvability. Although heuristic arguments have been made to explain this surprisingly slow thermalization process, a physically satisfactory theory is still missing.

Here, we apply our rigorous theory of metastability to the PXP model. First, we ask whether the CDW state is metastable: The answer is no, because flipping a single qubit already maps the state to an orthogonal one $|01000101\dots\rangle$ with the same energy 0. In other words, $(H, |010101\dots\rangle)$ is not (Δ, R) metastable for any $\Delta > 0, R \geq 0$. [104] Nevertheless, one can ask if there is any metastable state for H beyond the ground state (which numerical simulations find to be gapped [105]). Intriguingly, we find

$$|0-\rangle := |0-0-0-\dots\rangle \quad (75)$$

to be metastable, together with its translated cousin $|-0\rangle := |-0-0-0\dots\rangle$, which share a similar form with the CDW states. Here, $X|\pm\rangle = \pm|\pm\rangle$ on a single site. Note that the overlap $\langle -0|0-\rangle = 2^{-N/2}$ vanishes quickly in the thermodynamic limit. We fix N to be even for simplicity.

The intuition for the product state $|0-\rangle$ being metastable is as follows. Observe that $|0-\rangle$ minimizes all terms with even i in Eq. (74), and the odd terms have zero expectation. To gain (lower) energy from the odd terms, one needs to add $|1\rangle$ components to odd sites. However, the Hamiltonian (74) is “constrained”: To excite an odd site, one needs to first put its two even-site neighbors from $|-\rangle$ to $|0\rangle$, which requires an operator of size $R+1 \geq 3$. Furthermore, this

process necessarily loses a large energy ≈ 2 from the two even PXP terms, with $|0-0-00-00-0-\rangle$ being a concrete example. So to gain enough energy that compensates the loss, one needs to apply a very nonlocal change to the state $|0-\rangle$, leading to a large metastability range R .

Although Definition 2 formally does apply to Eq. (75), the name “metastability” might be misleading: Although this state is excited and has a finite energy density relative to the ground state, it is locally close to the ground state, so in the sense of Proposition 4, the “metastability” of the gapped ground state can, in turn, imply “metastability” of $|0-\rangle$. This is not, then, necessarily a metastable state which is “locally far” from the ground state, like our previous examples of the false vacuum or helical state. Nevertheless, since we have shown that our theory of metastability has a number of useful implications, it is worth investigating the dynamics of $|0-\rangle$ a little further.

Exact diagonalization (ED) verifies in Fig. 5(a) that $|0-\rangle$ has metastability range $R = 7$ in the bulk ($|-\rangle$ is the same due to symmetry). More precisely, for each R , we diagonalize a “metastable Hamiltonian” on $R+1$ sites with boundary conditions determined by the metastable product state, to minimize the energy difference in Eq. (23) (with $|0\rangle$ replaced by $|0-\rangle$); the result is labeled by Δ . The maximal metastability range is the largest R with a positive Δ . [106] Δ for odd R turns out to equal that for the even $R-1$; the reason is that the left boundary condition for $|0-0-\dots\rangle$ fixes the leftmost qubit to 0 to minimize energy. In Fig. 5(a), we also plot the minimal energy difference when changing the boundary condition from the metastable Hamiltonian to the usual open and periodic boundaries, where Δ becomes negative already before $R < 4$. This comparison emphasizes the fact that, by treating the interactions among different parts (i.e., boundary conditions) more carefully, a system can be much more metastable than what finite-size ED would naively suggest.

The metastable state $|0-\rangle$ has energy density -0.5 at large N , which is noticeably larger than the ground state energy density -0.6034 that has been estimated numerically using the density matrix renormalization groups [97,105]. Therefore, although it is at relatively low energy comparing to the CDW state (which lies exactly in the middle of the spectrum—i.e., at infinite temperature), it is still probing physics at a nonvanishing energy density or the physics at a low but positive temperature. We confirm this by diagonalizing the lowest energy levels of the PXP Hamiltonian up to $N = 30$ in the two symmetry sectors where $|0-\rangle$ is supported [the inversion-even zero-momentum sector ($\mathcal{I} = +1, k = 0$) and the inversion-odd π -momentum sector ($\mathcal{I} = -1, k = \pi$)] and show in Fig. 5(b) that the overlaps of the metastable state with the ground state and the first excited state decay exponentially with N . Moreover, the single-particle gap is $\tilde{\Delta} \approx 0.9682$ in the $k = \pi$ momentum sector [105], so for $N \geq 20$ the

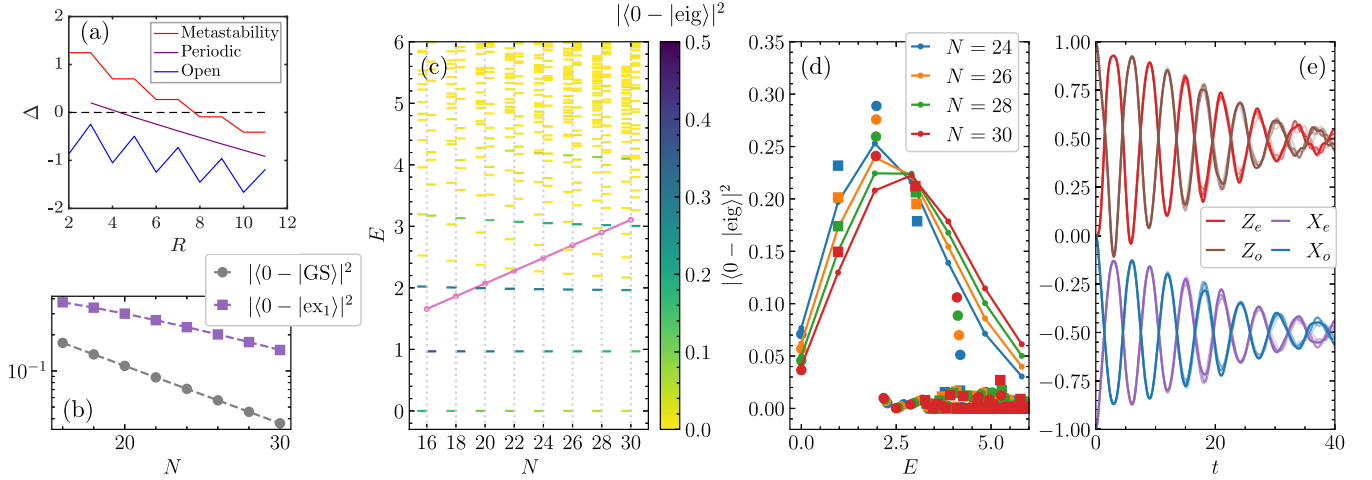


FIG. 5. Metastability of state $|\psi_0\rangle = |0-\rangle$ in the PXP model with periodic boundaries from ED. (a) For a given R , the energy difference (23) is minimized to obtain Δ , which shows the metastability range $R = 7$. To this end, an $(R + 1)$ -site metastable Hamiltonian with suitable boundary conditions is diagonalized. The data are compared with more usual boundary conditions, where the periodic one applies only to odd R . (b) Overlap of the $|0-\rangle$ state with the ground state and first excited state. Both overlaps decay exponentially with the system size N . (c) Low energy spectrum (dots) E of H in the constrained subspace $\mathcal{H}_{\text{PXP}}$, with the ground state energy set to 0. For each N , the energy levels in the $\mathcal{I} = +1, k = 0$ sector ($\mathcal{I} = -1, k = \pi$ sector) are plotted on the left (right) of the dashed line. The energy of $|0-\rangle$ (pink solid line) is deep in a continuum of excited states at large N . The color of each eigenstate represents its overlap with $|0-\rangle$, showing that $|0-\rangle$ is approximately supported on a minority of eigenstates. (d) An alternative view of the eigenstate overlap of $|0-\rangle$, showing that the state is predominantly supported on a handful of eigenstates of approximately equal energy spacing (circle markers for $\mathcal{I} = +1, k = 0$ sector, square markers for $\mathcal{I} = -1, k = \pi$, no line). For comparison, we have shown (solid line) what the spectrum would look like if the state was made of a Poisson-distributed collection of noninteracting “quasiparticles” $\sum \alpha^{n/2} e^{-\alpha n/2} (n!)^{-1/2} \cdot |n\rangle$, where $|n\rangle$ denotes the state with n quasiparticles at momentum $k \approx \pi$. The similarity in the spectra is consistent with a quasiparticle interpretation for the metastable state. (e) We observe slow thermalization and persistent oscillations in a quench from the metastable state, as evidenced by $Z_e = \langle 2N^{-1} \sum_{i \text{ even}} Z_i \rangle$, for example. Data for $N = 24, 26, 28, 30$ are plotted (they almost exactly overlap), with color intensity increasing with N .

energy of the $|0-\rangle$ state is above the two-particle threshold, in a “dense” continuum of excited states.

Numerically, we construct the exact low energy eigenstates for PXP chains up to $N = 30$ and find that the overlap of the metastable $|0-\rangle$ state is concentrated on only a handful of states: Fig. 5(c). Intriguingly, one possible explanation for this is that the metastable state is “close to” a superposition of weakly interacting quasiparticles at lattice momentum $k = \pi$. A similar quasiparticle picture is expected to be quite general and has been applied to low-energy-density states in other models, demonstrating long-lived nonthermal behavior [107,108]. The underlying intuition is that states with low energy density correspond to a low density of quasiparticles, which, being dilute, interact only weakly. Although the system will eventually thermalize due to quasiparticle scattering, this process occurs over a long timescale. Our framework offers a systematic method to justify and rigorously establish such intuitive quasiparticle descriptions. While our approach applies only to a low energy density state (the $|0-\rangle$ state), in the PXP model such quasiparticles have been argued to be responsible for the slow dynamics even at very high energy density (i.e., starting from the infinite temperature CDW state) [58,63,97,103].

To argue that the quasiparticle interpretation applies to the $|0-\rangle$ state, in Fig. 5(d) we plot the energies of the eigenstates with which $|0-\rangle$ has high overlap and compare them to what we would predict if we considered a cartoon model where the metastable state took the form

$$|0-\rangle \sim \sum_{n=0}^{\infty} \sqrt{\frac{\alpha^n e^{-\alpha n}}{n!}} |n \text{ quasiparticles at } k = \pi\rangle, \quad (76)$$

where the “ $|n \text{ quasiparticles}\rangle$ ” state is a cartoon (not actual) quantum state which is approximately a superposition of n noninteracting $k = \pi$ quasiparticles and which is assumed to obey $H|n\rangle = \tilde{\Delta}n|n\rangle$, where $\tilde{\Delta}$ is the single-particle gap above. The α parameter is chosen so the average energy of the cartoon state (76) is equal to $\langle 0 - | H | 0 - \rangle$. There is a surprisingly good fit between this crude model and the numerical data, which suggests that our metastable state is close to a state built out of these quasiparticles and that there exist exact eigenstates of the PXP model which do admit (approximately) an interpretation in terms of these quasiparticles.

From this perspective, our theory of metastability can then provide a rigorous justification for the intuitive

argument that these weakly interacting quasiparticles' slow decay is responsible for the slow thermalization of the $|0-\rangle$ state in the PXP model. We investigate the slow thermalization from the initial state $|0-\rangle$ in Fig. 5(e), where we show the presence of persistent oscillations in the expectation value of operators Z and X on even and odd sublattices—in the $|0-\rangle$ (or $| -0\rangle$) states, such expectation values show large differences between even and odd sites, whereas the differences will be zero in a typical thermal state. The persistence of large oscillations confirms that the metastable state appears athermal over long timescales. In contrast to the expectations from Theorem 9, we do *not* see these correlation functions approximately constant at short timescales. This is not, of course, a contradiction to our rigorous result, but a consequence of finite $R = 7$ and—far more importantly—the very tiny metastability gap Δ , which is not large enough to guarantee slow dynamics. In principle, we could adjust the couplings in the PXP Hamiltonian on odd and even sites to make the $|0-\rangle$ state a metastable state with as large of an R as we wish. For sufficiently large R (albeit small enough R that Δ/ϵ is not too small), Theorem 9 can provide strong bounds on the thermalization time scale that differ from Fermi's golden rule.

Ultimately, the theory of metastability guides us to discover initial states for which we can argue, a little more “carefully,” for slow dynamics in the strict thermodynamic limit. As noted above, we can certainly modify the Hamiltonian to make such metastability-derived bounds increasingly strict, if desired. Hence, our methodology provides a new path toward obtaining controlled models of slow thermalization and constrained dynamics, which we expect could be applied to many other models, in which exact diagonalization for medium-size (R) systems could help to identify metastable short-range-entangled states, for which Theorem 9 can imply slow thermalization.

In some cases, a quasiparticle interpretation can accurately describe the quench dynamics also away from the ground state, as was shown, for example, for an initial state close to the false vacuum in the mixed field quantum Ising chain [31]. Our results hint that, more broadly, metastable states with sufficiently low energy density might have long-lived quasiparticles as well, irrespective of whether they are close to the global ground state. These quasiparticles may be of very different nature from the low energy ones. This is an intriguing possibility as it is far from obvious that such a quasiparticle description would be valid so far from the ground state. Although such quasiparticle interpretations have been proposed to describe the physics of quantum many-body scars in the PXP model by many authors [58,63,97,103], there is not a rigorous understanding of how and why such quasiparticles arise, especially as they seem to survive to very high energies. Our observation that $|0-\rangle$ is metastable does not answer this question, yet our numerical results in Figs. 5(c) and 5(d) suggest that our

theory of metastability may provide valuable new insight into the origin of quasiparticles.

In the literature, nonthermal dynamics in the PXP model is often justified by trying to deform H to a Hamiltonian H_{deform} that has exact scars [63], with $H - H_{\text{deform}}$ a sum of small local terms. However, we are not aware of any systematic algorithm to obtain H_{deform} for general models and further whether there would be any guarantee that the dynamics of H_{deform} would be robust when deforming back to PXP [109,110]; indeed, one might expect Fermi's golden rule to lead to perturbative decay rates at second order in the size of $H - H_{\text{deform}}$. Our Theorem 5 provides useful insights in this regard, which guarantees, e.g., the proximity of H to some other Hamiltonian that stabilizes $|0-\rangle$, along with robustness of dynamics to far higher orders in perturbation theory.

C. Two-dimensional Ising model

In Sec. VA, we construct one-dimensional models with exponentially many finite energy-density metastable states. However, we need to tune the local Hilbert space dimension to get an asymptotically long metastable timescale. We do not have an explicit construction of such a proliferation of metastable states in one dimension in a local qubit model. In contrast, in two or more dimensions, the canonical Ising model H'_0 (58) is itself a system with exponentially many metastable states with asymptotically long lifetimes.

To show this explicitly, we place the qubits on the faces of a two-dimensional square lattice of linear size L (with open boundary conditions) and consider a state $|\psi_0\rangle$ consisting of stripes of all-zero and all-one regions, as shown in Fig. 6. The stripes all have width $\geq \ell$ with a tunable parameter ℓ , and neighboring stripes are separated by a straight line of domain walls (DWs). H'_0 conserves the

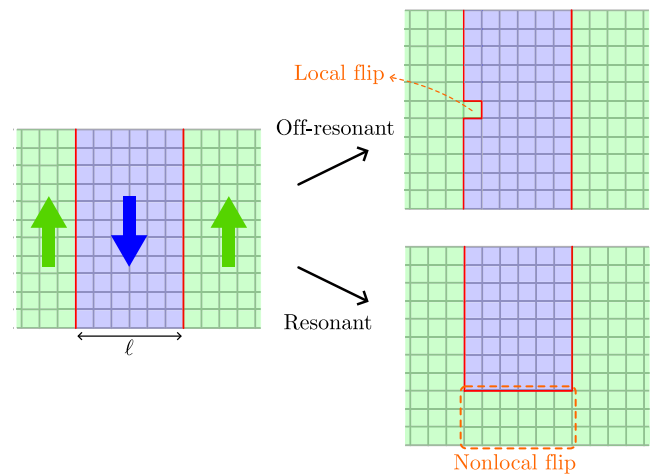


FIG. 6. Left: a metastable state in the 2d Ising model. Right: local operators always raise the energy, consistent with our definition of metastability. A nonlocal operator of radius ℓ is required to find any many-body resonance.

number of DWs, which becomes a crucial dynamical constraint for the system under perturbation, indicated by numerics and perturbative arguments [73,74,111–114]. Here, we provide rigorous lifetime bounds for the particular stripe states. Note that the setup looks similar to inert states in Hilbert space shattering with dipole conservation [68].

With respect to H'_0 , $|\psi_0\rangle$ is metastable with range $R = \ell$: As shown in Fig. 6, any local operator will always create more DWs and increase the energy; in order to preserve energy, one has to flip a region that connects two stripes of the same polarization, whose diameter is at least ℓ . Theorem 9 then implies that under a generic perturbation of local norm ϵ , $|\psi_0\rangle$ has lifetime $t_* = \min(\exp(\Omega(\ell)), \exp[\Omega((\Delta/\epsilon)^{1/3})])$.

In fact, since H'_0 is commuting, we expect Theorem 10 can be generalized to show that the stripe state has $t_* = \exp[\Omega(\ell_{\text{resonant}}^2)]$ with $\ell_{\text{resonant}} = \min(\ell, \Delta/\epsilon)$. This is because the smallest resonant bubble flips $\sim \ell_{\text{resonant}}^2$ qubits: If $\Delta/\epsilon \ll \ell$, the smallest bubbles are false-vacuum-like that melt the false-vacuum stripes from inside or by flipping regions adjacent to the walls; otherwise, if $\Delta/\epsilon \gg \ell$, the dynamics would be dominated by processes in the bottom right in Fig. 6, which connects neighboring stripes of the same polarization by flipping rectangles. Furthermore, if we require that the perturbation preserves the Ising $\mathbb{Z}_2$ symmetry, there is no longer a false vacuum decay as in Sec. IV, so we expect (but do not prove) that $t_* = \exp[\Omega(\ell^2)]$ coming from the latter channel alone, so long as one starts from the genuine metastable state where each stripe corresponds to a “true ground state” of the symmetric Hamiltonian.

If ℓ divides L , we can consider a cartoon where each set of ℓ adjacent columns in the lattice is chosen to be either all 0 or all 1. The metastability radius is at least ℓ , as above, for every such state, and clearly there are $2^{L/\ell}$ such metastable states $|\psi_0\rangle$. This is an undercount of the number of actual metastable states of this kind, as we could consider a more complex mixture of domains, with widths that are not constrained to integer multiples of ℓ . Larger ℓ thus leads to fewer (but still exponential in the linear system size L) metastable states. Unlike the case of shattering from dipole conservation [68], here the DWs have to form straight lines in $|\psi_0\rangle$ and cannot be roughened; otherwise, the DWs are free to move locally, and, thus, by our definition, the state is not metastable. There are exponential-in-volume $\sim 2^{L^2}$ number of such roughened states, and they may belong to a “metastable manifold,” but again, such a generalization is beyond the scope of this work.

VI. CONCLUSION

In this paper, we have developed a mathematical framework for defining metastable pure states in many-body quantum systems with finite-dimensional Hilbert space. Definition 2 says, loosely, that a state $|\psi\rangle$ is

metastable with respect to Hamiltonian H if all few-body perturbations acting on $\leq R$ sites either stabilize $|\psi\rangle$ or raise the average energy of a state by a finite amount. This definition is inspired by a natural definition of metastability in classical spin systems. We showed that this definition of metastability implies a number of remarkable properties. Theorem 5 proves that all short-range-entangled metastable states, obtained from product states in sufficiently short time evolution, are exact eigenstates of a Hamiltonian $H_0 = H - V$, where V is perturbatively small in the metastability radius R . Theorem 9 proved that all metastable states exhibit prethermalization and, in particular, that local correlation functions in the metastable state do not decay until times $t \sim \exp[R^q]$, which are nonperturbatively large in the small parameter R^{-1} .

This theory of metastability has a number of profound implications. Theorem 10 proves that the lifetime of the false vacuum of an Ising-like model in d dimensions is $\exp(\epsilon^{-d})$, which saturates the famous semiclassical prediction of Ref. [7], using quantum field theory. This provides a mathematically rigorous justification for the long lifetime of the false vacuum and establishes a precise quantum mechanical theory of nucleation. We also saw a number of one-dimensional models with finite energy density metastable states, such as the PXP model and a “helical” spin chain where we, in fact, proved the existence of exponentially many metastable states.

There are a number of exciting possible extensions and generalizations of our metastability theory. First, our metastability theory applies only to pure states, but there are certainly metastable mixed states, such as the thermal state describing a supercooled liquid. While such states are presumably well described in practice by the classical theory of metastability [1], it could be formally desirable to understand a quantum theory of such mixed states, to understand whether there are any intrinsically quantum notions of mixed state metastability. We note that recent work [15] may provide a useful starting point for such analyses. A theory of mixed state metastability might provide a rigorous framework to study quantum dielectric breakdown (see, e.g., the cartoon models [115,116]).

Second, to prove the slow false vacuum decay, it was important to use a stronger notion of metastability than Definition 2, which accounts for the “surface tension” of the domain wall between true and false vacua. More generally, on physical grounds, we expect that a state deserves to be called metastable if it lies in a “metastable basin” of states, even when low-energy single-spin flips do exist. Generalizing our formalism to account for such states is a worthwhile endeavor, and it would be interesting if it is relevant for understanding infinite-temperature (approximate) quantum scarring in the PXP model.

Another interesting motivation for extending the theory of metastability along these lines comes from classical spin glasses (and related models) perturbed by quantum

fluctuations. It has recently been proved that such models have localized many-body eigenstates [117] (see also Refs. [118,119]), which ensure the complete breakdown of thermalization; states are no longer just metastable, but genuinely stable. In the proof of Ref. [117], macroscopic energy barriers between different bit strings play an essential role, even though it takes many bit flips to reach the top of the energy barrier. A more general metastability theory that takes into account the paths through configuration space needed to escape local energy barriers could perhaps play an intriguing role in building a more systematic theory for slow dynamics in finite-dimensional quantum glasses and in ergodicity-breaking systems robust to perturbations up to a system-size-dependent order [71,72].

Third, our theory of metastability did not invoke any notions of symmetry, yet symmetry and/or perturbatively weak symmetry breaking do often play important roles in understanding why metastability arises in nature. For example, the slow false vacuum decay arises because the Hamiltonian H is close to one with a spontaneously broken discrete symmetry. Can such symmetry be used to prove stronger bounds? Do models with a “symmetry-derived” metastability have a parametrically stronger notion of metastability than more general models, such as the PXP or helical models we studied? How much of our theory persists if one only has symmetry-protected metastable states (i.e., where we require in the definition of metastability that $\mathcal{O}$ is invariant under a certain symmetry group)?

Fourth, one intriguing result of Ref. [44] is the approximate conservation of quasiparticle number in the Hubbard model with strong on-site interactions and weak kinetic interactions. We expect that an analogous story exists in models with weakly interacting quasiparticle excitations, in which the single quasiparticle dispersion relation is gapped: Whether or not interactions formally preserve quasiparticle number, in sufficiently low particle or energy density states, quasiparticles should be long-lived. Indeed, we saw some numerical evidence for this in the PXP model in Fig. 5. Such metastability of quasiparticles was also used recently to understand slow thermalization dynamics in Floquet driven systems [120], where certain states are extremely hard to heat up, in contrast to typical states. A simple obstruction to using Definition 2 directly is that it is not obvious how to guarantee that no Hermitian operator exists which can remove quasiparticles from a candidate metastable state (which is likely to lower the energy and, thus, render the state not metastable). [121] We expect that operators which change the quasiparticle number either raise or lower the energy by a finite amount, however, and so it may be possible to generalize the spectral filtering techniques we developed for metastability to tackle this problem as well.

It will be desirable to extend our metastability theory to systems with long-range interactions. The presence of sufficiently long-range interactions is known to drastically

modify the theory of phase transitions [122,123] and nucleation [124,125]. Moreover, intriguing slow dynamics enabled by long-range interactions has recently been proposed in many-body quantum systems [126–131]. Despite the important progress in developing Lieb-Robinson bounds for systems with power-law interactions [132–134], generalizing the methods we have developed here to power-law interacting models appears to be an extremely challenging technical problem in full generality. Nevertheless, for special cases where H_0 is commuting, we expect known results on Floquet prethermalization in long-range interacting systems [45,46,135] could be adapted to establish metastability (see Refs. [44,51] for relations between static and Floquet prethermalization), where the crucial step of using Lieb-Robinson bounds to control SWT from evolution under H_0 becomes much simpler.

Finally, an interesting open question concerns the extent to which our results might generalize to models of interacting bosons. Like the case of power-law interactions, there are recent Lieb-Robinson bounds [136–139], especially in one dimension, which are strong enough to suggest that some of our key ideas may generalize to bosonic models, so long as the boson number is a conserved quantity. For more general bosonic models, strong locality bounds are not possible [140], and we expect entirely new techniques would be needed to develop a theory of metastability for these systems.

ACKNOWLEDGMENTS

We thank Alessio Lerose, Olexei Motrunich, Pablo Sala, and Dong Yuan for useful discussions. Tensor network calculations were performed using the TeNPy Library (version 1.0.0) [93]. This work was supported by the Alfred P. Sloan Foundation under Grant No. FG-2020-13795 (A.L.), the Heising-Simons Foundation under Grant No. 2024-4848 (A.L.), and the Department of Energy under Quantum Pathfinder Grant No. DE-SC0024324 (C. Y. and A. L.). F. M. S. acknowledges support provided by the U.S. Department of Energy Office of Science, Office of Advanced Scientific Computing Research (DE-SC0020290); DOE National Quantum Information Science Research Centers, Quantum Systems Accelerator; and by Amazon Web Services, AWS Quantum Program. F. M. S. and A. L. also thank the Kavli Institute for Theoretical Physics, which is supported by the National Science Foundation under Grant No. PHY-1748958, for hospitality as this work was initiated.

APPENDIX A: FINDING A “PRETHERMAL DECOMPOSITION”

This appendix proves Theorem 5. To set the stage for the proof, given the metastable state $|\mathbf{0}\rangle$ and $q = 2$ local Hilbert space dimension, we can take the Hamiltonian H and separate it into Pauli Z strings and others that involve

Pauli- X . It is natural to put Z strings in H into H_0 because they stabilize $|\mathbf{0}\rangle$, while the others into V so that $H = H_0 + V$. The challenge is to show V is small locally.

Here, we illustrate the idea behind our proof with the simple 1D example $H_0 = -\sum_i Z_i Z_{i+1}$, $V = -\epsilon \sum_i X_i$ (the following does not rely on the integrability of $H = H_0 + V$). Because of the X field, $|\mathbf{0}\rangle$ is not the exact ground state of H , and one *gains* energy from V by slightly rotating a single site i by angle $\theta \ll 1$: $V[|\theta\rangle \otimes |\mathbf{0}\rangle_{\text{rest}}] - V[|\mathbf{0}\rangle] \approx -2\epsilon\theta$, where

$$|\theta\rangle := \cos(\theta)|0\rangle + \sin(\theta)|1\rangle. \quad (\text{A1})$$

This rotation also *costs* energy from H_0 : $H_0[|\theta\rangle \otimes |\mathbf{0}\rangle_{\text{rest}}] - H_0[|\mathbf{0}\rangle] \approx 4\theta^2$. Therefore, in order to gain net energy for H , one should choose

$$\theta \approx \epsilon/4, \quad (\text{A2})$$

so that $H[|\theta\rangle \otimes |\mathbf{0}\rangle_{\text{rest}}] - H[|\mathbf{0}\rangle] \approx -\epsilon^2/4$. This does not violate the metastability condition, because $|\theta\rangle \otimes |\mathbf{0}\rangle_{\text{rest}}$ is not orthogonal to $|\mathbf{0}\rangle$: They actually have large overlap $\langle 0|\theta\rangle \approx 1 - \theta^2/2$. However, in a size- R region, one can slightly rotate every other qubit and consider the state $|\phi\rangle = (|\theta\rangle_1 \otimes |\theta\rangle_3 \otimes \cdots \otimes |\theta\rangle_{R-1}) \otimes |\mathbf{0}\rangle_{\text{rest}}$, which now have small overlap

$$\langle \mathbf{0}|\phi\rangle \approx (1 - \theta^2/2)^{R/2}. \quad (\text{A3})$$

On the other hand, $|\phi\rangle$ gains energy from every length-3 motif $|0\theta 0\rangle$ and so has energy smaller than $H[|\mathbf{0}\rangle]$. Therefore, the metastability condition implies that R cannot be too large, because otherwise $|\phi\rangle$ becomes almost orthogonal to $|\mathbf{0}\rangle$. According to Eq. (A3), this requires $R \lesssim \theta^{-2}$, which yields the desired $\epsilon \lesssim R^{-1/2}$ due to Eq. (A2).

1. Proof of Theorem 5

Proof of Theorem 5. For any local term H_S , it can be written as

$$H_S = |\mathbf{0}\rangle_S \langle \mathbf{0}| H_S |\mathbf{0}\rangle_S \langle \mathbf{0}| + (I - |\mathbf{0}\rangle_S \langle \mathbf{0}|) H_S (I - |\mathbf{0}\rangle_S \langle \mathbf{0}|) + [(I - |\mathbf{0}\rangle_S \langle \mathbf{0}|) H_S |\mathbf{0}\rangle_S \langle \mathbf{0}| + \text{H.c.}], \quad (\text{A4})$$

where the projector $|\mathbf{0}\rangle_S \langle \mathbf{0}|$ is a local operator that acts only inside S . The first two terms in Eq. (A4) stabilize $|\mathbf{0}\rangle$. Furthermore, we can expand $(I - |\mathbf{0}\rangle_S \langle \mathbf{0}|) H_S |\mathbf{0}\rangle_S \langle \mathbf{0}| = \sum_{F \subset S: F \neq \emptyset} \epsilon_{S,F} |\varphi_{S,F}\rangle_F \otimes |\mathbf{0}\rangle_{S \setminus F}$, where F can be disconnected, $\epsilon_{S,F}$ is a complex number, and the normalized $|\varphi_{S,F}\rangle_F = \sum_{\mathbf{z} \in \{1, \dots, q-1\}^F} \tilde{c}_{S,F,\mathbf{z}} |\mathbf{z}\rangle_F$ contains computational basis states that are not $|\mathbf{0}\rangle_i$ if and only if $i \in F$. The third term in Eq. (A4) then becomes

$$\begin{aligned} (I - |\mathbf{0}\rangle_S \langle \mathbf{0}|) H_S |\mathbf{0}\rangle_S \langle \mathbf{0}| + \text{H.c.} &= \sum_{F \subset S: F \neq \emptyset} \epsilon_{S,F} |\varphi_{S,F}\rangle_F \langle \mathbf{0}| \otimes |\mathbf{0}\rangle_{S \setminus F} \langle \mathbf{0}| + \text{H.c.} \\ &= \sum_{F \subset S: F \neq \emptyset} \epsilon_{S,F} |\varphi_{S,F}\rangle_F \langle \mathbf{0}| - \epsilon_{S,F} |\varphi_{S,F}\rangle_F \langle \mathbf{0}| \otimes (1 - |\mathbf{0}\rangle_{S \setminus F} \langle \mathbf{0}|) + \text{H.c.} \\ &= \sum_{F \subset S: F \neq \emptyset} \epsilon_{S,F} |\varphi_{S,F}\rangle_F \langle \mathbf{0}| \otimes |\mathbf{0}\rangle_{B_{i(F)} \setminus F} \langle \mathbf{0}| - \epsilon_{S,F} |\varphi_{S,F}\rangle_F \langle \mathbf{0}| \otimes (|\mathbf{0}\rangle_{B_{i(F)} \setminus F} \langle \mathbf{0}| - |\mathbf{0}\rangle_{S \setminus F} \langle \mathbf{0}|) \\ &\quad + \text{H.c.} \end{aligned} \quad (\text{A5})$$

The last line here comes from the following manipulation: For each F that can appear in Eq. (A5) for any S with $\text{diam} S \leq r$ [a parameter that will be chosen as $r \sim \ln R$ later in Eq. (A10a)], $\text{diam} F \leq r$ so it is included in some ball $B_i := B(i, r)$ centered at i with radius r . We choose a function $i(F)$ (independent of S) such that $F \subset B_{i(F)}$ for any such F . This grouping is for later convenience, as we will explain.

We can then combine the first terms (and their conjugates) in Eq. (A5) into such balls by defining

$$\begin{aligned} V_{B_j} &:= \sum_{S: \text{diam} S \leq r} \sum_{F \subset S: F \neq \emptyset, i(F)=j} \epsilon_{S,F} |\varphi_{S,F}\rangle_F \langle \mathbf{0}| \otimes |\mathbf{0}\rangle_{B_{i(F)} \setminus F} \langle \mathbf{0}| \\ &\quad + \text{H.c.} \\ &=: \epsilon_j (|\varphi_j\rangle_{B_j} \langle \mathbf{0}| + \text{H.c.}), \end{aligned} \quad (\text{A6})$$

which is supported in B_j , and $|\varphi_j\rangle$ is normalized and orthogonal to $|\mathbf{0}\rangle$, whose phase makes $\epsilon_j \geq 0$. See Fig. 7(a) for an illustration. Note that we apply translation invariance at the very end of this proof, so for now ϵ_j can be nonuniform. We combine terms in Eq. (A6) so that (i) the number of local terms in the final perturbation $V = \sum_S V_S$ is significantly reduced (number of size- r local terms overlapping a site is $\sim r^d$ instead of $\sim 2^r$). (ii) Because $|\varphi_{S,F}\rangle_F \otimes |\mathbf{0}\rangle_{F^c}$ for different F are orthogonal to each other, so are the $|\varphi_j\rangle$ for different j :

$$(\langle \varphi_i |_{B_i} \otimes \langle \mathbf{0} |_{B_i^c}) (|\varphi_j\rangle_{B_j} \otimes |\mathbf{0}\rangle_{B_j^c}) = 0 \quad \forall i \neq j. \quad (\text{A7})$$

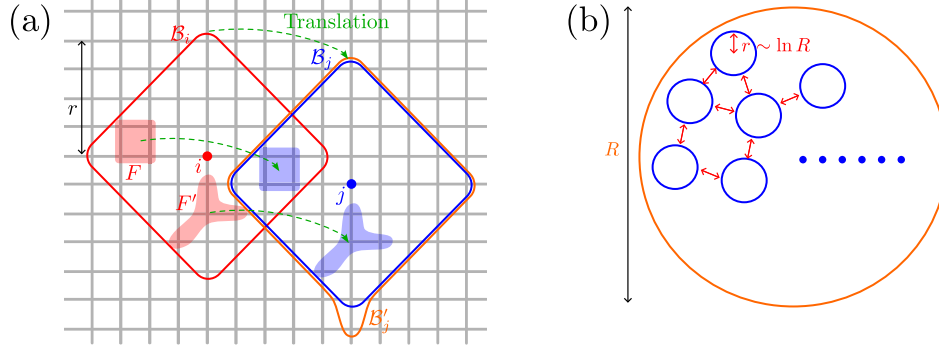


FIG. 7. (a) Terms in H that do not stabilize $|\mathbf{0}\rangle$ are $\propto |\varphi\rangle_F \langle \mathbf{0}|$ (and their conjugates) for some set F , where the string $|\varphi\rangle$ contains no 0 elements. We combine such terms (shaded regions) with sufficiently small support F into balls V_{B_j} of radius r , centered at site j . The red and blue colors represent two balls and their corresponding representative F 's. This procedure can be done in a way respecting translation symmetry. In particular, since the red square F is assigned to B_i , its translated version, the blue square, is assigned to B_j even if it is also a subset of B_i . We also assign a neighboring vertex to each B_j so that they combine to B'_j , in a translation-invariant way. (b) For a ball-like region of size R , we can fill it by smaller balls of radius r where the balls have distance $\geq r$ to each other. The number of balls can be chosen to scale as $\gtrsim (R/r)^d$ while respecting these constraints.

Indeed, since each set F was associated to a unique ball B_i , although $\langle \varphi_i |_{B_i}$ is generically an entangled state, it will have no basis vectors in common with $\langle \varphi_j |_{B_j}$ in the computational basis.

Having defined V_S for the case $S = B_j$ (for some j), the remaining local terms of V are defined by

$$V_S = \begin{cases} (I - |\mathbf{0}\rangle_S \langle \mathbf{0}|) H_S |\mathbf{0}\rangle_S \langle \mathbf{0}| + \text{H.c.}, & S \neq B'_j (\forall j). \\ (I - |\mathbf{0}\rangle_S \langle \mathbf{0}|) H_S |\mathbf{0}\rangle_S \langle \mathbf{0}| + (I - |\mathbf{0}\rangle_{B_j} \langle \mathbf{0}|) H_{B_j} |\mathbf{0}\rangle_{B_j} \langle \mathbf{0}| + \text{H.c.}, & S = B'_j \end{cases} \quad (\text{A8})$$

for any S with $\text{diam} S > r$ and $S \neq B_j$. Here, we have defined B'_j for each j such that B'_j consists of B_j and one extra neighboring site (so $\text{diam} B'_j \leq 2r + 1$). For concreteness, we choose the extra site to be the one right below B_j 's "south pole," as shown in Fig. 7(a). We also dilate the original $S = B_j$ term which can, in principle, arise in Eq. (A4) to B'_j , so that $S = B_j$ do not appear twice in $V = \sum_S V_S$ [V_{B_j} is already defined in Eq. (A6)].

The above procedure of organizing terms in V respects translation symmetry, so that if H is translation invariant (i.e., $H_S = H_{T_x(S)}$ for any S and any translation T_x by

vector x), so is V : Any F appearing in Eq. (A6) can be labeled by its shape $\tilde{F}$ and coordinate x , so F is just $T_x(\tilde{F})$. We first assign a ball of center $i_{\tilde{F}}$ for each shape $\tilde{F}$, and then F is assigned to the ball centered at the $T_x(i_{\tilde{F}})$. The way to assign B'_j for B_j is also translation invariant. The remaining part $H_0 = H - V$ with decomposition $H_{0,S} = H_S - V_S$ then satisfies Eq. (29), because it contains the first two terms in Eq. (A4) along with the second term in Eq. (A5) that all stabilizes $|\mathbf{0}\rangle$.

Our main goal is to bound the local norm of V : For any $\kappa < 2\mu$,

$$\begin{aligned} \|V\|_\kappa &= \max_i \left(\sum_{j: B_j \ni i} e^{\kappa(\text{diam} B_j)^\alpha} \|V_{B_j}\| + \sum_{j: B'_j \ni i} e^{\kappa \text{diam} B'_j} \|V_{B'_j}\| + \sum_{S \ni i: \text{diam} S > r: S \neq B_j, B'_j} e^{\kappa \text{diam} S} \|V_S\| \right) \\ &\leq \max_i \left(\sum_{j \in B_i} e^{\kappa(2r)^\alpha} \epsilon_j + e^\kappa \sum_{S \ni i: \text{diam} S > r} e^{\kappa \text{diam} S} \|H_S\| \right) \\ &\leq c_d (2r)^d e^{\kappa(2r)^\alpha} \max_j \epsilon_j + e^\kappa \max_i \sum_{S \ni i: \text{diam} S > r} e^{(\kappa - 2\mu)(r+1)} e^{2\mu \text{diam} S} \|H_S\| \\ &\leq c_d (2r)^d e^{\kappa(2r)^\alpha} \max_j \epsilon_j + e^\kappa e^{(\kappa - 2\mu)(r+1)} h. \end{aligned} \quad (\text{A9})$$

Here, we treat the two kinds of terms (A6) and (A8) separately and use $i \in \mathcal{B}_j$ is equivalent to $j \in \mathcal{B}_i$ together with $|\mathcal{B}_i| \leq c_d(\text{diam}\mathcal{B}_i)^d \leq c_d(2r)^d$ for the first kind. The second kind is bounded using $\|V_S\| \leq \|H_S\|$ for Eq. (A8) with $S \neq \mathcal{B}'_j$ and the fact that it contains only large support terms in H that decay with exponent 2μ . The special $S = \mathcal{B}'_j$ case is bounded by adding an extra factor e^κ because $e^{\kappa \text{diam}\mathcal{B}'_j} \leq e^\kappa e^{\kappa \text{diam}\mathcal{B}_j}$. As a result, we choose

$$r = \lfloor (2\mu)^{-1} \ln(4hR^d/c_d\Delta) \rfloor + 1, \quad (\text{A10a})$$

$$\kappa = \mu, \quad (\text{A10b})$$

so that the second term in Eq. (A9) is upper bounded by $\frac{1}{2}\sqrt{c_d h \Delta} R^{-d/2} \leq \frac{1}{2}\sqrt{c_d} h R^{-d/2}$ using Eq. (25). Therefore, to show Eq. (30b) it remains to deal with the first term.

To bound $\max_i e_i$, for any j , consider the ball $B(j, R/2)$ of diameter $\leq R$, which contains many radius- r balls $\mathcal{B}_i \subset B(j, R/2)$. Pick a subset of these balls $\{\mathcal{B}_{i_m} \subset B(j, R/2) : m = 1, \dots, M\}$ that are mutually faraway with distance $\gtrsim r$:

$$d(\mathcal{B}_{i_m}, \mathcal{B}_{i_{m'}}) \geq r, \quad \forall m \neq m'. \quad (\text{A11})$$

There always exists such a set of balls of number

$$M \geq c_{\text{ball}} R^d / [c_d(2r)^d], \quad (\text{A12})$$

with constant $0 < c_{\text{ball}} < c_d$ determined by d ; see Fig. 7(b) for illustration.

We aim to show that $\sum_m e_{i_m}^2$ (roughly speaking) cannot be large; otherwise, the metastability condition on H would be violated. In particular, consider state

$$|\phi\rangle := |\mathbf{0}\rangle_{S^c} \otimes_{m=1}^M |\phi_m\rangle, \quad (\text{A13})$$

where $S = \cup_{m=1}^M \mathcal{B}_{i_m}$ is disconnected, and

$$|\phi_m\rangle := C_m \left(|\mathbf{0}\rangle_{\mathcal{B}_{i_m}} - \frac{\epsilon_{i_m}}{\Delta_m} |\varphi_{i_m}\rangle_{\mathcal{B}_{i_m}} \right), \quad (\text{A14})$$

where

$$\Delta_m = H \left[|\varphi_{i_m}\rangle_{\mathcal{B}_{i_m}} \otimes |\mathbf{0}\rangle_{\mathcal{B}_{i_m}^c} \right] \geq \Delta. \quad (\text{A15})$$

Here, C_m is an overall normalization constant

$$C_m = \left(1 + \left(\frac{\epsilon_{i_m}}{\Delta_m} \right)^2 \right)^{-1/2}, \quad (\text{A16})$$

and we assume $H[|\mathbf{0}\rangle] = 0$ without loss of generality. Equation (A14) is approximately the ground state in the two-dimensional subspace $\text{span}(|\mathbf{0}\rangle_{\mathcal{B}_{i_m}}, |\varphi_{i_m}\rangle_{\mathcal{B}_{i_m}}) \otimes |\mathbf{0}\rangle_{\mathcal{B}_{i_m}^c}$ at small ϵ_{i_m} : H restricted to this subspace is

$$H_{\text{restricted}} = \begin{pmatrix} 0 & \epsilon_{i_m} \\ \epsilon_{i_m} & \Delta_m \end{pmatrix} + V_{>r,m}. \quad (\text{A17})$$

Here, all $V_{\mathcal{B}_i}$ with $i \neq i_m$ do not contribute $\langle \mathbf{0} | V_{\mathcal{B}_i} | \varphi_{i_m} \rangle_{\mathcal{B}_{i_m}} \otimes |\mathbf{0}\rangle_{\mathcal{B}_{i_m}^c} = 0$, thanks to the orthogonality condition (A7). $V_{>r,m}$ is the contribution from large-support terms (A8), bounded by

$$\begin{aligned} \|V_{>r,m}\| &\leq \sum_{S' \neq \mathcal{B}_i (\forall i) : S' \cap \mathcal{B}_{i_m} \neq \emptyset, \text{diam} S' > r} \|V_{S'}\| \\ &\leq \sum_{i \in \mathcal{B}_{i_m}} \sum_{S' \ni j : \text{diam} S' > r} \|H_{S'}\| \\ &\leq c_d(2r)^d h e^{-2\mu(r+1)}, \end{aligned} \quad (\text{A18})$$

similarly as the second term in Eq. (A9). Therefore,

$$\begin{aligned} H[|\phi_m\rangle \otimes |\mathbf{0}\rangle_{\mathcal{B}_{i_m}^c}] &\leq C_m^2 \begin{pmatrix} 1 & -\frac{\epsilon_{i_m}}{\Delta_m} \end{pmatrix} \begin{pmatrix} 0 & \epsilon_{i_m} \\ \epsilon_{i_m} & \Delta_m \end{pmatrix} \begin{pmatrix} 1 \\ -\frac{\epsilon_{i_m}}{\Delta_m} \end{pmatrix} \\ &\quad + \|V_{>r,m}\| \leq 0 + \|V_{>r,m}\| \\ &\leq c_d(2r)^d h e^{-2\mu(r+1)}. \end{aligned} \quad (\text{A19})$$

Since the balls $\mathcal{B}_{i_m}$ are chosen to be far $\gtrsim r$ from each other, we expect the flips $|\mathbf{0}\rangle \rightarrow |\phi_m\rangle$ in them change the energy in an almost *additive* way:

$$\begin{aligned} H[|\phi\rangle] &\approx \sum_m H[|\phi_m\rangle \otimes |\mathbf{0}\rangle_{\mathcal{B}_{i_m}^c}] \leq M c_d(2r)^d h e^{-2\mu(r+1)} \\ &\leq c_{\text{ball}} R^d h e^{-2\mu(r+1)} \leq \frac{\Delta}{4}, \end{aligned} \quad (\text{A20})$$

where we use Eq. (A10a). The error comes from terms in H that act on more than one ball:

$$\begin{aligned} H[|\phi\rangle] &- \sum_m H[|\phi_m\rangle \otimes |\mathbf{0}\rangle_{\mathcal{B}_{i_m}^c}] \\ &\leq \sum_{S' : S' \text{ intersects } \geq 1 \text{ balls } \mathcal{B}_{i_m}} \|H_{S'}\| \leq \sum_{i \in S} \sum_{S' \ni i : \text{diam} S' > r} \|H_{S'}\| \\ &\leq c_d R^d h e^{-2\mu(r+1)} \leq \frac{\Delta}{4}. \end{aligned} \quad (\text{A21})$$

For technical simplicity, we have overcounted in the above expression by including $H_{S'}$ that intersect only one ball, as long as $\text{diam} S' > r$; we have also counted each $\|H_{S'}\|$ in every ball that it intersects. As a result,

$$H[|\phi\rangle] \leq \frac{\Delta}{2}. \quad (\text{A22})$$

So in order for metastability to hold, the overlap between $|\phi\rangle$ and $|\mathbf{0}\rangle$ should be large.

We now write

$$|\phi\rangle = C_\phi|\mathbf{0}\rangle + \sqrt{1 - C_\phi^2}|\mathbf{0}\rangle^\perp, \quad (\text{A23})$$

where $|\mathbf{0}\rangle^\perp$ is orthogonal to $|\mathbf{0}\rangle$ and $C_\phi = \prod_m C_m$. According to Eq. (23),

$$H[|\mathbf{0}\rangle^\perp] \geq \Delta. \quad (\text{A24})$$

Combining this with Eq. (A22) and $H[|\mathbf{0}\rangle] = 0$,

$$\begin{aligned} \frac{\Delta}{2} &\geq H[|\phi\rangle] \\ &= 0 + (1 - C_\phi^2)H[|\mathbf{0}\rangle^\perp] + C_\phi\sqrt{1 - C_\phi^2}(\langle\mathbf{0}|H|\mathbf{0}\rangle^\perp + \text{H.c.}) \\ &\geq (1 - C_\phi^2)\Delta - 2c_d h C_\phi R^d, \end{aligned} \quad (\text{A25})$$

where we use

$$|\langle\mathbf{0}|H|\mathbf{0}\rangle^\perp| \leq \left\| \sum_{S': S' \cap S \neq \emptyset} H_{S'} \right\| \leq \sum_{i \in S} \sum_{S' \ni i} \|H_{S'}\| \leq c_d R^d h. \quad (\text{A26})$$

Obviously, Eq. (A25) cannot hold for $C_\phi = 0$; solving a quadratic equation, we see that any solution obeys

$$\begin{aligned} C_\phi &\geq \sqrt{\eta^2 + \frac{1}{2}} - \eta = \eta \left(\sqrt{1 + \frac{1}{2\eta^2}} - 1 \right) \geq \eta \times \frac{2}{5} \times \frac{1}{2\eta^2} \\ &= \frac{\Delta}{5c_d h R^d}, \end{aligned} \quad (\text{A27})$$

where

$$\eta = c_d h R^d / \Delta \geq c_d R^d \geq 1, \quad (\text{A28})$$

where $2/5$ is from monotonicity of function $(\sqrt{1+x} - 1)/x \geq 2(\sqrt{1+1/2} - 1) > 2/5$ at $x \leq 1/2$. Using C_m in Eq. (A14), Eq. (A27) becomes

$$\begin{aligned} 2 \ln \frac{5c_d h R^d}{\Delta} &\geq \sum_{m=1}^M \ln \left[1 + \left(\frac{\epsilon_{i_m}}{\Delta_m} \right)^2 \right] \\ &\geq \sum_{m=1}^M \ln \left(1 + \frac{\epsilon_{i_m}^2}{[c_d(2r)^d h]^2} \right), \end{aligned} \quad (\text{A29})$$

where we use $\Delta_m \leq c_d(2r)^d h$ because only $H_{S'}$'s that intersect with $\mathcal{B}_{j_m}$ contribute similar to Eq. (A26).

If the system is translation invariant (the procedure to define V respects translation invariance, as we have explained), $\epsilon_{i_m} := \epsilon$ with

$$\ln \left(1 + \frac{\epsilon^2}{c_d^2(2r)^{2d} h^2} \right) \leq \frac{2c_d(2r)^d}{c_{\text{ball}} R^d} \ln \frac{5c_d h R^d}{\Delta} \leq \ln 3. \quad (\text{A30})$$

Here, we use the bound (A12) on M , along with the fact that $\sim R^{-d}(\ln R)^{d+1}$ is bounded by $\ln 3$ at sufficiently large $R \geq c_R$, with c_R determined by $d, h/\Delta, \mu$. We then deduce

$$\begin{aligned} \epsilon^2 &\leq 2c_d^2(2r)^{2d} h^2 \frac{2c_d(2r)^d}{c_{\text{ball}} R^d} \ln \frac{5c_d h R^d}{\Delta} \\ &\leq c_{\text{TI}} h^2 (\ln R)^{3d+1} R^{-d}, \end{aligned} \quad (\text{A31})$$

where we use $\ln(1+x) \geq x/2$ if $x \leq 2$, with constant c_{TI} determined by $d, h/\Delta, \mu$. Plugging Eq. (A31) into Eq. (A9) with Eq. (A10b) leads to Eq. (30b), because we have bounded the second term of Eq. (A9) above, while the first term uses $(\ln R)^{(5d+1)/2} e^{c(\ln R)^\alpha} = O(e^{\delta \ln R})$ for any constant $\delta > 0$. This also bounds the local norm of H_0 :

$$\begin{aligned} \sum_{S \ni i} \|H_{0,S}\| e^{2\mu_0 \text{diam} S} &\leq \sum_{S \ni i} \|H_S\| e^{2\mu_0 \text{diam} S} + \sum_{S \ni i} \|V_S\| e^{2\mu_0 \text{diam} S} \\ &\leq h + c_d(2r)^d e^{2\mu_0(2r)} \max_j \epsilon_j + h \leq c_H h, \end{aligned} \quad (\text{A32})$$

for some constant c_H determined by $d, h/\Delta, \mu$, where we generalize Eq. (A9) to bound the V contribution and use $\mu_0 = \mu/5$ so that the second term in the second line scales as $(\ln R)^{(5d+1)/2} R^{2d/5} R^{-d/2} = O(1)$. Finally, $(H_0, |\mathbf{0}\rangle)$ is $(\Delta/2, c'_\delta R^{1/2-\delta/d})$ metastable for $c'_\delta = (\Delta/2c_d c_\delta h)^{1/d}$, because for any S with $\text{diam} S \leq c'_\delta R^{1/2-\delta/d}$, and any orthogonal state $|\phi\rangle_S$,

$$\begin{aligned} H_0[|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}] &\geq H[|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}] - V[|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}] \\ &\geq \Delta - \sum_{S': S' \cap S \neq \emptyset} \|V_{S'}\| \geq \Delta - |S| \|V\|_\mu \\ &\geq \frac{\Delta}{2}, \end{aligned} \quad (\text{A33})$$

where we use $|S| \leq \Delta/(2c_\delta h) R^{d/2-\delta}$ and Eq. (30b). ■

2. Relaxing the assumption of translation invariance

In the above proof, translation invariance is used only to simplify Eq. (A29). Therefore, the conclusion generalizes directly as long as the *strength* is uniformly bounded $\epsilon_{i_m} \leq \epsilon$, even if the form of perturbation can be highly translational asymmetric. Without translation invariance, it is not necessarily clear how to guarantee such uniform boundedness. Still, even without such uniform boundedness, Eq. (A29) guarantees that, although some $\epsilon_{i_m} \gtrsim \Delta$ can be large, the number of these large ones in the size- R ball is bounded by $O(\ln R) \ll R^d$. In other words, any perturbation to H_0 that is strong must itself be isolated from other strong perturbations.

We can consider a contrived example in $d = 1$ to illustrate the points above. Consider

$$H = - \sum_{i=1}^{NR-1} Z_i Z_{i+1} - \epsilon \sum_{i=1}^N X_{Ri}, \quad (\text{A34})$$

where we do not require that $|\epsilon| \ll 1$, since it acts once every R sites. As this Hamiltonian commutes with Z_i if i is not divisible by R , it is exactly solvable. Proceeding without this observation, we observe that $(H, |\mathbf{0}\rangle)$ is $(1, R)$ metastable—focusing, e.g., on $i = R$, while a rotation by $O(\epsilon)$ angle clearly lowers the energy, any *orthogonal* state raises the energy by $O(1)$ —recall the discussion around Eq. (A3). Any other orthogonal states are also of higher energy, because $|\mathbf{0}\rangle$ has $> 1/2$ fidelity with the true gapped ground state $|\theta\rangle_i \otimes |\mathbf{0}\rangle_{\text{other}}$. However, dynamically the qubit at R can be flipped to negative polarization due to the large X field. In this case, one should use $(|\theta\rangle_R \otimes |\theta\rangle_{2R} \otimes \dots) \otimes |\mathbf{0}\rangle_{\text{rest}}$ for the metastable state in the first place, where the parameter θ for Eq. (A1) minimizes the “local energy” of the three-qubit state $|0\theta\theta\rangle_{R-1,R,R+1}$, measured by $-h(Z_{R-1}Z_R + Z_RZ_{R+1}) + \epsilon X_R$. Indeed, this state is $(2, R)$ metastable and also dynamically stable because it is an eigenstate of H . In other words, we expect that, beyond uniform boundedness, more general cases of metastable H can be captured by the perturbation scenario $H = H_0 + V$ with small V , by choosing carefully the metastable state.

3. On the metastability radius in Theorem 5

In Theorem 5, the metastability radius for H_0 is shrunk from the metastability radius R of the original H , to a smaller $R_0 \sim \sqrt{R}$ (setting $\delta \rightarrow 0$). There is a similar square root in the perturbation strength (30b). Here, we show that such shrinking is, unfortunately, physical by presenting a simple example that saturates this scaling.

Consider the qubit-chain Hamiltonian

$$H = \sum_{i=1}^{N-1} (1 - P_{00,++})_{i,i+1} + \epsilon \sum_{i=1}^N (Z_i + X_i), \quad (\text{A35})$$

where

$$P_{00,++} := |00\rangle\langle 00| + \frac{(|01\rangle + |10\rangle + |11\rangle)(\langle 01| + \langle 10| + \langle 11|)}{3} \quad (\text{A36})$$

is a two-qubit projector onto the two-dimensional subspace $\text{span}(|00\rangle, |++\rangle)$, where $|+\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle)$. If $\epsilon = 0$, H is frustration-free and the ground states of H can be found exactly (see Theorem 3 in Ref. [141]) to be $\text{span}(|\mathbf{0}\rangle, |++\dots+\rangle)$. Moreover, the two-dimensional ground subspace is gapped, featuring a spontaneously

broken $\mathbb{Z}_2$ symmetry generated by Hadamard gates (i.e., flipping $Z_i \leftrightarrow X_i$).

We focus on the metastable state $|\mathbf{0}\rangle$ when turning on ϵ . The metastability condition (23) can be thought of as restricting Hamiltonian (A35) to a length- $(R+2)$ chain with a fixed boundary condition that the two boundary qubits are set to 0 and asking whether $|\mathbf{0}\rangle$ has large overlap with the ground state in this subspace and whether there is a gap above.

Since the $Z + X$ perturbation preserves the $\mathbb{Z}_2$ symmetry, for periodic boundaries the gap and the degenerate ground states are expected to be stable if ϵ is smaller than some constant. More precisely, the perturbed ground states are $e^A|\mathbf{0}\rangle$ and $e^A|+\dots+\rangle$, where A is quasilocal with local norm $\sim \epsilon$. The fixed boundary condition for the length- $(R+2)$ chain picks the single ground state $e^A|\mathbf{0}\rangle$, because the other one violates the boundary terms. The metastability radius, i.e., the maximal R such that $|\mathbf{0}\rangle$ has large overlap with $e^A|\mathbf{0}\rangle$, is then expected to scale as

$$R \sim \epsilon^{-2}, \quad (\text{A37})$$

where the power of 2 is analogous to the coefficient of θ^2 in Eq. (A3). In Fig. 8(a), we verify this scaling numerically by exact diagonalization up to $R = 22$.

Since the only term in Eq. (A35) that does not stabilize $|\mathbf{0}\rangle$ is the X term, it is natural to separate $H = H_0 + V$ with $V = \epsilon \sum_i X_i$, so that H_0 stabilizes $|\mathbf{0}\rangle$. Indeed, the procedure in the above proof finds this separation with $r = 0$ because the X term is just one-local. Therefore, $\|V\|_\kappa \sim 1/\sqrt{R}$, the scaling in Eq. (30b), is saturated. On the other hand, H_0 is perturbing the $\epsilon = 0$ Hamiltonian with ϵZ that explicitly breaks the $\mathbb{Z}_2$ symmetry, so that $|\mathbf{0}\rangle$ becomes the false vacuum. The metastability radius of H_0 is then

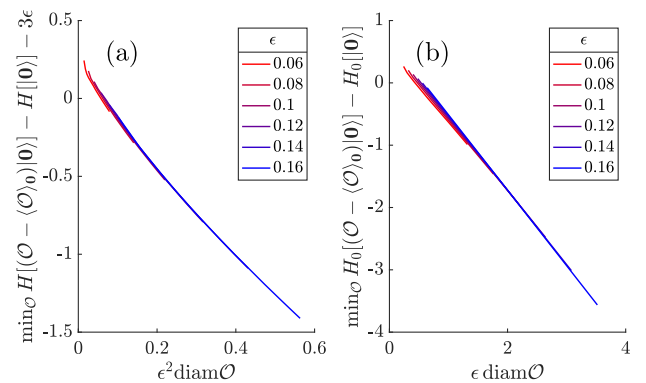


FIG. 8. An 1D example (A35) saturating $R_0 \sim \sqrt{R}$ and $\|V\|_\kappa \sim R^{-d/2}$ when decomposed by $H = H_0 + V$ according to Theorem 5. In particular, each color represents a different ϵ , and the collapse of the different colors indicate scalings (A37) (a) and (A38) (b). To make the collapse more visible in (a), the energy difference has offset -3ϵ , which vanishes at small ϵ so does not change the conclusion.

$$R_0 \sim \epsilon^{-1}, \quad (\text{A38})$$

because bubbles containing all-+ of this length scale become resonant to $|\mathbf{0}\rangle$. This satisfies $R_0 \sim \sqrt{R}$ and is verified numerically in Fig. 8(b).

To summarize this example, the shrinking of $R_0 \sim \sqrt{R}$ comes from the fact that, the part of the perturbation $V(\epsilon X)$ that would naively evolve $|\mathbf{0}\rangle$, actually restores the $\mathbb{Z}_2$ Hadamard symmetry, such that this perturbation actually increases the metastability radius from R_0 to R . This example also highlights that our Theorem 9 does not capture stability in the thermodynamic limit: It proves $t_* \sim e^{-R_0} \sim e^{-1/\epsilon}$ for this example, while the physical expectation is actually $t_* = \infty$ because of the robustness against the symmetry-preserving perturbation $X + Z$.

APPENDIX B: DIAGONALIZING METASTABLE SYSTEMS VIA SCHRIEFFER-WOLFF TRANSFORMATIONS

This appendix is organized around the proof of Theorem 9, which proves that the metastability condition in Definition 2 implies nonperturbatively slow dynamics in local correlation functions, where the metastable state is a product state. As discussed in Sec. II, this is equivalent (up to a quasilocal change of basis) to any model where $|\psi_0\rangle$ is SRE. We first present the proof, with subsequent subsections used to prove technical propositions for some intermediate steps. In the final subsection, we generalize the results to general $|\psi_0\rangle$, which requires further technical modifications to a few steps.

1. Proof of Theorem 9

Proof of Theorem 9. We assume $|\psi_0\rangle = |\mathbf{0}\rangle$ without loss of generality. We use an iterative SWT procedure to find a quasilocal unitary U that makes $(H_0, V, |\mathbf{0}\rangle)$ $(v_*, \mathbb{d}_*, a_*, \kappa_*)$ locally diagonalizable, where

$$v_* = \tilde{c}_* \epsilon \left(2^{-k_*} + \sum_{k=1}^{k_*-1} 2^{-k} e^{-\frac{\kappa_k}{2}(R/2)^\alpha} \right), \quad (\text{B1a})$$

$$\mathbb{d}_* = c_D \epsilon, \quad (\text{B1b})$$

$$a_* = c_A \epsilon. \quad (\text{B1c})$$

Here,

$$k_* := \left\lceil c_* \left(\frac{\Delta}{\epsilon} \right)^{\frac{2\alpha-1}{2\alpha-1}} \right\rceil, \quad (\text{B2})$$

and κ_k is defined below. In fact, the unitaries constructed below are exactly the same as those in Ref. [16] that proves prethermalization for perturbing a gapped H_0 ; the crucial

observation here is that such SWT is useful for the weaker metastable condition instead of a global gap. To be self-contained, we describe the iterative procedure as follows.

Suppose at the k th step, we have constructed a unitary U_{k-1} such that the Hamiltonian is rotated to

$$U_{k-1}^\dagger H U_{k-1} = H_0 + D_k + V_k. \quad (\text{B3})$$

Here, D_k is the effective Hamiltonian at this order that roughly stabilizes $|\mathbf{0}\rangle$ (as we see), and V_k is the remaining perturbation that is roughly suppressed to the order of e^k . For example, $k = 1$ corresponds to the original Hamiltonian with $D_1 = 0$, $U_0 = I$, and $V_1 = V$. For $k = k_*$, we have $U = U_{k_*-1}$ and $H_* + V_* = H_0 + D_{k_*} + V_{k_*}$. Note that $V_* \neq V_{k_*}$ as we move terms around in the end of the proof.

Defining superoperator $\mathcal{L}_0$ by

$$\mathcal{L}_0 \mathcal{O} := i[H_0, \mathcal{O}], \quad (\text{B4})$$

we choose

$$A_k = \mathbb{A} V_k := i \int_{-\infty}^{\infty} dt W(t) e^{t \mathcal{L}_0} V_k \quad (\text{B5})$$

[$\mathbb{A}$ is a superoperator that acts on operators, while the function $W(t)$ will be defined shortly to rotate to the next order so that $U_k = U_{k-1} e^{A_k}$], and

$$\begin{aligned} e^{-A_k} (H_0 + D_k + V_k) \\ = H_0 + D_{k+1} + V_{k+1}, \quad \text{where } A_k = [A_k, \cdot]. \end{aligned} \quad (\text{B6})$$

Here,

$$D_{k+1} := D_k + \mathbb{P} V_k, \quad (\text{B7})$$

and V_{k+1} is defined by Eq. (B6). Here, the superoperator $\mathbb{P}$ acting on any operator $\mathcal{O}$ is defined similarly as $\mathbb{A}$ in Eq. (B5):

$$\mathbb{P} \mathcal{O} := \int_{-\infty}^{\infty} dt w(t) e^{t \mathcal{L}_0} \mathcal{O}. \quad (\text{B8})$$

The real kernel functions $w(t)$ and $W(t)$ are the same as Ref. [16] with the only difference of rescaling $\Delta \rightarrow \Delta/2$; for example,

$$\begin{aligned} w(t) &:= c_\Delta \frac{\Delta}{2} \prod_{n=1}^{\infty} \left(\frac{\sin a_n t}{a_n t} \right)^2, \quad \text{where } a_1 = c_1 \Delta/2, \\ a_n &= \frac{a_1}{n \ln^2 n}, \quad \forall n \geq 2. \end{aligned} \quad (\text{B9})$$

Here, $c_1 \approx 0.161$ is chosen such that $\sum_{n=1}^{\infty} a_n = \Delta/4$, and $c_\Delta \in (1/(2\pi), 1/\pi)$ is a pure number chosen so that the

function is normalized $\int_{-\infty}^{\infty} dt w(t) = 1$. $w(t)$ decays almost exponentially at large t :

$$w(t) \leq \frac{e^2 \Delta^2}{2} t e^{-\frac{2}{7 \ln^2(\Delta t/2)}}, \quad \text{if } t \geq 2e^{1/\sqrt{2}}/\Delta, \quad (\text{B10})$$

and has a compact Fourier transform $\hat{w}(E)$:

$$\hat{w}(E) = 0, \quad \forall |E| \geq \Delta/2. \quad (\text{B11})$$

The odd function $W(t)$ is then defined as an integral over $w(t)$:

$$W(t) = -W(-t) := \int_t^{\infty} ds w(s) \quad (t > 0). \quad (\text{B12})$$

As one can verify from Eqs. (B8) and (B11), $\mathbb{P}\mathcal{O}$ annihilates all matrix elements $\langle E|\mathcal{O}|E' \rangle$ of $\mathcal{O}$ that connect eigenstates whose energies $|E - E'| \leq \Delta/2$. We refer to $\mathbb{P}$ as a superprojector. On the other hand, Eq. (B12) leads to

$$[H_0, A_k] + (1 - \mathbb{P})V_k = 0; \quad (\text{B13})$$

namely, at first order, the rotated H_0 , $(e^{-A_k} - 1)H_0 \approx [H_0, A_k]$, cancels the off-resonant perturbation $(1 - \mathbb{P})V_k$.

The perturbation at next order is then

$$\begin{aligned} V_{k+1} &= e^{-A_k}(H_0 + D_k + V_k) - H_0 - (D_k + \mathbb{P}V_k) \\ &= (e^{-A_k} - 1)H_0 + (1 - \mathbb{P})V_k + (e^{-A_k} - 1)(D_k + V_k) \\ &= \int_0^1 ds (e^{-sA_k} - 1)(\mathbb{P} - 1)V_k + (e^{-A_k} - 1)(D_k + V_k). \end{aligned} \quad (\text{B14})$$

To bound the locality of V_k , etc., in the above procedure, we invoke the following two propositions adapted from Ref. [16].

Proposition 11 (Proposition 5 in Ref. [16]). Suppose H_0 has (μ, u) -LRB. Let $0 < \kappa' < \kappa$, with $\delta\kappa = \kappa - \kappa'$. If

$$\kappa \leq \min(\mu/5, 1/e), \quad (\text{B15})$$

then

$$\|\mathbb{P}\mathcal{O}\|_{\kappa'} \Delta \|\mathbb{A}\mathcal{O}\|_{\kappa'} \leq c_w (-\ln \delta\kappa)^{d-1} \|\mathcal{O}\|_{\kappa}, \quad (\text{B16})$$

where c_w is determined by $\alpha, d, \mu, u/\Delta$.

Below, when applying this proposition, the condition (B15) will always be satisfied because we use only $\kappa \leq \kappa_1$ and κ_1 is assumed to be bounded (45).

Proposition 12 (Proposition 6 in Ref. [16]). If

$$\|A\|_{\kappa} \leq (\delta\kappa)^{\frac{3d+1}{\alpha}}, \quad (\text{B17})$$

with $\delta\kappa < \kappa \leq 1/e$, then

$$\|e^{-A}\mathcal{O} - \mathcal{O}\|_{\kappa'} \leq c_- (\delta\kappa)^{-\frac{2d+1}{\alpha}} \|A\|_{\kappa} \|\mathcal{O}\|_{\kappa}, \quad (\text{B18})$$

where $\kappa' = \kappa - \delta\kappa$ and c_- is determined by α, d .

Note that c_w and c_- do not depend on κ_1 anymore comparing to the statements in Ref. [16], because we have explicit upper bound $\kappa \leq 1/e$, and c_w, c_- do not blow up at small κ_1 by tracing the proof. One can see this latter point for c_- more directly just from the statement (B18) in Ref. [16]: It holds regardless of how small κ is with respect to κ_1 , which does not appear on both sides.

With the above two propositions, we follow Ref. [16] to define

$$\begin{aligned} \kappa_k &:= \frac{\kappa_1}{2} \left(1 + \frac{1}{1 + \ln k} \right) > \frac{\kappa_1}{2} =: \kappa_*, \quad 2\delta\kappa_k := \kappa_{k-1} - \kappa_k, \\ \mathbb{d}_k &= \|D_k\|_{\kappa_k}, \quad v_k := \|V_k\|_{\kappa_k}, \quad \tilde{v}_k = c_w (-\ln \delta\kappa_{k+1})^{d-1} v_k, \end{aligned} \quad (\text{B19})$$

and ultimately obtain a iteration relation: If

$$\tilde{v}_{k-1} \leq c_a (k \ln^2 k)^{-\frac{3d+1}{\alpha}} \Delta \quad (\text{B20})$$

holds, then

$$\mathbb{d}_k \leq \mathbb{d}_{k-1} + \tilde{v}_k, \quad (\text{B21a})$$

$$\begin{aligned} \tilde{v}_k &\leq \frac{\tilde{c}_v}{\Delta} (k \ln^2 k)^{\frac{2d-1}{\alpha}} (\ln k)^{d-1} \tilde{v}_{k-1} (\mathbb{d}_{k-1} + \tilde{c}_w \tilde{v}_{k-1}) \\ &\leq \frac{c_v}{\Delta} k^{\frac{2d-1}{2\alpha-1}} \tilde{v}_{k-1} (\mathbb{d}_{k-1} + \tilde{c}_w \tilde{v}_{k-1}). \end{aligned} \quad (\text{B21b})$$

Here, all constants $c_a, \tilde{c}_v, \tilde{c}_w$, and c_v are determined by $\kappa_1, c_-,$ and c_w . In Eq. (B21b), we have replaced the two power functions of $\ln k$ by power functions of k that multiply to $k^{\frac{2d-1}{2\alpha-1} \frac{2d-1}{\alpha}}$, with the price of adjusting the prefactor. Along the way, we fix a minus-sign typo in the exponent of the corresponding Eq. (S54) in Ref. [16].

One can verify that if Eq. (46) holds for a sufficiently small constant c_v determined by c_a and c_v , then at the initial step $k = 2$, Eq. (B20) is satisfied and Eq. (B21b) becomes

$$\tilde{v}_2 \leq \tilde{v}_1/2. \quad (\text{B22})$$

By iteration,

$$\tilde{v}_k \leq \tilde{v}_{k-1}/2 \leq \dots \leq 2^{1-k} \tilde{v}_1, \quad (\text{B23})$$

as long as $k < k_*$ where the power of k factor in Eq. (B21b) has not grown too large comparing to the suppression factor

$$\mathbb{d}_{k-1} \leq \sum_{k'=1}^{k-1} \tilde{v}_k \leq 2\tilde{v}_1. \quad (\text{B24})$$

In this regime, $\tilde{v}_k$ decays faster than the right-hand side of Eq. (B20), so it is automatically satisfied for all k (as long as the constant c_V is chosen sufficiently small). Equations (B23) and (B24) at the last step prove Eq. (B1b). Similarly, Eq. (B1c) comes from Eqs. (B16) and (B23):

$$a_* \leq \max_{k=1, \dots, k_*-1} 2^k \|A_k\|_{\kappa_k - \delta \kappa_{k+1}} \leq \max_{k=1, \dots, k_*-1} 2^k \tilde{v}_k \leq 2\tilde{v}_1. \quad (\text{B25})$$

Here, we can take maximum over k , because the total unitary U can be written as

$$U = \mathcal{T} \exp \left[\int_0^1 ds A(s) \right], \quad \text{where}$$

$$A(s) = \begin{cases} 0, & s < 2^{1-k_*} \\ 2^k A_k, & 2^{-k} \leq s < 2^{1-k}, \quad (k = 1, \dots, k_* - 1). \end{cases} \quad (\text{B26})$$

It remains to define V_* and show Eq. (B1a). Since

$$D_{k_*} = \mathbb{P}(V_1 + \dots + V_{k_*-1}) \quad (\text{B27})$$

does not stabilize $|\mathbf{0}\rangle$, we need to extract such contribution and put it into V_* . We use the following proposition, proven later, for each $\mathbb{P}V_k$.

Proposition 13. Suppose H_0 satisfies Eq. (44) and has (μ, u) -LRB (17). Suppose $(H_0, |\mathbf{0}\rangle)$ is (Δ, R) metastable where H_0 stabilizes $|\mathbf{0}\rangle$ as Eq. (29), and $R \geq c_R$ is sufficiently large with c_R determined by $d, \mu, \Delta/h_0$, and Δ/u . For any κ satisfying Eq. (B15) and any Hermitian V , there exists Hermitian V^P that stabilizes $|\mathbf{0}\rangle$, and

$$\|\mathbb{P}V - V^P\|_0 \leq 3c_w (-\ln(\kappa/2))^{d-1} e^{-\frac{\kappa}{2}(R/2)^\alpha} \|V\|_\kappa. \quad (\text{B28})$$

As a result, we define

$$H_* = H_0 + \sum_{k=1}^{k_*-1} V_k^P, \quad (\text{B29})$$

that satisfies Eq. (36), and

$$V_* := V_{k_*} + \sum_{k=1}^{k_*-1} (\mathbb{P}V_k - V_k^P), \quad (\text{B30})$$

with bound

$$\begin{aligned} \|V_*\|_0 &\leq \|V_{k_*}\|_{\kappa_{k_*}} + 3c_w \sum_{k=1}^{k_*-1} (-\ln(\kappa_k/2))^{d-1} e^{-\frac{\kappa_k}{2}(R/2)^\alpha} \|V_k\|_{\kappa_k} \\ &\leq \frac{c'_V}{2} \left(\tilde{v}_{k_*} + \sum_{k=1}^{k_*-1} e^{-\frac{\kappa_k}{2}(R/2)^\alpha} \tilde{v}_k \right) \\ &\leq c'_V \tilde{v}_1 \left(2^{-k_*} + \sum_{k=1}^{k_*-1} 2^{-k} e^{-\frac{\kappa_k}{2}(R/2)^\alpha} \right), \end{aligned} \quad (\text{B31})$$

for some constant c'_V , where we use Eqs. (B19) and (B23). This proves Eq. (B1a).

Finally, the bound (B1a) can be simplified by

$$\begin{aligned} v_* &\leq \tilde{c}_* \epsilon 2^{-k_*} + \tilde{c}_* \epsilon k_* \max_{1 \leq k \leq k_*} 2^{-k} e^{-\frac{\kappa_k}{2}(R/2)^\alpha} \\ &\leq \exp \left[-\Omega \left(\left(\frac{\Delta}{\epsilon} \right)^{\frac{2\alpha-1}{2d-1}} \right) \right] \\ &\quad + \epsilon^{\frac{2(d-\alpha)}{2d-1}} \exp \left[-\Omega \left(\frac{R^\alpha}{\ln[\min(R, \Delta/\epsilon)]} \right) \right], \end{aligned} \quad (\text{B32})$$

where we use that function $ck + R^\alpha / \ln k$ with constant c is minimized at $k/\ln^2 k \sim R^\alpha$ with value $\sim R^\alpha / \ln k \sim R^\alpha / \ln R$. Applying Proposition 8 with Eq. (B32) yields Eq. (47) and completes the proof. ■

2. Local operators after superprojection almost stabilizes the metastable state

The goal is now to prove Proposition 13. Since $\mathbb{P}V = \sum_S \mathbb{P}V_S$, we first focus on one local operator $\mathcal{O} = V_S$ in this subsection, showing an approximation $\mathcal{O}^P \approx \mathbb{P}\mathcal{O}$ that stabilizes the state, and combine different V_S terms in the end of Appendix B 4. The idea for a single local operator is as follows: We know $\mathbb{P}\mathcal{O}$ is quasilocal around the original $\mathcal{O}$ from Proposition 11. Suppose $\mathbb{P}\mathcal{O}$ is exactly localized in a region of diameter R , then $|\phi\rangle := (\mathbb{P}\mathcal{O} - \langle \mathbb{P}\mathcal{O} \rangle_0) |\mathbf{0}\rangle$ has energy $\geq \Delta$ from the metastability condition (assuming $|\mathbf{0}\rangle$ has energy 0). However, $\mathbb{P}\mathcal{O}|\mathbf{0}\rangle$ has no support on energies $> \Delta/2$ from Eq. (B11): The only resolution is then $|\phi\rangle = 0$ —i.e., $\mathbb{P}\mathcal{O}$ exactly stabilizes $|\mathbf{0}\rangle$. In reality, although $\mathbb{P}\mathcal{O}$ is not exactly localized inside this size- R region, the leakage outside is exponentially small in R ; this leads to an approximate stabilizer $\mathcal{O}^P \approx \mathbb{P}\mathcal{O}$ instead, with exponentially small error. This last step requires a technical lemma in terms of energy distributions of locally-excited states, which we prove in Appendix B 3.

For a single local operator $\mathcal{O}$ supported on S_0 , $\mathbb{P}\mathcal{O}$ can be decomposed to two parts, a nearby part and a faraway one with cutoff distance r :

$$\begin{aligned} \mathbb{P}\mathcal{O} &= \mathbb{Q}'_r \mathbb{P}\mathcal{O} + (1 - \mathbb{Q}'_r) \mathbb{P}\mathcal{O} \\ &= \mathbb{Q}'_r \mathbb{P}\mathcal{O} + \int_{-\infty}^{\infty} dt w(t) (1 - \mathbb{Q}'_r) (e^{t\mathcal{L}_0} \mathcal{O}), \end{aligned} \quad (\text{B33})$$

where the superoperator $\mathbb{Q}'_r$ is defined by

$$\mathbb{Q}'_r \mathcal{O} := \int_{\text{Haar outside } B(S_0, r)} dU U^\dagger \mathcal{O} U, \quad (\text{B34})$$

selects the part of the operator that acts on $B(S_0, r)$, i.e., sites of distance $\leq r$ to the original support S_0 . The faraway part will be negligible for sufficiently large r due to the locality of $\mathbb{P}$ [see, e.g., Eq. (B16)]. In the following, we show that the nearby part almost stabilizes $|\mathbf{0}\rangle$ from the metastability condition.

Proposition 14. Suppose H_0 satisfies Eq. (44) and has (μ, u) -LRB (17). Suppose $(H_0, |\mathbf{0}\rangle)$ is (Δ, R) metastable where $H_0|\mathbf{0}\rangle = 0$. [142] There exists a constant c_R determined by $d, \mu, \Delta/h_0$, and Δ/u , such that if

$$R \geq c_R, \quad (\text{B35})$$

the following holds.

For any local Hermitian operator $\mathcal{O}$ supported in S_0 with diameter satisfying

$$\max(1, \text{diam} S_0) =: s \leq R/2, \quad (\text{B36})$$

let

$$r = \lfloor (R - s)/2 \rfloor \geq R/4 - 1 \quad (\text{B37})$$

[where we use Eq. (B36)]. There exists a Hermitian operator $\mathcal{O}^P$ supported in $B(S_0, r)$ that stabilizes $|\mathbf{0}\rangle$, and

$$\|\mathbb{Q}'_r \mathbb{P} \mathcal{O} - \mathcal{O}^P\| \leq 8\|\mathcal{O}\| \frac{h_0 c_d R^d}{\Delta} \left[e^{-\mu r/3} + c_\delta e^{-(2\Delta r/u)^{1-\delta}} \right], \quad (\text{B38})$$

for any arbitrarily small $\delta > 0$, with c_δ determined solely by δ .

Proof. Lieb-Robinson bound (17) implies

$$\|(1 - \mathbb{Q}'_r)(e^{t\mathcal{L}_0} \mathcal{O})\| \leq 2\|\mathcal{O}\| \min(1, c_d c_{\text{LR}} s^{d-1} e^{\mu(u|t|-r)}), \quad (\text{B39})$$

which bounds the second term in Eq. (B33),

$$\begin{aligned} \|(1 - \mathbb{Q}'_r) \mathbb{P} \mathcal{O}\| &\leq 2\|\mathcal{O}\| \int_{-\infty}^{\infty} dt w(t) \min\left(1, c_d c_{\text{LR}} s^{d-1} e^{\mu(u|t|-r)}\right) \\ &\leq 2\|\mathcal{O}\| \left[2 \int_0^{t_r} dt w(t) c_d c_{\text{LR}} s^{d-1} e^{\mu(ut-r)} + 2 \int_{t_r}^{\infty} dt 2(e\Delta/2)^2 t e^{-\frac{1}{7\ln^2(\Delta t/2)} \frac{\Delta t}{2}} \right] \\ &\leq 2\|\mathcal{O}\| \left[c_d c_{\text{LR}} s^{d-1} \left(\max_{|t| \leq t_r} e^{\mu(ut-r)} \right) + \int_{t_r}^{\infty} dt (e\Delta)^2 t e^{-\frac{1}{7\ln^2(\Delta t/2)} \frac{\Delta t}{2}} \right] \\ &\leq 2\|\mathcal{O}\| \left[c_d c_{\text{LR}} s^{d-1} e^{-2\mu r/3} + c_\delta e^{-(2\Delta r/u)^{1-\delta}} \right] \leq 2\|\mathcal{O}\| \left[e^{-\mu r/3} + c_\delta e^{-(2\Delta r/u)^{1-\delta}} \right]. \end{aligned} \quad (\text{B40})$$

In the second line, we use the even parity of $w(t)$ and define

$$t_r := r/(3u) \geq 2e^{1/\sqrt{2}}/\Delta, \quad (\text{B41})$$

which holds because of Eqs. (B37) and (B35) for a sufficiently large c_R . Therefore, we can use bound (B10) in the second line of Eq. (B40). In the third line of Eq. (B40), we use $\int_0^\infty w(t) = 1/2$. In the last line of Eq. (B40), we use Eq. (B37) and introduce an arbitrarily small $\delta > 0$ that determines the constant c_δ .

According to decomposition (B33),

$$(\mathbb{P} \mathcal{O} - \langle \mathbb{P} \mathcal{O} \rangle_{\mathbf{0}}) |\mathbf{0}\rangle = |\psi_{\leq R}\rangle + |\psi_{> R}\rangle, \quad (\text{B42})$$

where

$$|\psi_{\leq R}\rangle = |\psi_{\leq R}\rangle_S \otimes |\mathbf{0}\rangle_{S^c} := (\mathbb{Q}'_r \mathbb{P} \mathcal{O} - \langle \mathbb{Q}'_r \mathbb{P} \mathcal{O} \rangle_{\mathbf{0}}) |\mathbf{0}\rangle \quad (\text{B43})$$

with $S = B(S_0, r)$ and $\text{diam} S \leq R$ from Eq. (B37), and

$$\begin{aligned} \|\psi_{> R}\rangle\| &= \|[(1 - \mathbb{Q}'_r) \mathbb{P} \mathcal{O} - \langle (1 - \mathbb{Q}'_r) \mathbb{P} \mathcal{O} \rangle_{\mathbf{0}}] |\mathbf{0}\rangle\| \\ &\leq \|(1 - \mathbb{Q}'_r) \mathbb{P} \mathcal{O}\|, \end{aligned} \quad (\text{B44})$$

due to

$$\|(A - \langle A \rangle_\psi) |\psi\rangle\| \leq \|A\|, \quad (\text{B45})$$

for any operator A and state ψ .

$|\psi_{\leq R}\rangle$ describes some local excitation above the all-zero state, which has the following nice property in terms of its energy distribution (proven in Appendix B 3).

Lemma 15. Suppose H_0 satisfies Eq. (44). Suppose $(H_0, |\mathbf{0}\rangle)$ is (Δ, R) metastable where $H_0|\mathbf{0}\rangle = 0$. Let

$$\Pi_{>E} := \sum_{E' > E} |E'\rangle \langle E'| \quad (\text{B46})$$

be the projection operator onto eigenstates $\{|E'\rangle\}$ of H_0 with energy larger than E . Then,

$$\|\Pi_{>\Delta/2} |\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}\| \geq \frac{\Delta}{4h_0|S|} \|\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}\|, \quad (\text{B47})$$

for any $|\phi\rangle_S \perp |\mathbf{0}\rangle_S$ with $\text{diam} S \leq R$.

Therefore, using Eq. (14) we have

$$\|\Pi_{>\Delta/2} |\psi_{\leq R}\rangle\| \geq \frac{\Delta}{4h_0 c_d R^d} \|\psi_{\leq R}\rangle\|. \quad (\text{B48})$$

To upper bound $\|\psi_{\leq R}\rangle\|$, it suffices to bound $\|\Pi_{>\Delta/2} |\psi_{\leq R}\rangle\|$ due to Eq. (B48). To this end, notice that $\mathbb{P}$ projects onto the $\leq \Delta/2$ energy subspace: Using Eq. (B11),

$$\begin{aligned} \Pi_{>\Delta/2} (\mathbb{P}\mathcal{O} - \langle \mathbb{P}\mathcal{O} \rangle_{\mathbf{0}}) |\mathbf{0}\rangle &= \Pi_{>\Delta/2} \mathbb{P}\mathcal{O} |\mathbf{0}\rangle \\ &= \sum_{E > \Delta/2} \hat{w}(E) \mathcal{O}_{E,0} |E\rangle = 0, \end{aligned} \quad (\text{B49})$$

where $\mathcal{O}_{E,0} := \langle E | \mathcal{O} | \mathbf{0} \rangle$. Therefore, Eqs. (B48) and (B42) lead to

$$\begin{aligned} \|\psi_{\leq R}\rangle\| &\leq \frac{4h_0 c_d R^d}{\Delta} \|\Pi_{>\Delta/2} |\psi_{\leq R}\rangle\| \\ &= \frac{4h_0 c_d R^d}{\Delta} \|\Pi_{>\Delta/2} |\psi_{>R}\rangle\| \leq \frac{4h_0 c_d R^d}{\Delta} \|\psi_{>R}\rangle\| \\ &\leq 8 \|\mathcal{O}\| \frac{h_0 c_d R^d}{\Delta} \left[e^{-\mu r/3} + c_\delta e^{-(2\Delta r/u)^{1-\delta}} \right], \end{aligned} \quad (\text{B50})$$

where the last step uses Eqs. (B44) and (B40).

Define the Hermitian operator

$$\mathcal{O}^P = \mathbb{Q}'_r \mathbb{P}\mathcal{O} - (|\psi_{\leq R}\rangle_S \langle \mathbf{0}| + \text{H.c.}), \quad (\text{B51})$$

supported in S . It stabilizes $|\mathbf{0}\rangle$ because

$$\mathcal{O}^P |\mathbf{0}\rangle = \mathbb{Q}'_r \mathbb{P}\mathcal{O} |\mathbf{0}\rangle - |\psi_{\leq R}\rangle = \langle \mathbb{Q}'_r \mathbb{P}\mathcal{O} \rangle_{\mathbf{0}} |\mathbf{0}\rangle, \quad (\text{B52})$$

from Eq. (B43); furthermore, Eq. (B50) leads to the bound (B38). ■

3. Proof of Lemma 15

Proof. Intuitively, we need only to show $|\phi \otimes \mathbf{0}\rangle := |\phi\rangle_S \otimes |\mathbf{0}\rangle_{S^c}$ has small weight on extremely large $\gtrsim |S|$ energies, so that the large average energy $H_0[|\phi \otimes \mathbf{0}\rangle]$ implies sufficient weight in range $[\Delta/2, O(|S|)]$.

To upper bound the weight on large energies, observe that

$$\begin{aligned} H_0 |\phi \otimes \mathbf{0}\rangle &= H_0 (|\phi\rangle_S \langle \mathbf{0}|) |\mathbf{0}\rangle \\ &= [H_0, |\phi\rangle_S \langle \mathbf{0}|] |\mathbf{0}\rangle + (|\phi\rangle_S \langle \mathbf{0}|) H_0 |\mathbf{0}\rangle \\ &= \left[\sum_{S': S' \cap S \neq \emptyset} H_{0,S'}, |\phi\rangle_S \langle \mathbf{0}| \right] |\mathbf{0}\rangle, \end{aligned} \quad (\text{B53})$$

where we use $H_0 |\mathbf{0}\rangle = 0$ and only terms in H_0 that intersect S contribute to the commutator. The energy variance is then

$$\begin{aligned} \langle \phi \otimes \mathbf{0} | H_0^2 | \phi \otimes \mathbf{0} \rangle &= \|H_0 |\phi \otimes \mathbf{0}\rangle\|^2 \\ &\leq \left(2 \sum_{S': S' \cap S \neq \emptyset} \|H_{0,S'}\| \right)^2 \\ &\leq 4 \left(\sum_{i \in S} \sum_{S' \ni i} \|H_{0,S'}\| \right)^2 \\ &\leq 4h_0^2 |S|^2, \end{aligned} \quad (\text{B54})$$

using Eq. (44).

For any random variable Z , the Paley-Zygmund inequality holds (as a lower bound version of Markov inequality)

$$\Pr[Z > \theta E[Z]] \geq (1 - \theta)^2 \frac{E[Z]^2}{E[Z^2]}, \quad (\text{B55})$$

for any $\theta \in (0, 1)$. Let $Z = E$ be the energy distribution of $|\phi \otimes \mathbf{0}\rangle$ and $\theta E[Z] = \Delta/2$; we use energy expectation (23) due to $\text{diam} S \leq R$ and variance (B54) to get

$$\begin{aligned} \|\Pi_{>\Delta/2} |\phi \otimes \mathbf{0}\rangle\|^2 &\geq \left(1 - \frac{\Delta}{2H_0[|\phi \otimes \mathbf{0}\rangle]} \right)^2 \frac{H_0[|\phi \otimes \mathbf{0}\rangle]^2}{4h_0^2 |S|^2} \\ &\geq \left(\frac{\Delta}{4h_0 |S|} \right)^2, \end{aligned} \quad (\text{B56})$$

for any normalized state $|\phi \otimes \mathbf{0}\rangle$. Equation (B47) then follows. ■

4. Proof of Proposition 13

Proof. The constant c_R is just the one determined by Proposition 14. We decompose $V = V_{\leq R/2} + V_{>R/2}$ where $V_{\leq R/2}$ contains all local terms V_{S_0} with diameter $\text{diam} S_0 \leq R/2$, for which we can apply Proposition 14 and obtain a stabilizer $V_{S_0}^P$ of $|\mathbf{0}\rangle$ that approximates $\mathbb{Q}'_r \mathbb{P} V_{S_0}$ [here, $\mathbb{Q}'_r$ depends on S_0 with r given by Eq. (B37)]. They sum up to the desired total stabilizer defined by

$$V^P := \sum_{V_{S_0}: \text{diam} S_0 \leq R/2} V_{S_0}^P. \quad (\text{B57})$$

Therefore, we have

$$\begin{aligned} \mathbb{P}V - V^P &= \mathbb{P}V_{>R/2} + \sum_{V_{S_0}: \text{diam} S_0 \leq R/2} (1 - \mathbb{Q}'_r) \mathbb{P}V_{S_0} \\ &+ \sum_{V_{S_0}: \text{diam} S_0 \leq R/2} (\mathbb{Q}'_r \mathbb{P}V_{S_0} - V^P_{S_0}). \end{aligned} \quad (\text{B58})$$

$$\begin{aligned} \|\mathbb{P}V_{>R/2}\|_0 &\leq c_w (-\ln(\kappa/2))^{d-1} \|V_{>R/2}\|_{\kappa/2} \\ &\leq c_w (-\ln(\kappa/2))^{d-1} e^{-\frac{\kappa}{2}(R/2)^\alpha} \|V\|_\kappa. \end{aligned} \quad (\text{B59})$$

The first term in Eq. (B58) is bounded by Proposition 11:

The second term in Eq. (B58) is the diameter $\geq R$ part of $\mathbb{P}V_{\leq R/2}$ and bounded similarly

$$\begin{aligned} \left\| \sum_{V_{S_0}: \text{diam} S_0 \leq R/2} (1 - \mathbb{Q}'_r) \mathbb{P}V_{S_0} \right\|_0 &\leq e^{-\frac{\kappa}{2}R^\alpha} \|\mathbb{P}V_{\leq R/2}\|_{\kappa/2} \leq e^{-\frac{\kappa}{2}R^\alpha} c_w (-\ln(\kappa/2))^{d-1} \|V_{\leq R/2}\|_\kappa \\ &\leq c_w (-\ln(\kappa/2))^{d-1} e^{-\frac{\kappa}{2}R^\alpha} \|V\|_\kappa. \end{aligned} \quad (\text{B60})$$

To bound the final term in Eq. (B58), we use the fact that, for $(\mathbb{Q}'_r \mathbb{P}V_{S_0} - V^P_{S_0})$ to act on a site i , V_{S_0} should act on a site j with $d(i, j) \leq r \leq R/2$. Therefore,

$$\begin{aligned} \left\| \sum_{V_{S_0}: \text{diam} S_0 \leq R/2} (\mathbb{Q}'_r \mathbb{P}V_{S_0} - V^P_{S_0}) \right\|_0 &\leq \max_i \sum_{j: d(i, j) \leq R/2} \sum_{S_0 \ni j} \|\mathbb{Q}'_r \mathbb{P}V_{S_0} - V^P_{S_0}\| \\ &\leq c_d R^d \max_j \sum_{S_0 \ni j} 8 \|V_{S_0}\| \frac{h_0 c_d R^d}{\Delta} \left[e^{-\mu R/15} + c_\delta e^{-(2\Delta R/5u)^{1-\delta}} \right] \\ &\leq 8c_d^2 \|V\|_\kappa \frac{h_0}{\Delta} R^{2d} \left[e^{-\mu R/15} + c_\delta e^{-(2\Delta R/5u)^{1-\delta}} \right]. \end{aligned} \quad (\text{B61})$$

Here, in the second line, we use Eq. (14) to bound the number of j 's and Eq. (B38) with $r \geq R/5$ from Eq. (B37) and that R is sufficiently large.

Using the triangle inequality to combine Eqs. (B59), (B60) [which is bounded by the right-hand side of Eq. (B59)], and (B61), we have

$$\begin{aligned} \|\mathbb{P}V - V^P\|_0 &\leq \|V\|_\kappa \left\{ 2c_w (-\ln(\kappa/2))^{d-1} e^{-\frac{\kappa}{2}(R/2)^\alpha} + 8c_d^2 \frac{h_0}{\Delta} R^{2d} \left[e^{-\mu R/15} + c_\delta e^{-(2\Delta R/5u)^{1-\delta}} \right] \right\} \\ &\leq \|V\|_\kappa \cdot 3c_w (-\ln(\kappa/2))^{d-1} e^{-\frac{\kappa}{2}(R/2)^\alpha}, \end{aligned} \quad (\text{B62})$$

where we set $\delta = 1 - \alpha$ and use again the largeness of R , together with Eq. (B15). ■

5. Beyond short-range-entangled states

Before stating the generalized result, we introduce one more definition: We say an operator $\mathcal{O}$ is k -local, if there exists a decomposition $\mathcal{O} = \sum_S \mathcal{O}_S$ with all $|S| \leq k$ in the sum, where S need *not* be connected.

Theorem 16. Suppose the conditions of Theorem 9 hold with the metastable state $|\mathbf{0}\rangle$ replaced by any state $|\psi_0\rangle$ that may not be SRE, and H_0 satisfies one extra property that it is k_0 -local with a constant k_0 .

In particular, H_0 stabilizes $|\psi_0\rangle$. Then the conclusions of Theorem 9 still hold subject to a slight modification on the locally diagonalizable property: For any $\alpha \in (0, 1)$ and κ_1 satisfying Eq. (45) there exist constants c_R , c_* , c_V , c_A , and c_D determined by α , d , μ , Δ/h_0 , Δ/u , κ_1 , and k_0 (where c_R does not depend on α and κ_1), such that if $R \geq c_R$, for

any perturbation V with small local norm ϵ obeying Eq. (46), $(H_0, V, |\psi_0\rangle)$ is $(v_*, c_D \epsilon, c_A \epsilon, \kappa_1/2)$ locally diagonalizable*.

Here locally diagonalizable* means Definition 7, but with Eq. (37b) replaced by

$$\max_i \sum_{S \ni i} \|V_{*,S} |\psi_0\rangle\| \leq v_*, \quad (\text{B63})$$

and v_* is small so that $|\psi_0\rangle$ has lifetime t_* in Eq. (47) ($\delta > 0$ being any positive constant) in the sense of Eq. (43): Local correlations functions are preserved for a long time starting from a locally dressed state $|\tilde{\psi}_0\rangle$.

Note that, by our definition, k_0 -local Hamiltonians include, e.g., two-qubit interactions that decay exponentially with distance. Although Eq. (B63) is weaker than the original Eq. (37b) by bounding $\|V_{*,S} |\psi_0\rangle\| \leq \|V_{*,S}\|$, it is

still sufficient to imply long lifetime for the particular initial state, as we see in the proof.

Proof. We first generalize Proposition 8 and show that the weaker locally diagonalizable* condition also implies long lifetime; the proof of this result will appear at the end.

Proposition 17. If the triple $(H_0, V, |\psi_0\rangle)$ is $(v_*, \mathbb{d}_*, a_*, \kappa_*)$ locally diagonalizable* with quasilocal unitary U , and $\|H\|_{\kappa_1} \leq h$ for Eq. (26), then the conclusions 1 (locality of U) and 3 (lifetime of dressed $|\psi_0\rangle$) in Proposition 8 still hold.

It remains to establish the locally diagonalizable* condition with the extra k_0 property. Recall that the proof of Theorem 9 works by first performing SWTs that rotate the Hamiltonian to $H_0 + D_{k_*} + V_{k_*}$ and then separating $D_{k_*} = D_{k_*}^P + (D_{k_*} - D_{k_*}^P)$ to define $H_* = H_0 + D_{k_*}^P$ and $V_* = (D_{k_*} - D_{k_*}^P) + V_{k_*}$. The SWT part is always achievable because it deals only with operators and has nothing to do with any state being metastable. The second separation part is done by Proposition 14 (Proposition 13 is its immediate generalization, from one local operator to sums of them), which is the only place using the product structure of the metastable state.

The problem here for general $|\psi_0\rangle$ is that we no longer have a way to separate D_{k_*} such that the nonstabilizing part has small operator norms. So we do not separate the operator, define $H_* = H_0 + \langle D_{k_*} \rangle_{\psi_0}$,

$$V_* = D_{k_*} - \langle D_{k_*} \rangle_{\psi_0} + V_{k_*}, \quad (\text{B64})$$

and aim at the weaker Eq. (B63). This reduces to bounding

$$\begin{aligned} \|V_{*,S}|\psi_0\rangle\| &\leq \left\| (D_{k_*,S} - \langle D_{k_*,S} \rangle_{\psi_0})|\psi_0\rangle \right\| \\ &\quad + \|V_{k_*,S}|\psi_0\rangle\|, \end{aligned} \quad (\text{B65})$$

since the local operator norms of V_{k_*} are already bounded in the previous proof.

Actually, the proof of Proposition 14 for $|\psi_0\rangle = |\mathbf{0}\rangle$ follows immediately from the exponential upper bound (B50) on Eq. (B65), [143] by constructing the stabilizer $\mathcal{O}^P$ in Eq. (B51). It is this stabilizer construction step that lead to the stronger locally diagonalizable condition, but it is

unnecessary here for the weaker locally diagonalizable* condition. As a result, here it suffices to generalize Eq. (B50) to

$$\| |\psi_{\leq R}\rangle \| \leq 2c_{\text{LRE}} \|\mathcal{O}\| \frac{h_0 c_d R^d}{\Delta} \left[e^{-\mu r/3} + c_\delta e^{-(2\Delta r/u)^{1-\delta}} \right], \quad (\text{B66})$$

where $|\psi_{\leq R}\rangle = (\mathbb{Q}'_r \mathbb{P} \mathcal{O} - \langle \mathbb{Q}'_r \mathbb{P} \mathcal{O} \rangle_{\psi_0})|\psi_0\rangle =: \tilde{\mathcal{O}}|\psi_0\rangle$. Previously, $c_{\text{LRE}} = 4$, while here we determine this constant shortly in Eq. (B73); this change just modifies constants and does not influence the qualitative results of Theorem 16. Operator $\tilde{\mathcal{O}}$ is supported in S with $\text{diam} S = R$; furthermore,

$$\|\tilde{\mathcal{O}}\| \leq 2\|\mathbb{Q}'_r \mathbb{P} \mathcal{O}\| \leq 2\|\mathbb{P} \mathcal{O}\| \leq 2\|\mathcal{O}\|, \quad (\text{B67})$$

where the last inequality comes from $\int w(t)dt = 1$ and the triangle inequality.

Following the manipulations in Eq. (B50), we see that Eq. (B66) holds as long as Lemma 15 can be generalized to show

$$\|\Pi_{>\Delta/2}|\psi_{\leq R}\rangle\| \geq \frac{\Delta}{c_{\text{LRE}} h_0 |S|} \| |\psi_{\leq R}\rangle \|. \quad (\text{B68})$$

Note that we have set the energy zero to $H_0|\psi_0\rangle = 0$, without loss of generality. Tracing the proof of Lemma 15, Eq. (B68) follows from bounding the energy variance

$$\langle \psi_{\leq R} | H_0^2 | \psi_{\leq R} \rangle \leq (c_{\text{LRE}} h_0 |S|)^2 \| |\psi_{\leq R}\rangle \|^2. \quad (\text{B69})$$

With the k_0 -locality of H_0 , we can invoke Theorem 2.1 in Ref. [144] on the energy distribution of local operator $\tilde{\mathcal{O}}$ to get

$$\begin{aligned} \|\Pi_{>E} \tilde{\mathcal{O}}|\psi_0\rangle\| &\leq \|\Pi_{>E} \tilde{\mathcal{O}} \Pi_{\leq 0}\| \leq \|\tilde{\mathcal{O}}\| e^{-\frac{1}{2h_0 k_0}(E-2h_0|S|)} \\ &\leq 2\|\mathcal{O}\| e^{-\frac{1}{2h_0 k_0}(E-2h_0|S|)}. \end{aligned} \quad (\text{B70})$$

$\|\Pi_{<-E} \tilde{\mathcal{O}}|\psi_0\rangle\|$ obeys the same bound. For any E ,

$$\begin{aligned} \langle \psi_{\leq R} | H_0^2 | \psi_{\leq R} \rangle &\leq E^2 \| |\psi_{\leq R}\rangle \|^2 + \int_E^\infty dE' (E')^2 \left(\|\Pi_{>E} \tilde{\mathcal{O}}|\psi_0\rangle\|^2 + \|\Pi_{<-E} \tilde{\mathcal{O}}|\psi_0\rangle\|^2 \right) \\ &\leq E^2 \| |\psi_{\leq R}\rangle \|^2 + 8\|\mathcal{O}\|^2 \int_E^\infty dE' (E + E' - E)^2 e^{-\frac{1}{h_0 k_0}(E' - 2h_0|S|)} \\ &= E^2 \| |\psi_{\leq R}\rangle \|^2 + 8\|\mathcal{O}\|^2 h_0 k_0 e^{-\frac{1}{h_0 k_0}(E - 2h_0|S|)} [E^2 + 2Eh_0 k_0 + 2(h_0 k_0)^2]. \end{aligned} \quad (\text{B71})$$

Assuming that $\|\psi_{\leq R}\| \geq 8\|\mathcal{O}\|(h_0 c_d R^d/\Delta)e^{-\mu R/3}$, because otherwise Eq. (B66) holds trivially, and taking $E \geq 3h_0 k_0$,

$$\begin{aligned} \langle \psi_{\leq R} | H_0^2 | \psi_{\leq R} \rangle &\leq E^2 \|\psi_{\leq R}\|^2 + \|\psi_{\leq R}\|^2 \frac{1}{8} \left(\frac{\Delta}{h_0 c_d R^d} \right)^2 e^{2\mu R/3} h_0 k_0 e^{-\frac{1}{h_0 k_0}(E-2h_0|S|)} E^2 \left(1 + \frac{2}{3} + \frac{2}{9} \right) \\ &\leq E^2 \|\psi_{\leq R}\|^2 \left[1 + \frac{1}{4} \left(\frac{\Delta}{h_0 c_d R^d} \right)^2 e^{2\mu R/3} h_0 k_0 e^{-\frac{1}{h_0 k_0}(E-2h_0|S|)} \right]. \end{aligned} \quad (\text{B72})$$

Equation (B69) then follows by choosing

$$E = c_{\text{LRE}} h_0 |S| / \sqrt{2}, \quad \text{where } c_{\text{LRE}} = \sqrt{2}(2 + k_0/\mu + 3k_0) \quad (\text{B73})$$

(which satisfies the previous condition $E \geq 3h_0 k_0$), because the second term in Eq. (B71) scales as $\lesssim R^{-2d} e^{2\mu R/3 - \mu|S|} \leq e^{-\mu R/3}$ from $|S| \geq R$ and is, thus, upper bounded by 1 if R is sufficiently large given by the condition (B35).

To conclude, if R satisfies Eq. (B35) where c_R now also depends on k_0 , either $|\psi_{\leq R}\rangle = \tilde{\mathcal{O}}|\psi_0\rangle$ is already very small, or its relative weight at very large energy $\geq E$ is bounded by the locality of $\tilde{\mathcal{O}}$ in Eq. (B70), so that $|\psi_{\leq R}\rangle$ is still small, as given by Eq. (B66) following the previous proof. This implies that the effective Hamiltonians $D_k = \mathbb{P}V_k$ obtained from SWT still almost stabilize the state $|\psi_0\rangle$, with Eq. (B65) being exponentially small in R . Because of Proposition 17, this is sufficient to imply a long lifetime of $|\psi_0\rangle$. The conclusions of Theorem 9 thus generalize, where the constants gain extra dependence on k_0 . ■

We expect the restriction of Theorem 16 to strictly local Hamiltonians is a technical issue. To overcome this problem potentially requires a generalization of imaginary-time operator growth bounds used in Ref. [144] beyond strictly local Hamiltonians. We finally prove the generalization on implications of local diagonalizability.

Proof of Proposition 17. The locality of U is independent of the change from Eq. (37b) to Eq. (B63), so follows immediately.

Although Conclusion 2 no longer holds, a weaker version still holds that replaces the operator norm in Eq. (40) with expectation in $|\tilde{\psi}_0\rangle$:

$$\begin{aligned} &|\langle \tilde{\psi}_0 | (e^{itH} \mathcal{O} e^{-itH} - U e^{itH_*} U^\dagger \mathcal{O} U e^{-itH_*} U^\dagger) | \tilde{\psi}_0 \rangle| \\ &\leq c_{\text{oper}} v_* |S| (|t| + 1)^{d+1} \|\mathcal{O}\|. \end{aligned} \quad (\text{B74})$$

To show this, we follow the proof in Ref. [16] of its Corollary 4, starting from Eq. (S69).

Define superoperators

$$\begin{aligned} \mathcal{L} &= i[H, \cdot], & \tilde{\mathcal{L}} &= i[U^\dagger H U, \cdot], \\ \tilde{\mathcal{L}}' &= i[H_*, \cdot], & \mathcal{U} &= U^\dagger \cdot U. \end{aligned} \quad (\text{B75})$$

The left-hand side of Eq. (B74) is then

$$\begin{aligned} \langle \tilde{\psi}_0 | (e^{t\tilde{\mathcal{L}}} - \mathcal{U}^\dagger e^{t\tilde{\mathcal{L}}'} \mathcal{U}) \mathcal{O} | \tilde{\psi}_0 \rangle &= \langle \psi_0 | \mathcal{U}^\dagger (e^{t\tilde{\mathcal{L}}} - e^{t\tilde{\mathcal{L}}'}) \mathcal{U} \mathcal{O} | \psi_0 \rangle \\ &= \langle \psi_0 | (e^{t\tilde{\mathcal{L}}} - e^{t\tilde{\mathcal{L}}'}) \mathcal{U} \mathcal{O} | \psi_0 \rangle, \end{aligned} \quad (\text{B76})$$

where we use $|\tilde{\psi}_0\rangle = U|\psi_0\rangle$. Using the Duhamel identity

$$e^{t\tilde{\mathcal{L}}} - e^{t\tilde{\mathcal{L}}'} = \int_0^t dt' e^{(t-t')\tilde{\mathcal{L}}'} (\tilde{\mathcal{L}} - \tilde{\mathcal{L}}') e^{t'\tilde{\mathcal{L}}}, \quad (\text{B77})$$

Eq. (B76) is further bounded by

$$\begin{aligned} &|\langle \psi_0 | (e^{t\tilde{\mathcal{L}}} - e^{t\tilde{\mathcal{L}}'}) \mathcal{U} \mathcal{O} | \psi_0 \rangle| \\ &\leq \int_0^t dt' |\langle \psi_0 | (\tilde{\mathcal{L}} - \tilde{\mathcal{L}}') e^{t'\tilde{\mathcal{L}}} \mathcal{U} \mathcal{O} | \psi_0 \rangle| \\ &\leq |t| \max_{0 \leq t' \leq t} |\langle \psi_0 | [V_*, \mathcal{O}(t')] | \psi_0 \rangle| \\ &\leq |t| \max_{0 \leq t' \leq t} \sum_r |\langle \psi_0 | \left[\sum_{S \cap S_r \neq \emptyset} V_{*,S}, \mathcal{O}_r(t') \right] | \psi_0 \rangle|. \end{aligned} \quad (\text{B78})$$

Here in the first line, we use H_* stabilizes $|\psi_0\rangle$ and Eq. (35). We then define $\mathcal{O}(t') := U^\dagger (e^{t'\mathcal{L}} \mathcal{O}) U = \sum_r \mathcal{O}_r(t')$ and expand it into $\mathcal{O}_r(t')$ supported in a radius- r ball S_r around the initial operator $\mathcal{O}$. The commutator in Eq. (B78) does not vanish only when the support of V_* overlaps with S_r .

In the original proof, we have bound (37b) on the operator norms of V_* , so Eq. (B78) is bounded by

$$\begin{aligned} &|\langle \psi_0 | \left[\sum_{S \cap S_r \neq \emptyset} V_{*,S}, \mathcal{O}_r(t') \right] | \psi_0 \rangle| \leq 2 \sum_{S \cap S_r \neq \emptyset} \|V_{*,S}\| \|\mathcal{O}_r(t')\| \\ &\leq 2|S_r| v_* \|\mathcal{O}_r(t')\|. \end{aligned} \quad (\text{B79})$$

Here, although we have only the weaker Eq. (B63), the result of Eq. (B79) still holds by just replacing the factor of $\|V_{*,S}\|$ by $\|V_{*,S}|\psi_0\rangle\|$, because V_* acts directly on $|\psi_0\rangle$. This is essentially the only proof step that needs modification.

The rest of the proof then follows verbatim from Ref. [16]: Eq. (B74) holds by combining Eqs. (B78) and (B79) and constraining the locality of $\mathcal{O}_r(t')$. Finally, the result for local density matrix (43) holds as it is equivalent

to Eq. (B74): The dynamics dropping off V_* does not evolve the initial state, so by choosing all final local operators $\mathcal{O}$, the final state $\rho_S(t)$ is close to the initial one. ■

APPENDIX C: SLOW FALSE VACUUM DECAY

This appendix contains our proof of (a more generalized version of) Theorem 10. As the discussion is rather lengthy, we begin by first summarizing a more general setting for the problem, where we prove our formal results.

1. Setup of the solvable Hamiltonian

We consider a commuting projector Hamiltonian

$$H'_0 = \sum_{f: \text{diam} f \leq \ell_0} H'_{0,f}, \quad \text{where } H'_{0,f} = \lambda_f P_f^\perp, \quad P_f^\perp = I - P_f, \quad (C1)$$

that has finite range ℓ_0 and local strength

$$h'_0 = \max_i \sum_{f \ni i} |\lambda_f|. \quad (C2)$$

Here, each local projector is labeled by a “factor” $f = (S, \eta)$, where $S \subset \Lambda$ is the support of P_f , and $\eta = 1, 2, \dots$ means multiple factors can act on a given subset S . Nevertheless, it is useful to still view $f = (S, \eta)$ as a subset, so we define $\text{diam} f = \text{diam} S$, and let $i \in f$ ($f \subset S'$) denote $i \in S$ ($S \subset S'$), etc. The terminology “commuting projector” means

$$P_f^2 = P_f, \quad [P_f, P_{\tilde{f}}] = 0, \quad \forall f, \tilde{f}. \quad (C3)$$

We assume

$$\lambda_f \geq \Delta' \quad \forall f, \quad (C4)$$

and H'_0 is frustration-free: A projector onto the ground state subspace is $P = \prod_f P_f \neq 0$ that minimizes all local terms in H simultaneously. Note that any gapped and frustration-free H made out of commuting terms can always be put in the form of Eq. (C1), as any operator can be decomposed into a sum of projectors onto its eigenstates.

We work with such a general H'_0 (C1), but it is useful to keep in mind a canonical example, the Ising Hamiltonian

$$H_{\text{Ising}} = \frac{1}{2} \sum_{i \sim j} \lambda_{i \sim j} (1 - Z_i Z_j), \quad \text{where } \lambda_{i \sim j} \geq \Delta', \quad (C5)$$

which sums over nearest-neighboring sites $i \sim j$ on a d -dimensional lattice. Z_i is the Pauli- Z on site i , and the coupling coefficients can be inhomogeneous. For this example, each factor f has its own subset.

Given set F , let

$$P_F := \prod_{f \in F} P_f, \quad P_F^\perp := I - P_F \quad (C6)$$

be the local ground state projector in that set. In particular, $P = P_\Lambda$. Because the projectors commute, we have

$$P_F^\perp H'_{0,F} P_F^\perp \geq \Delta', \quad \text{where } H'_{0,F} := \sum_{f \in F} H'_{0,f}, \quad (C7)$$

because any state in $P_F^\perp$ must violate at least one “check” $2P_f - I$, i.e., the check operator acting on the state returns -1 . In particular, setting $F = \Lambda$ in Eq. (C7) implies H'_0 has a global gap at least Δ' .

2. Operator decomposition based on syndromes

Based on the commuting “Pauli” checks $\{2P_f - I\}$, it is useful to arrange operators based on their commutation relations with the checks. Since $2P_f - I$ has eigenvalues ± 1 , any operator can be decomposed into a sum of terms that either commute or anticommute with each “Pauli” individually (this can be shown by a simple calculation in the eigenbasis of the checks). For an operator $\mathcal{O}$, if it either commutes or anticommutes with any check $2P_f - I$ [see the precise statement (C9)], then we say it has syndrome $\mathbf{s} = \{f : \{\mathcal{O}, 2P_f - I\} = 0\}$. In other words, the syndrome is the set of factors that anticommute with the operator (the factors not included in the syndrome commute with $\mathcal{O}$). We then expand any operator $\mathcal{O}$ that we encounter (except H'_0 itself) by

$$\mathcal{O} = \sum_S \mathcal{O}_S, \quad \mathcal{O}_S = \sum_{\mathbf{s}: f \in \mathbf{s} \Rightarrow f \subset S} \mathcal{O}_{S,\mathbf{s}}, \quad (C8)$$

where S is connected as before, and the second equation further expands each local term to terms with different syndrome $\mathbf{s}$. Here, $\mathcal{O}_{S,\mathbf{s}}$ is supported in S and has syndrome $\mathbf{s}$:

$$\mathcal{O}_{S,\mathbf{s}}(2P_f - I) = \begin{cases} -(2P_f - I)\mathcal{O}_{S,\mathbf{s}}, & \forall f \in \mathbf{s} \\ (2P_f - I)\mathcal{O}_{S,\mathbf{s}}, & \forall f \notin \mathbf{s}. \end{cases} \quad (C9)$$

To gain more intuition, note that this is equivalent to either $P_f \mathcal{O}_{S,\mathbf{s}} P_f = P_f^\perp \mathcal{O}_{S,\mathbf{s}} P_f^\perp = 0$ ($f \in \mathbf{s}$) or $P_f^\perp \mathcal{O}_{S,\mathbf{s}} P_f = P_f \mathcal{O}_{S,\mathbf{s}} P_f^\perp = 0$ ($f \notin \mathbf{s}$). In the summation over syndrome (C8), we further require that $\mathcal{O}_{S,\mathbf{s}}$ is “strongly supported” [49] in S , in the sense that it only anticommutes with factors that are completely included in S :

$$[P_f, \mathcal{O}_{S,\mathbf{s}}] = 0 \quad \forall f \not\subset S. \quad (C10)$$

Note that P_f commutes with $\mathcal{O}_{S,\mathbf{s}}$ trivially for f that does not intersect with S ; Eq. (C10) also requires P_f to commute

if f is not fully contained in S . A more explicit construction of $\mathcal{O}_{S',s}$ is given by

$$\mathcal{O}_S = \sum_{s_1, s_2} \tilde{P}_{s_1} \mathcal{O}_S \tilde{P}_{s_2} = \sum_s \mathcal{O}_{S,s},$$

$$\text{where } \mathcal{O}_{S,s} = \sum_{s_1, s_2: s_1 + s_2 = s} \tilde{P}_{s_1} \mathcal{O}_S \tilde{P}_{s_2}. \quad (\text{C11})$$

Here, $s \in \{0, 1\}^{\{f: f \subset S'\}}$ is viewed as a binary string over the set of f 's contained in S' , and $s_1 + s_2$ returns another binary string that is the elementwise summation of the two binary strings (modulo 2). One can verify that Eq. (C9) is then satisfied.

Later on, we use decomposition $\mathcal{O} = \sum_{S,s} \mathcal{O}_{S,s}$ (C8) with the strongly supported condition $f \in s \Rightarrow f \subset S$ being implicit. We see that this condition is useful to deal with operator growth by the finite-range H'_0 in SWTs. Based on this decomposition, we define the local K -norm of $\mathcal{O}$ that cares about the syndromes

$$\|\mathcal{O}\|_K := \max_{i \in \Lambda} \sum_{S \ni i, s} e^{K|S|} \|\mathcal{O}_{S,s}\|. \quad (\text{C12})$$

Equation (C12) implies stronger locality than the previous κ -norm (18) that is based on the support *diameter*, because the operator needs to decay exponentially with the support *volume* in order for $\|\mathcal{O}\|_K$ to be finite. Slightly abusing notation, we differentiate these two local norms simply by the subscript of $\|\cdot\|_{\kappa/K}$. We exclusively use the volume local norm in this section, while we use only the diameter local norm elsewhere.

One can always massage an operator into decomposition of the form (C8), with a mild increase in local norm.

Proposition 18. For any operator $\mathcal{O} = \sum_S \mathcal{O}_S$ with

$$\max_i \sum_{S \ni i} \|\mathcal{O}_S\| e^{K|S|} \leq 1, \quad (\text{C13})$$

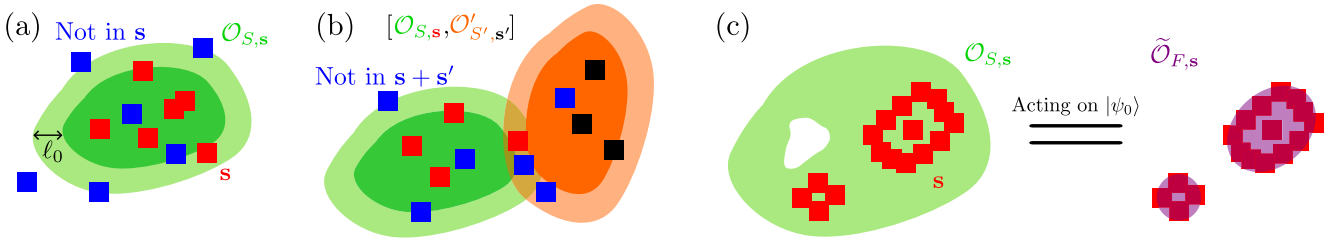


FIG. 9. (a) Illustration of a local term $\mathcal{O}_{S,s}$ in decomposition (C8). Blue (red) boxes show examples of checks $2P_f - I$ that commute (anticommute) with $\mathcal{O}_{S,s}$. $\mathcal{O}_{S,s}$ is strongly supported in the larger (lighter) green region S , so that the red syndrome can only be inside S ; i.e., they must intersect with the smaller darker green region. Any local operator can be “fattened” in its support to satisfy this condition. (b) Illustration that any commutator $[\mathcal{O}_{S,s}, \mathcal{O}'_{S',s'}]$ is strongly supported in $S \cup S'$. This commutator also has syndrome $s + s'$. For example, the blue check overlapping with the two darker regions can be in both syndromes s and s' so that it is not included in $s + s'$. (c) Illustration of the locally nondegenerate condition Definition 19. Here, S is always connected in decomposition (C8) (which can have “holes” as shown by the white region), while F can be disconnected. The condition guarantees that when acting on $|\psi_0\rangle$, the local operator can shrink its support to F (purple regions) such that its boundary ∂F are covered by syndrome.

one can always decompose it to the form (C8), which induces local norm $\|\mathcal{O}\|_{K'} \leq c_d (2\ell_0)^d$ with $K' = c_{\ell_0} K - c_{\text{factor}}$, where constant c_{ℓ_0} is determined by d, ℓ_0 , and constant $c_{\text{factor}} = \ln(2) \times \max_i |\{f: f \ni i\}|$ (the maximal number of factors that act on a given site).

Note that c_{ℓ} does not depend explicitly on the local Hilbert space dimension q . Before the proof, we present the idea of the syndrome decomposition using the Ising example: Each local term $\mathcal{O}_S$ can be expanded into Pauli strings (products of Z_i 's and X_j 's) contained in S , and each Pauli string has syndrome s containing all edges $i \sim j$ with exactly one Pauli- X on either i or j . $\mathcal{O}_{S,s}$ is then the sum over Pauli strings of the given syndrome s .

Proof. To satisfy Eq. (C10), we first fatten the support of each $\mathcal{O}_S$ by a buffer region of finite thickness [see Fig. 9(b)], viewing it as a term that is strongly supported in $S' = B(S, \ell_0)$ instead, so that $\mathcal{O} = \sum_{S'} \mathcal{O}_{S'}$. Since $|B(S, \ell_0)| \leq |S|/c_{\ell_0}$ for some constant c_{ℓ_0} determined by d, ℓ_0 , we have

$$\begin{aligned} \max_i \sum_{S' \ni i} \|\mathcal{O}_{S'}\| e^{c_{\ell_0} K |S'|} &\leq \max_i \sum_{j: d(i,j) \leq \ell_0} \sum_{S \ni j} \|\mathcal{O}_S\| e^{c_{\ell_0} K |S'|} \\ &\leq c_d (2\ell_0)^d \max_j \sum_{S \ni j} \|\mathcal{O}_S\| e^{K |S|} \\ &\leq c_d (2\ell_0)^d. \end{aligned} \quad (\text{C14})$$

Here, we use that if $i \in B(S, \ell_0)$, then there exists $j \in S$ within distance ℓ_0 to i . The number of such j 's is bounded by $c_d (2\ell_0)^d$ because they form a subset of diameter $\leq 2\ell_0$. Note that a simple bound is $c_{\ell_0} \geq 1/c_d (2\ell_0)^d$.

Then we decompose each $\mathcal{O}_{S'}$ by syndromes as follows. The projectors $\{P_f: f \subset S'\}$ can be simultaneously diagonalized by a set of orthogonal subspaces $\{\tilde{P}_s\}$ labeled by syndrome s that span the Hilbert space of the subsystem S' , where $P_f \tilde{P}_s = \tilde{P}_s$ (otherwise = 0) if and only if $f \in s$. Expand the operator $\mathcal{O}_{S'}$ onto this basis (C11). Since

$$\|\mathcal{O}_{S',s}\| = \left\| \sum_{s_1} \tilde{P}_{s_1} \mathcal{O}_{S'} \tilde{P}_{s_1+s} \right\| = \max_{s_1} \|\tilde{P}_{s_1} \mathcal{O}_{S'} \tilde{P}_{s_1+s}\| \leq \|\mathcal{O}_{S'}\|, \quad (\text{C15})$$

we have

$$\begin{aligned} \|\mathcal{O}\|_{K'} &= \max_i \sum_{S' \ni i} e^{K'|S'|} \sum_s \|\mathcal{O}_{S',s}\| \\ &\leq \max_i \sum_{S' \ni i} e^{K'|S'|} \|\mathcal{O}_{S'}\| e^{c_{\text{factor}}|S'|} \\ &= \max_i \sum_{S' \ni i} \|\mathcal{O}_{S'}\| e^{c_{\ell_0} K|S'|} \leq c_d (2\ell_0)^d, \end{aligned} \quad (\text{C16})$$

where we use that the number of s for S' is $2^{|\{f: f \subset S'\}|} \leq 2^{|S'| \times \max_i |\{f: f \ni i\}|} = e^{c_{\text{factor}}|S'|}$ and invoke Eq. (C14) in the end. ■

Let $|\psi_0\rangle$ be one ground state of H'_0 (i.e., $P|\psi_0\rangle = |\psi_0\rangle$). We assume that this state has the following property.

Definition 19. We say the ground state $|\psi_0\rangle$ of H'_0 is R' -locally nondegenerate, if the following holds for any operator $\mathcal{O}_{S,s}$ supported in $S \subset \Lambda$ with $|S| \leq c_d(R')^d$ that has syndrome s : Suppose all $f \in s$ overlaps (has a common site) with a smaller subset $F \subset S$ (including the empty set meaning $\mathcal{O}_{S,s}$ commutes with all projectors) such that all connected components of subset $S \setminus F$ are connected with S^c . Then there exists an operator $\tilde{\mathcal{O}}_{F,s}$ supported in the smaller region F with the same syndrome s , such that $\|\tilde{\mathcal{O}}_{F,s}\| \leq \|\mathcal{O}_{S,s}\|$ and

$$\mathcal{O}_{S,s}|\psi_0\rangle = \tilde{\mathcal{O}}_{F,s}|\psi_0\rangle. \quad (\text{C17})$$

See Fig. 9(a) for illustration. In particular, let $\mathcal{O}_{S,s}$ be a local operator with $|S| \leq c_d(R')^d$ that commutes with all P_f so that $s = \emptyset$, the above condition implies that

$$\mathcal{O}_{S,\emptyset}|\psi_0\rangle \propto |\psi_0\rangle. \quad (\text{C18})$$

This criterion forbids trivial ground state degeneracies (which explains the name “locally nondegenerate” of this condition) like a dangling qubit that no Hamiltonian term acts on, where the two local states can be flipped to each other easily by perturbation. The parameter R' will end up rather unimportant, as it will generally be much larger than the metastability length scale R . Indeed, the Ising Hamiltonian (C5) satisfies Definition 19 with R' scaling with the system size: $\mathcal{O}_{S,s}$ acts on $|\mathbf{0}\rangle$ as if it is all-identity in region $S \setminus F$, because the qubits in $S \setminus F$ are forced to be all-zero in $\mathcal{O}_{S,s}|\mathbf{0}\rangle$. To see this, consider a qubit $i \in \partial S$, which is connected by a ZZ check to an outside $|\mathbf{0}\rangle$ state that is untouched by the operator. Since this check is not violated by $\mathcal{O}_{S,s}$, i is still in the $|\mathbf{0}\rangle$ state after the operator action. Then one can turn to qubit $j \in S$ connected to i and repeat the same argument to prove j remains to be $|\mathbf{0}\rangle$ if $\{i, j\} \notin s$,

so on and so forth. An alternative way is to thinking about $\mathcal{O}_{S,s}$ as a superposition of Pauli strings, where all X 's are contained in F . Outside F , there can be Z 's but they act trivially on $|\mathbf{0}\rangle$.

Definition 19 can be viewed as a formal condition for (spontaneous) discrete symmetry breaking in the ground state sector. It is interesting to compare this with the topological order condition in Ref. [89], which enables a stability proof of commuting topological order. Indeed, the topological order condition also implies the nontrivial ground state degeneracy (C18) with R' of the order of the system size, and our later proof shares the similar SWT techniques with their stability proof.

3. Preprocessing with SWTs

Consider perturbing H'_0 by

$$V'_1 = \sum_{S: \text{diam}(S) \leq \ell_V} \sum_s V'_{1,S,s}, \quad (\text{C19})$$

with finite range ℓ_V . Here, we have expanded V'_1 into syndromes without loss of generality, as explained above. In order to prove a long-time stability for this system, it turns out useful to first preprocess the problem by a *finite* round of SWTs, which is summarized in the following.

Proposition 20. Let ℓ_V be a finite number. There exist constants c_V , c'_A , c'_D , c'_* , and ℓ'_D determined by d and ℓ_V such that the following holds.

For any perturbation (C19), if

$$\|V'_1\|_0 =: \epsilon' \leq c_V \Delta' \quad (\text{C20})$$

($\|\cdot\|_0$ means $\|\cdot\|_K$ with $K = 0$; similar below), then there exists a quasilocal unitary $U_{\text{finite}} = e^{\int_0^1 ds A_{\text{finite}}(s)}$ that makes

$$U_{\text{finite}}^\dagger (H'_0 + V'_1) U_{\text{finite}} = H'_0 + D' + V_1, \quad (\text{C21})$$

where

$$D' = \sum_{S: \text{diam} S \leq \ell'_D} \sum_s D'_{S,s} \quad (\text{C22})$$

has finite range ℓ'_D and

$$\begin{aligned} \|D'\|_0 &\leq c'_D \epsilon', & \|A_{\text{finite}}(s)\|_1 &\leq \frac{c'_A \epsilon'}{\Delta'}, \\ \|V_1\|_1 &\leq c'_* \epsilon' \left(\frac{\epsilon'}{\Delta'} \right)^{d-2/3}. \end{aligned} \quad (\text{C23})$$

Furthermore, D' is locally block diagonal with respect to H'_0 .

Here, we define the following.

Definition 21. We say D' is locally block diagonal with respect to H'_0 , if its decomposition (C22) satisfies

$$[D'_{S,s}, P_S] = 0 \quad \forall S, s. \quad (\text{C24})$$

Here, P_S is defined in Eq. (C6) and includes all $f \subset S$. Note that $D'_{S,s}$ also commutes with P_f for any $f \not\subset S$ because it is strongly supported in S . As an example, for 1D Ising $H'_0, D'_{S,s} = X_2(1 - Z_1 Z_3)$ with $s = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle\}$ satisfies Eq. (C24) because it annihilates local patterns $|000\rangle_{123}, |111\rangle_{123}$ with no domain walls (DWs). In words, D' does not create local excitations out of the “vacuum” (however, the action on excited states which have ≥ 1 syndrome flip is unconstrained). As a result, Definition 21 implies D' stabilizes $|\psi_0\rangle$ locally:

$$\begin{aligned} D'_{S,s}|\psi_0\rangle &= D'_{S,s}P_S|\psi_0\rangle = P_S D'_{S,s}|\psi_0\rangle \\ &= P_S D'_{S,s}P_S|\psi_0\rangle \propto |\psi_0\rangle, \end{aligned} \quad (\text{C25})$$

if $|\psi_0\rangle$ is a R' -locally nondegenerate ground state of H'_0 with $R' \geq \ell'_D$. The last step of Eq. (C25) utilizes $[P_S D'_{S,s} P_S, P_f] = 0$, $\forall f$ and Eq. (C18). In the proof of Proposition 20, we see that this locally block diagonal property holds because D' is the effective Hamiltonian (up to some SWT order) generated by the perturbation V'_1 .

Proposition 20 suppresses the local strength of the perturbation to high order, from the original $V'_1 \sim \epsilon'$ to $V_1 \ll (\epsilon')^d$. Later, we will treat the new perturbation problem $H_0 + V_1$ where

$$H_0 = H'_0 + D', \quad (\text{C26})$$

using the idea [145] that H_0 is metastable up to operator volume R^d with $R \sim \Delta'/\epsilon'$. Here, we first show how to obtain D' and V_1 .

Proof of Proposition 20. We perform a finite k_d [determined later in Eq. (C45)] number of SWT steps, which are formally the same as Eqs. (B3), (B6), (B13), and (B14), except that we add primes over H_0 , V_k , and $\mathbb{P}$. For example, Eq. (B13) now becomes

$$[H'_0, A_k] + V'_k = \mathbb{P}' V'_k =: D_{k+1} - D_k. \quad (\text{C27})$$

Instead of using the filter function, here $A_k, \mathbb{P}' V'_k$ [with their syndrome decompositions (C8)] are constructed in a way that explicitly exploits the solvable structure of H'_0 , such that $\mathbb{P}' V'_k$ is locally block diagonal, and the resulting V'_{k+1} is at higher order of ϵ' comparing to V'_k . More precisely, we prove in Appendix C 5 the following.

Lemma 22. For any V , there exist $A, \mathbb{P}' V$ such that

$$[H'_0, A] + V = \mathbb{P}' V, \quad (\text{C28})$$

where $\mathbb{P}' V$ is locally block diagonal with respect to H'_0 , and then

$$\max \left(\Delta' \|A\|_K, \frac{1}{2} \|\mathbb{P}' V\|_K \right) \leq \|V\|_K, \quad (\text{C29})$$

for any K . In particular, if V has finite range ℓ_V , then $A, \mathbb{P}' V$ has the same finite range ℓ_V .

Furthermore, suppose D' is of finite range ℓ'_D , locally block diagonal, and

$$\mathfrak{d}' = \|D'\|_0, \quad (\text{C30})$$

satisfies (C54) and determines R by Eq. (C55) with sufficiently small constants c_V and c_R determined by d, ℓ_0 , and ℓ'_D . Then if V contains only V_S with $|S| \leq c_d R^d$,

$$\|[D', A]\|_K \leq \frac{1}{3} \|V\|_K. \quad (\text{C31})$$

We describe how to find the local terms $A_{S,s}, (\mathbb{P}' V)_{S,s}, ([D', A])_{S,s}$ in the proof. Equation (C31) is not needed for the current proof but will play an important role later.

For the first SWT step, the above lemma implies $D_1 = \mathbb{P}' V'_1, A_1$ has finite range ℓ_V and local norm

$$\|D_1\|_0 \leq 2\epsilon', \quad \|A_1\|_0 \leq \frac{\epsilon'}{\Delta'}. \quad (\text{C32})$$

Technicalities arise for V'_2 : It is not finite range anymore. To tackle this, we define

$$K'_1 := \frac{1}{3c_d \ell_V^d} \log \frac{\Delta'}{\epsilon'}, \quad K'_k := \frac{K'_1}{1 + \ln k}, \quad \delta K'_{k+1} = K'_k - K'_{k+1}, \quad (\text{C33})$$

so that

$$\begin{aligned} \|V'_1\|_{K'_1} &= \max_i \sum_{S \ni i, s} \|V'_{1,S,s}\| e^{K'_1 |S|} \leq \max_i \sum_{S \ni i, s} \|V'_{1,S,s}\| e^{K'_1 c_d \ell_V^2} \\ &\leq \epsilon' e^{K'_1 c_d \ell_V^d} = \epsilon' \left(\frac{\epsilon'}{\Delta'} \right)^{-1/3}. \end{aligned} \quad (\text{C34})$$

$D_1/2, \Delta' A_1$ have exactly the same bound.

We then invoke Lemma 4.1 in Ref. [44].

Lemma 23 (adapted from Ref. [44]). Consider two operators A and $\mathcal{O}$ with local decompositions (C8). If

$$\|A\|_K \leq \frac{\delta K}{3} := \frac{K - K'}{3}, \quad (\text{C35})$$

$$\begin{aligned} & \| (e^{-A} - 1) \mathcal{O} \|_{K'} \leq 2 \left\| \int_0^1 ds (e^{-sA_k} - 1) \mathcal{O} \right\|_{K'} \\ & \leq \frac{18}{K' \delta K} \|A\|_K \|\mathcal{O}\|_K, \end{aligned} \quad (\text{C36a})$$

$$\|e^{-A} \mathcal{O}\|_{K'} \leq \left(1 + \frac{18}{K' \delta K} \|A\|_K\right) \|\mathcal{O}\|_K. \quad (\text{C36b})$$

Proof. We define the local term of, e.g., Eq. (C36b) by

$$\begin{aligned} (e^{-A} \mathcal{O})_{S,s} &= \sum_{m=0}^{\infty} \sum_{\substack{S_0, S_1, \dots, S_m: \\ S_0 \cup S_1 \cup \dots \cup S_m = S_0 + S_1 + \dots + S_m = S}} \sum_{\substack{S_0, S_1, \dots, S_m: \\ S_0 \cup S_1 \cup \dots \cup S_m = S_0 + S_1 + \dots + S_m = S}} a_{S_0, S_1, \dots, S_m} \\ &\times [A_{S_m, S_m}, \dots, [A_{S_2, S_2}, [A_{S_1, S_1}, \mathcal{O}_{S_0, S_0}]] \dots], \end{aligned} \quad (\text{C37})$$

where the coefficient $a_{S_0, S_1, \dots, S_m}$ is explicitly zero if, e.g., $S_1 \cap S_0 = \emptyset$ such that the multicommutator vanishes. The syndromes are added as in Eq. (C11). One can verify that (i) summing Eq. (C37) over S, s returns the whole $e^{-A} \mathcal{O}$. (ii) Equation (C37) is strongly supported in S as inherited from $A, \mathcal{O}$ [see Fig. 9(c) for illustration]. (iii) It also has syndrome s , which is the elementwise summation of the binary strings $s_1, \dots, s_m \in \{0, 1\}^{\{f\}}$ (modulo 2). This is because one can commute $2P_f - I$ through $\mathcal{O}_{S_0, S_0}, A_{S_1, S_1}$, etc., one by one by adding $\pm$ phases; the final phase is $-$ if and only if $f \in s$.

Taking the operator norm of Eq. (C37) and summing over s , we have

$$\begin{aligned} \sum_s \left\| (e^{-A} - 1) \mathcal{O} \right\|_{S,s} &\leq \sum_{m=1}^{\infty} \sum_{\substack{S_0, S_1, \dots, S_m: \\ S_0 \cup S_1 \cup \dots \cup S_m = S}} \sum_{S_0, S_1, \dots, S_m} |a_{S_0, S_1, \dots, S_m}| (2\|A_{S_m, S_m}\|) \dots (2\|A_{S_1, S_1}\|) \|\mathcal{O}_{S_0, S_0}\| \\ &= \sum_{m=1}^{\infty} \sum_{\substack{S_0, S_1, \dots, S_m: \\ S_0 \cup S_1 \cup \dots \cup S_m = S}} |a_{S_0, S_1, \dots, S_m}| \left(2 \sum_{S_m} \|A_{S_m, S_m}\|\right) \dots \left(2 \sum_{S_1} \|A_{S_1, S_1}\|\right) \left(\sum_{S_0} \|\mathcal{O}_{S_0, S_0}\|\right). \end{aligned} \quad (\text{C38})$$

Viewing, e.g., $(\sum_{S_0} \|\mathcal{O}_{S_0, S_0}\|)$ as $\|Z_{S_0}\|$ in Eq. (5.1) in Ref. [44], the proof there then applies straightforwardly to yield Eqs. (C36b) and (C36a) for the first argument. The key point here is that our extra summation over syndromes can be absorbed into our syndrome-dependent local norms $\|A\|_K, \|\mathcal{O}\|_K$.

The second argument (the integral) in Eq. (C36a) also holds by the following trick:

$$2 \int_0^1 ds (e^{-sA_k} - 1) \mathcal{O} = \sum_{m=1}^{\infty} \left(2 \int_0^1 (-s)^m ds\right) \sum_{S_0, S_1, S_2, \dots, S_m} \sum_{S_0, \dots, S_m} a_{S_0, S_1, \dots, S_m} [A_{S_m, S_m}, \dots, [A_{S_2, S_2}, [A_{S_1, S_1}, \mathcal{O}_{S_0, S_0}]] \dots], \quad (\text{C39})$$

which is again decomposed to local terms labeled by S, s using the rules in Eq. (C37). Since each local term in Eq. (C39) is proportional to the corresponding term in $(e^{-A} - 1) \mathcal{O}$ with coefficient $|(2 \int_0^1 (-s)^m ds)| \leq 1$, they satisfy the same bound (C36a). ■

Using this lemma, V'_2 [expressed in Eq. (B14) with primes added] is then bounded by

$$\|V'_2\|_{K'_2} \leq \frac{18}{K'_2 \delta K'_2} \left(\frac{1}{2} \times 3 + 1\right) \frac{\|V'_1\|_{K'_1}^2}{\Delta'} \leq c'_1 \epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{1/3}, \quad (\text{C40})$$

for some constant c'_1 determined by d, ℓ_V and c_V in Eq. (C20). Here, V'_2 does not scale as $(\epsilon')^2$ due to the way we invoke the K' norm. Nevertheless, V'_2 is still suppressed by some power of ϵ' comparing to V'_1 , and this is sufficient. Note that the condition (C35) for Lemma 23 is always satisfied in this proof, because K'_k s are large $\sim \log(\Delta'/\epsilon')$.

At higher but finite order $2 \leq k \leq k_d$, the D_k term in Eq. (B14) dominates: Assuming

$$\|D_k\|_{K'_k} \leq 2\|D_1\|_{K'_1} \leq 4\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{-1/3}, \quad (\text{C41})$$

we have

$$\begin{aligned} \|V'_{k+1}\|_{K'_{k+1}} &\leq \frac{18}{K'_{k+1} \delta K'_{k+1} \Delta'} 4\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{-1/3} \|V'_k\|_{K'_k} \\ &\leq c'_k \left(\frac{\epsilon'}{\Delta'}\right)^{2/3} \|V'_k\|_{K'_k}, \end{aligned} \quad (\text{C42})$$

which implies that

$$\|V'_{k+1}\|_{K'_{k+1}} \leq \tilde{c}'_k \epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{\frac{2k-1}{3}}, \quad (\text{C43})$$

for constants c'_k and $\tilde{c}'_k$ determined by d and ℓ_V . Consistently, Eq. (C42) guarantees Eq. (C41) because each

D_{k+1} is suppressed by a factor $(\epsilon'/\Delta')^{2/3}$ comparing to D_k , e.g.,

$$\|D_2\|_{K'_2} \leq 2\|V'_2\|_{K'_2} \leq 2c'_1\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{1/3}, \quad (\text{C44})$$

so that D_1 dominates D_k for sufficiently small c_V .

According to Eq. (C42), we define

$$k_d := \left\lfloor \frac{3d}{2} \right\rfloor, \quad (\text{C45})$$

which implies $2k_d \geq 3d - 1$, so that after order k_d ,

$$\|V'_{k_d+1}\|_{K'_{k_d+1}} \leq \tilde{c}'_{k_d}\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{d-2/3}. \quad (\text{C46})$$

At this order, the Hamiltonian is $H'_0 + D_{k_d+1} + V'_{k_d+1}$, where D_{k_d+1} is not quite the desired D' because it is not finite range. Therefore, we truncate the operator support to define

$$D' := \sum_{S: \text{diam} S \leq \ell'_D} \sum_s D_{k_d+1, S, s}, \quad (\text{C47})$$

which is locally block diagonal because the original $\mathbb{P}'V'_k$ s are, where ℓ'_D is chosen such that

$$V_1 := V'_{k_d+1} + D_{k_d+1} - D', \quad (\text{C48})$$

is sufficiently small. More precisely, let $K_1 = 1$, and

$$\ell'_D := \ell_V[(6d-2)(1+\ln(k_d+1))]^{1/d}, \quad (\text{C49})$$

we have

$$\begin{aligned} \|D_{k_d+1} - D'\|_{K_1} &\leq \|D_{k_d+1}\|_{K'_{k_d+1}} e^{-(K'_{k_d+1}-1)c_d(\ell'_D)^d} \\ &\leq 4\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{-1/3} e^{-K'_{k_d+1}c_d(\ell'_D)^d/2} \\ &= 4\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{-1/3 + \frac{(\ell'_D/\ell_V)^d}{6(1+\ln(k_d+1))}} \\ &\leq 4\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{d-2/3}. \end{aligned} \quad (\text{C50})$$

Here, the first inequality comes from reducing the decay exponent from K'_{k_d+1} to K_1 so that the contribution to the local norm from each local term in $D_{k_d+1} - D'$ is suppressed exponentially, because they all have large support [see the similar later (C80)]. We then use Eq. (C41) and $K'_{k_d+1} \geq 2$ that holds for sufficiently small $c_V \geq \epsilon'/\Delta'$. Combining with Eq. (C46), we have

$$\epsilon = \|V_1\|_{K_1} \leq c'_*\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{d-2/3}, \quad (\text{C51})$$

with c'_* determined by d and ℓ_V .

On the other hand,

$$\begin{aligned} \|D'\|_0 &\leq \|D_1\|_0 + \|D_{k_d+1} - D_1\|_{K'_{k_d+1}} \leq 2\epsilon' + 2\|D_2\|_{K'_2} \\ &\leq 2\epsilon' + 2c'_1\epsilon' \left(\frac{\epsilon'}{\Delta'}\right)^{1/3} \leq c'_D\epsilon', \end{aligned} \quad (\text{C52})$$

for some constant c'_D determined by c'_1 and c_V . Here, we have separated D_1 out of D_{k_d+1} to have a stronger bound than Eq. (C41). Similarly,

$$\begin{aligned} \|A_{\text{finite}}(s)\|_{K_1} &\leq \|A_1\|_{K_1} + 2\|A_2\|_{K_1} \\ &\leq \|A_1\|_0 e^{c_d\ell_V^d} + \frac{2\|V'_2\|_{K_1}}{\Delta'} \leq \frac{c'_A\epsilon'}{\Delta'}, \end{aligned} \quad (\text{C53})$$

with constant c'_A determined by d and ℓ_V , and we have again separated A_{finite} to its A_1 contribution and the rest that is subdominant $\leq 2\|A_2\|_{K_1}$ like Eq. (C52). ■

4. Proof of Theorem 10

We are now ready to prove a generalized version of Theorem 10. First, we state the following important result.

Theorem 24. Let $|\psi_0\rangle$ be a ground state of H'_0 that is R' -locally nondegenerate. Suppose D' is locally block diagonal with respect to H'_0 [and thus stabilizes $|\psi_0\rangle$ due to Eq. (C25)] and has finite range ℓ'_D and local norm (C30). Define H_0 by Eq. (C26) and $H = H_0 + V_1$, where V_1 has local norm $\epsilon := \|V_1\|_{K_1}$ ($K_1 \leq 1$). There exist constants c_R , c_V , and c_* determined by d , ℓ_0 , ℓ'_D , and K_1 , such that the following holds: Suppose

$$\max(\mathfrak{d}', \epsilon) \leq c_V\Delta', \quad (\text{C54})$$

and define R by

$$R = c_R\Delta'/\mathfrak{d}'. \quad (\text{C55})$$

If $R < R'$, then $(H_0, V_1, |\psi_0\rangle)$ is $(v_*, \mathfrak{d}_*, a_*, K_1/6)$ locally diagonalizable (see Definition 7), where

$$v_* = \epsilon 2^{2-k_*}, \quad \mathfrak{d}_* = 4\epsilon, \quad a_* = 2\epsilon/\Delta', \quad (\text{C56})$$

and

$$k_* := \left\lfloor c_* \min\left(\frac{\Delta'/\epsilon}{\ln^2(\Delta'/\epsilon)}, R^d\right) \right\rfloor. \quad (\text{C57})$$

Here, we assume $K_1 \leq 1$ (the same for any K below) in the same spirit of Eq. (A10b), as one can always decrease K_1 for free. D' and V_1 in the above Theorem 24 does not

need to come from Proposition 20, so we treat the parameters $\mathfrak{d}'$ and ϵ as independent [instead of $\epsilon \sim (\mathfrak{d}')^{d+1/3}$ from Eq. (C23)]. Nevertheless, combining Theorem 24 with the preprocessing Proposition 20 proves the parametrically tighter bound (62) for false vacuum decay near the solvable point.

Corollary 25 (general version of Theorem 10). Let $|\psi_0\rangle$ be a ground state of H'_0 that is R' -locally nondegenerate. Let ℓ_V be a finite number. There exist constants c_V, c_A, c_D, c_* , and $\tilde{c}_*$ determined by d, ℓ_0 , and ℓ_V such that the following holds: For any perturbation V'_1 (C19) with finite range ℓ_V and sufficiently small local norm (C20), $(H'_0, V'_1, |\psi_0\rangle)$ is $(\tilde{c}_*\epsilon'2^{-k_*}, c_D\epsilon', c_A\epsilon', 1/6)$ locally diagonalizable (see Definition 7), where

$$k_* = \lfloor c_*(\Delta'/\epsilon')^d \rfloor. \quad (\text{C58})$$

Proof. For sufficiently small c_V , Proposition 20 rotates the Hamiltonian to $H'_0 + D' + V_1$ that satisfies all conditions of Theorem 24 with $K_1 = 1$. We can then invoke Theorem 24 and get

$$\begin{aligned} U^\dagger U_{\text{finite}}^\dagger (H'_0 + V'_1) U_{\text{finite}} U &= U^\dagger (H'_0 + D' + V_1) U \\ &= H_* + V_*, \end{aligned} \quad (\text{C59})$$

where U is the SWT unitary when invoking Theorem 24 and $(H'_0 + D', V_1, |\mathbf{0}\rangle)$ is $(\epsilon 2^{1-k_*}, 5\epsilon, 2\epsilon/\Delta', K_*)$ locally diagonalizable, where $K_* = 1/6$ and k_* is given by Eq. (C58). Here, we use the first entry in Eq. (C57) $\sim \Delta'/\epsilon \sim (\Delta'/\epsilon')^{d+1/3}$ is larger than the second one $\sim R^d \sim (\Delta'/\epsilon')^d$. Finally, we combine the two unitaries in Eq. (C59) to get that the original problem $(H'_0, V'_1, |\mathbf{0}\rangle)$ is $(v_*, \mathfrak{d}_*, a_*, K_*)$ locally diagonalizable. Here, $v_* = \epsilon 2^{2-k_*} \leq \tilde{c}_*\epsilon'2^{-k_*}$ is the same one in Eq. (C56) and further bounded by replacing $\epsilon \rightarrow \epsilon'$ with an constant $\tilde{c}_*$. Like Eq. (B26), the total generator $A_{\text{total}}(s)$ for $U_{\text{finite}} U$ can be chosen as $2A(2s)$ [with $A(s)$ the generator for U] for $s \leq 1/2$ and $2A_{\text{finite}}(2s-1)$ for $s > 1/2$, so that

$$\begin{aligned} a_* &= \max_s \|A_{\text{total}}(s)\|_{K_*} \leq 2 \max(\|A_{\text{finite}}(s)\|_{K_*}, 2\epsilon/\Delta') \\ &\leq c_A \epsilon'/\Delta', \end{aligned} \quad (\text{C60})$$

for some constant c_A , where we use $K_* \leq 1$, (C23) and (C56). Similarly,

$$\begin{aligned} \mathfrak{d}_* &= \|H_* + V_* - H'_0\|_{K_*} \leq \|D'\|_{K_*} + \|H_* + V_* - H_0\|_{K_*} \\ &\leq \|D'\|_{K_*} + 4\epsilon \leq c_D \epsilon'. \end{aligned} \quad (\text{C61})$$

This finishes the proof. $\blacksquare$

Proof of Theorem 24. We still use the SWT framework, where at the k th order suppose we have rotated the Hamiltonian to (note the difference between D' and D)

$$U_{k-1}^\dagger H U_{k-1} = H_0 + D_k + V_k + E_k, \quad (\text{C62})$$

and we want to rotate one step further by

$$e^{-\mathcal{A}_k}(H_0 + D_k + V_k + E_k) = H_0 + D_{k+1} + V_{k+1} + E_{k+1}, \quad (\text{C63})$$

aiming at suppressing the local norm of V_k . Note that we have added a “garbage” term E_k comparing to Eq. (B3) in previous proofs, whose meaning will become clear shortly. As in the finite SWT rounds, here we keep track of the syndrome decompositions (C8) of operators.

At each step, we use Lemma 22 to find $A_k, \mathbb{P}'V_{k,\leq}$ such that

$$[H'_0, A_k] + V_{k,\leq} = \mathbb{P}'V_{k,\leq}, \quad (\text{C64})$$

where $V_{k,\leq} = \sum_{S: |S| \leq c_d R^d} V_{k,S}$ neglects very nonlocal terms in V_k and

$$\mathbb{P}'V_{k,\leq} =: D_{k+1} - D_k \quad (\text{C65})$$

stabilizes $|\psi_0\rangle$ so that D_{k+1} is also a stabilizer (D_k already stabilizes $|\psi_0\rangle$ from previous iterations). The only difference to the finite SWT steps (C27) is that we do not rotate very nonlocal terms in V_k because their support exceeds the maximal size $c_d R^d$ for metastability; see the similar size cutoff (B36) in the previous approach.

Since H'_0 is only part of H_0 , now we have extra contributions comparing to Eq. (B14) to the perturbation at next order. Writing $V_{k,>} = V_k - V_{k,\leq}$, we have (B14)

$$E_{k+1} = e^{-\mathcal{A}_k} E_k + e^{-\mathcal{A}_k} V_{k,>} \quad (\text{C66})$$

and

$$\begin{aligned} V_{k+1} &= e^{-\mathcal{A}_k}(H'_0 + D_k + V_{k,\leq}) - H'_0 - D_{k+1} + (e^{-\mathcal{A}_k} - 1)D' \\ &= \int_0^1 ds (e^{-s\mathcal{A}_k} - 1)(\mathbb{P}' - 1)V_{k,\leq} \\ &\quad + (e^{-\mathcal{A}_k} - 1)(D_k + V_{k,\leq}) \\ &\quad + \int_0^1 ds (e^{-s\mathcal{A}_k} - 1)[D', A_k] + [D', A_k]. \end{aligned} \quad (\text{C67})$$

Here, the first two terms come from Eq. (B14) by substituting $V_k \rightarrow V_{k,\leq}$, and the last two terms are the higher-order and first-order contributions to the last term in the first line. We want to iterate this procedure up to order $k_* \sim \min(R^d, \Delta'/\epsilon)$, so that for $k \leq k_*$, the local norm of Eq. (C67) is bounded by, e.g., $1/2 \times v_k$, where v_k is the local norm of V_k . If so, then $v_k \sim 2^{-k}$. At the same time, we show that the garbage term remains small $\|E_{k_*}\|_0 = \exp[-\Omega(R^d)] = \exp[-\Omega(k_*)]$ because it basically contains only operators of large support $|S| \gtrsim R^d$, and we are now

counting the number of sites on which operators act, not simply the diameter of the set on which they act. As a result, we can get the desired $v_* = v_{k_*} + \|E_{k_*}\|_0$ at an order k_* limited by $k_* \lesssim R^d$. To show $v_{k+1} \leq 1/2 \times v_k$, let us estimate or preview the local norms in Eq. (C67). Assuming $A_k \sim v_k$, the first and the third terms scale as $v_k^2 \ll v_k$ and so are also subdominant. The second is dominated by the $[D_k, A_k] \sim [D_1, A_k]$ term and scales as $\epsilon v_k \times (\text{support volume of } A_k) \sim \epsilon v_k k$, where we use $D_1 \sim \epsilon$, and volume of A_k roughly grows linearly with k . This second term is thus subdominant up to $k \sim 1/\epsilon$, corresponding to the first argument in Eq. (C57). These terms above are easy to bound by Eq. (C29) and Lemma 23. The main problem is the last term in Eq. (C67), which naively scales as

$$\mathbb{d}' v_k \times (\text{support volume of } A_k) \sim \mathbb{d}' v_k k. \quad (\text{C68})$$

This would lead to a final $k_* \lesssim 1/\mathbb{d}' \sim R \ll R^d$, which violates the second argument in Eq. (C57).

To tackle the last term in Eq. (C67), we use Eq. (C31) of Lemma 22, which is stronger than the naive scaling (C68). This is our key for promoting $R \rightarrow R^d$ dependence in k_* . Here, $k_* \sim R^d$ is the order at which E_k becomes dominant, so that we have to stop the iteration. In a nutshell, to show Eq. (C31) we exploit the fact that large support operators in A_k can be much smaller in norm than the corresponding terms of V_k , because V_k violates $\gg 1$ ground state projectors P_f . Note that in order for Lemma 22 to hold, the constants c_V and c_R do not need to depend on K_1 . We use the same c_R independent of K_1 below but choose a c_V determined by all the constants d, ℓ_0, ℓ'_D , and K_1 ; for this new c_V , Lemma 22 will still hold because it is still sufficiently small.

After sketching the idea, we now start to bound the local norms. Define $v_1 = \epsilon$, $\mathbb{d}_k := \|D_k\|_{K_k}$, $v_k := \|V_k\|_{K_k}$, where for $k \geq 2$ [146]

$$K_k := \frac{K_1}{2} \left(1 + \frac{1}{1 + \ln k} \right),$$

$$\delta K_k := K_{k-1} - K_k = \frac{K_1 \ln(k/(k-1))}{2(1 + \ln k)[1 + \ln(k-1)]} \geq \frac{K_1}{6k \ln^2 k}, \quad (\text{C69})$$

where the 6 in the denominator was deduced by numerically plotting the function. Note that our choice of K_k improves upon $K_k = K_1/(1 + \ln k)$ used in the previous work [44], where our $K_k \geq K_1/2$ and does not vanish at large k . This ensures that the decay properties in terms of the locality of the Hamiltonian remain almost unchanged when going to high orders in perturbation theory. Furthermore, this improvement enables us to get a pure R^d scaling without logarithmic factors in Eq. (C57). The only price to pay is making each δK_k smaller by a constant

factor of 2, which does not change the qualitative results, as we see.

From the definition of the K -norm in Eq. (C12), it is manifest that $\|V_{k,\leq}\|_{K_k} \leq \|V_k\|_{K_k} = v_k$, because the sum over support S is simply truncated to contain fewer terms. Applying Lemma 22 to Eq. (C64) then yields

$$\begin{aligned} \|\mathbb{P}' V_{k,\leq}\|_K &\leq 2v_k, & \|A_k\|_K &\leq v_k/\Delta', \\ \|[D', A_k]\|_{K_k} &\leq v_k/3, & \mathbb{d}_{k+1} &\leq \mathbb{d}_k + 2v_k, \end{aligned} \quad (\text{C70})$$

using Eq. (C65). Combining this with Eq. (C67) and Lemma 23, if at step k ,

$$v_k \leq \Delta' \delta K_{k+1}/6, \quad (\text{C71})$$

then

$$\begin{aligned} \frac{v_{k+1}}{v_k} &\leq \frac{18}{\Delta' K_{k+1} \delta K_{k+1}} \left[\frac{1}{2} (2+1) v_k + \mathbb{d}_k + v_k + \frac{1}{2} \cdot \frac{1}{3} v_k \right] + \frac{1}{3} \\ &\leq \frac{18}{\Delta' K_{k+1} \delta K_{k+1}} (\mathbb{d}_k + 3v_k) + \frac{1}{3}. \end{aligned} \quad (\text{C72})$$

It then remains to iterate up to some order k_* . This part is similar to previous proofs. At small k , Eq. (C72) is bounded by $1/2$, because the first term $\propto \mathbb{d}_k + 3v_k \sim v_1 \ll 1$. Indeed, suppose

$$v_{k+1} \leq v_k/2, \quad k \leq k_* - 1, \quad (\text{C73})$$

with k_* to be determined shortly, we have

$$\begin{aligned} v_k &\leq v_{k-1}/2 \leq \dots \leq v_1 2^{1-k} = 2^{1-k} \epsilon, \\ \mathbb{d}_k + 3v_k &\leq 2(v_1 + v_2 + \dots + v_{k-1} + v_k + v_k) \leq 4v_1 = 4\epsilon, \end{aligned} \quad (\text{C74})$$

for $k \leq k_*$ from Eq. (C70). For sufficiently small ϵ/Δ' , the condition (C71) is also satisfied consistently because it holds at $k=1$, and the left-hand side decays exponentially faster $\sim 2^{-k}$ than the right-hand side $\sim 1/(k \ln^2 k)$.

To determine the final step k_* , we require

$$\frac{18}{\Delta' K_k \delta K_k} 4\epsilon + \frac{1}{3} \leq \frac{1}{2} \quad \forall k \leq k_*. \quad (\text{C75})$$

Since the left-hand side increases with k monotonically, it is sufficient if Eq. (C75) holds at $k = k_*$. If we choose k_* by Eq. (C57) for some sufficiently small constant c_* determined by K_1 and d , Eq. (C75) is satisfied, because the first term

$$\begin{aligned} \frac{18}{\Delta' K_* \delta K_{k_*}} 4\epsilon &\leq \frac{72 \times 2 \times 6\epsilon}{K_1^2 \Delta'} k_* \ln^2 k_* \\ &= O\left(\frac{\ln^2 k_*}{\ln^2(\Delta'/\epsilon)}\right) \ll 1, \end{aligned} \quad (\text{C76})$$

in terms of scaling with $\epsilon/\Delta' \rightarrow 0$ at sufficiently small c_V .

To summarize the iteration, using the key Lemma 22 to bound the last term in Eq. (C67), for sufficiently small constant c_V determined by d, ℓ_0, ℓ'_D , and K_1 , the condition (C71) for Lemma 23 and the resulting inductive relation (C73) hold together up to k_* given by Eq. (C55).

At the last step, the Hamiltonian is rotated to $H_0 + D_{k_*} + V_*$, where only $V_* = V_{k_*} + E_{k_*}$ does not stabilize $|\psi_0\rangle$. Let $K_* := K_1/6 < K_{k_*}$, and Eq. (C74) leads to

$$\begin{aligned} \|V_{k_*}\|_{K_*} &\leq v_{k_*} \leq \epsilon 2^{1-k_*}, \\ a_* &\leq (v_1 + v_2 + \dots)/\Delta' \leq 2\epsilon/\Delta', \end{aligned} \quad (\text{C77})$$

where we use Eq. (C70) at last. From iteration relation (C66), the garbage term is

$$E_{k_*} = \sum_{k=1}^{k_*-1} e^{-\tilde{A}_k} V_{k,>}, \quad (\text{C78})$$

where $e^{-\tilde{A}_k} = e^{-A_{k_*-1}} \dots e^{-A_k}$ [strictly speaking, $\tilde{A}_k$ is time-dependent like Eq. (B26), while we ignore this for notational simplicity] satisfies

$$\|\tilde{A}_k\|_{2K_*} \leq 2\epsilon/\Delta', \quad (\text{C79})$$

similarly as Eq. (C77). For truncating the operator size of V_k , we have

$$\begin{aligned} \|V_{k,>}\|_{2K_*} &= \max_i \sum_{S \ni i, |S| > c_d R^d, s} \|V_{k,S,s}\| e^{2K_*|S|} \\ &\leq \max_i \sum_{S \ni i, |S| > c_d R^d, s} \|V_{k,S,s}\| e^{-K_* c_d R^d} e^{K_*|S|} \\ &\leq v_k e^{-K_* c_d R^d}. \end{aligned} \quad (\text{C80})$$

Here, in the second step, we use the fact that $|S| \geq c_d R^d$ and $K_k - 2K_* > K_1/2 - 2K_* = K_*$. Because Eq. (C35) is satisfied by Eq. (C79) and $K = 2K_*, K' = K_*$ for sufficiently small constant c_V , we can apply Eqs. (C36b) and (C80) to bound Eq. (C78):

$$\begin{aligned} \|E_{k_*}\|_{K_*} &\leq \left(1 + \frac{18}{K_*^2}\right) \frac{2\epsilon}{\Delta'} \sum_{k=1}^{k_*-1} v_k e^{-K_* c_d R^d} \\ &\leq \frac{4\epsilon^2}{\Delta'} \left(1 + \frac{3 \times 6^3}{K_1^2}\right) e^{-K_* c_d R^d} \leq \epsilon 2^{1-k_*}, \end{aligned} \quad (\text{C81})$$

where the last inequality holds for suitable small constants c_V and c_* determined by K_1 and d . Combining this with Eq. (C77), we have

$$\begin{aligned} v_* &= \|V_{k_*} + E_{k_*}\|_0 \leq \|V_{k_*}\|_{K_*} + \|E_{k_*}\|_{K_*} \leq \epsilon 2^{2-k_*}, \\ \mathbb{D}_* &\leq \|D_{k_*}\|_{K_*} + \|V_{k_*} + E_{k_*}\|_{K_*} \leq \mathbb{D}_{k_*} + 2v_{k_*} \leq 4\epsilon, \end{aligned} \quad (\text{C82})$$

where we use Eq. (C74) at last. This finishes the proof. ■

5. Proof of Lemma 22

Proof. Since Eq. (C64) is linear, it is sufficient to first focus on one local term $V_{S,s}$ and find the corresponding $A, \mathbb{P}'V$; the whole A and $\mathbb{P}'V$ are then defined by just summing over S, s : e.g., $A = \sum_{S,s} A_{S,s}$.

As we see, we need only to cancel the part

$$P_S^\perp V_{S,s} |\psi_0\rangle \langle \psi_0| + \text{H.c.} = P_S^\perp \tilde{V}_{F,s} |\psi_0\rangle \langle \psi_0| + \text{H.c.}, \quad (\text{C83})$$

of $V_{S,s}$, where we replace $V_{S,s}$ by $\tilde{V}_{F,s}$ on a potentially smaller support $F \subset S$ with $\|\tilde{V}_{F,s}\| \leq \|V_{S,s}\|$ using the locally nondegenerate condition (Definition 19), such that

$$[P_f, \tilde{V}_{F,s}] = 0 \quad \forall f: f \cap F = \emptyset. \quad (\text{C84})$$

Furthermore, we can assume that

$$\forall i \in \partial F, \exists f \ni i \quad \text{such that} \quad \{2P_f - I, \tilde{V}_{F,s}\} = 0 \quad (\text{C85})$$

(i.e., any boundary site has a violated check for $\tilde{V}_{F,s}$), because otherwise $[P_f, \tilde{V}_{F,s}] = 0, \forall f \ni i$, so that we can further shrink F to $F \setminus \{i\}$ as a new smaller F .

After shrinking, we follow the standard SWT and define an anti-Hermitian operator

$$A_{S,s}(\bar{F}) = P_{\bar{F}} \tilde{V}_{F,s} P_{\bar{F}}^\perp (H'_{0,\bar{F}})^{-1} - (H'_{0,\bar{F}})^{-1} P_{\bar{F}}^\perp \tilde{V}_{F,s} P_{\bar{F}}, \quad (\text{C86})$$

as the corresponding A term for $V_{S,s}$, which is viewed as supported in S , but actually acts only nontrivially on

$$\bar{F} := F \cup \{i \in \Lambda : \exists f \ni i \quad \text{such that} \quad \{2P_f - I, \tilde{V}_{F,s}\} = 0\} \subset S. \quad (\text{C87})$$

This is emphasized by the notation $A_{S,s}(\bar{F})$, which is useful later for bounding $[D', A]$. Here, $\bar{F} \subset S$ comes from the fact that $V_{S,s}$ is strongly supported in S and that $\tilde{V}_{F,s}$ has the same syndrome as $V_{S,s}$. Note that the (pseudo)inverse in Eq. (C86) is well defined because of Eq. (C7): It is always multiplied by $P_{\bar{F}}^\perp$ that projects out the kernel. We verify that the resulting

$$\begin{aligned}\mathbb{P}'V_{S,s} &= [H'_{0,\bar{F}}, A_{S,s}(\bar{F})] + V_{S,s} \\ &= -P_{\bar{F}}^\perp \tilde{V}_{F,s} P_{\bar{F}} - P_{\bar{F}} \tilde{V}_{F,s} P_{\bar{F}}^\perp + V_{S,s}\end{aligned}\quad (\text{C88})$$

stabilizes $|\psi_0\rangle$:

$$\begin{aligned}\mathbb{P}'V_{S,s}|\psi_0\rangle &= -P_{\bar{F}}^\perp \tilde{V}_{F,s} P_{\bar{F}}|\psi_0\rangle + \tilde{V}_{F,s}|\psi_0\rangle = P_{\bar{F}} \tilde{V}_{F,s}|\psi_0\rangle \\ &= P \tilde{V}_{F,s}|\psi_0\rangle \propto |\psi_0\rangle.\end{aligned}\quad (\text{C89})$$

Here, in Eq. (C88), we use

$$[H'_{0,f}, A_{S,s}(\bar{F})] = 0 \quad \forall f: f \not\subset \bar{F}, \quad (\text{C90})$$

from the definition of $\bar{F}$ (C87) and the fact that $H'_{0,f}$ commutes with all projectors and H'_0 in Eq. (C86), and only possibly fails to commute with $\tilde{V}_{F,s}$ if $f \cap F \neq \emptyset$. In Eq. (C89), the second term of Eq. (C88) annihilates $|\psi_0\rangle$, and we use the locally nondegenerate condition for the last term. In Eq. (C89), we then absorb $P_{\bar{F}}$ into $|\psi_0\rangle$ and again

use the locally nondegenerate condition that the ground state component of $\tilde{V}_{F,s}|\psi_0\rangle$ is just $\propto |\psi_0\rangle$.

The operators constructed are all good decompositions: $\bar{F} \subset S$ and Eq. (C90) guarantees $A_{S,s}(\bar{F})$ is strongly supported in S (we do not want to use $\bar{F}$ as the support because it can be disconnected). $A_{S,s}(\bar{F})$ also has syndrome $\mathbf{s}$ because for each of the two terms in Eq. (C86), one can commute $2P_f - I$ through by adding the same $\pm$ sign when going through $\tilde{V}_{F,s}$. Finally, the properties also hold for $\mathbb{P}'V_{S,s}$ due to Eq. (C64), where $\mathbb{P}'V_{S,s} = (\mathbb{P}'V)_{S,s}$ is strongly supported in S because $[H'_0, A_{S,s}(\bar{F})]$ is also.

Then we prove the bounds in Eq. (C29). Here, we work with the original extensive operators

$$V = \sum_{S,s} V_{S,s}, \quad A = \sum_{S,s} A_{S,s}(\bar{F}), \quad \mathbb{P}'V = \sum_{S,s} (\mathbb{P}'V)_{S,s}. \quad (\text{C91})$$

Since

$$\|A_{S,s}(\bar{F})\| \leq \max \left[\|P_{\bar{F}} \tilde{V}_{F,s} P_{\bar{F}}^\perp (H'_{0,\bar{F}})^{-1}\|, \|(H'_{0,\bar{F}})^{-1} P_{\bar{F}}^\perp \tilde{V}_{F,s} P_{\bar{F}}\| \right] \leq \|\tilde{V}_{F,s}\|/\Delta' \leq \|V_{S,s}\|/\Delta', \quad (\text{C92})$$

from Eq. (C7), we have

$$\begin{aligned}\|A\|_K &= \max_i \sum_{S: S \ni i, s} \|A_{S,s}(\bar{F})\| e^{K|S|} \\ &\leq \frac{1}{\Delta'} \max_i \sum_{S: S \ni i, s} \|V_{S,s}\| e^{K|S|} = \|V\|_K/\Delta'.\end{aligned}\quad (\text{C93})$$

Note that the strongly supported condition of $V_{S,s}$ is crucial for this equation to hold: Otherwise, A can be supported in a region S' with $|S'| - |S| \gg 1$. Using Eq. (C88),

$$\begin{aligned}\|\mathbb{P}'V_{S,s}\| &\leq \|P_{\bar{F}}^\perp \tilde{V}_{F,s} P_{\bar{F}} + P_{\bar{F}} \tilde{V}_{F,s} P_{\bar{F}}^\perp\| + \|V_{S,s}\| \\ &\leq \|\tilde{V}_{F,s}\| + \|V_{S,s}\| \leq 2\|V_{S,s}\|.\end{aligned}\quad (\text{C94})$$

The second inequality of Eq. (C94) follows from the similar Eq. (C92). Similar to Eq. (C93), this leads to $\|\mathbb{P}'V\|_K \leq 2\|V\|_K$ by summing S, s , which proves Eq. (C29).

The proof until now applies for any V . From now on, we turn to operator $[D', A]$ and aim to show Eq. (C31) by the constraint that all local terms $V_{S,s}$ have support smaller than some metastability size:

$$|S| \leq c_d R^d. \quad (\text{C95})$$

We assign the local terms as

$$([D', A])_{S'', s''} = \sum_{S,s} \sum_{\substack{S': S \cup S' = S'' \\ S' \cap \bar{F} \neq \emptyset, \\ s': s \cup s' = s'', s+s'=s''}} [D'_{S', s'}, A_{S,s}(\bar{F})], \quad (\text{C96})$$

where each term is strongly supported in $S' \cup S$ (instead of the smaller set $S' \cup \bar{F}$, which can be disconnected) and has syndrome $\mathbf{s} + \mathbf{s}'$. Crucially, only S' that overlaps with $\bar{F}$ contributes because $A_{S,s}(\bar{F})$ acts as a nonidentity operator only in $\bar{F}$. Hence, Eq. (C96) is a valid decomposition of the form (C8). To get Eq. (C31), we need a stronger bound than the naive (C92). Because of Eq. (C85), $\tilde{V}_{F,s} P_{\bar{F}}|\psi\rangle$ for any $|\psi\rangle$ violates at least $|\partial F|/(c_d \ell_0^d)$ number of terms $H'_{0,f}$, because a single term covers at most $c_d \ell_0^d$ sites. As a result, $P_{\bar{F}}^\perp$ in Eq. (C86) can actually be replaced by $P_{\bar{F}}^{\geq |\partial F|}$ that projects onto the subspace of energy (measured by $H'_{0,\bar{F}}$) larger than (or equal to) $\Delta' |\partial F|/(c_d \ell_0^d)$. Using

$$P_{\bar{F}}^{\geq |\partial F|} H'_{0,\bar{F}} P_{\bar{F}}^{\geq |\partial F|} \geq \frac{\Delta' |\partial F|}{c_d \ell_0^d}, \quad (\text{C97})$$

instead of Eq. (C7), Eq. (C92) becomes

$$\|A_{S,s}(\bar{F})\| \leq \frac{c_d \ell_0^d \|V_{S,s}\|}{\Delta' |\partial F|}. \quad (\text{C98})$$

This leads to

$$\begin{aligned}
\sum_{S': S' \cap \bar{F} \neq \emptyset, S'} \|[D'_{S', S'}, A_{S, S}(\bar{F})]\| e^{K|S' \cup S|} &\leq \sum_{i \in \bar{F}} \sum_{S' \ni i, S'} 2\|D'_{S', S'}\| \|A_{S, S}(\bar{F})\| e^{K|S| + c_d(\ell'_D)^d} \\
&\leq 2c_d \ell'_0 e^{c_d(\ell'_D)^d} \frac{d'}{\Delta'} \|V_{S, S}\| \frac{|F|}{|\partial F|} e^{K|S|} \leq c_{\text{DA}} \frac{d'}{\Delta'} \|V_{S, S}\| R e^{K|S|} \\
&\leq \frac{1}{4} \|V_{S, S}\| e^{K|S|}, \tag{C99}
\end{aligned}$$

for some constant c_{DA} determined by d , ℓ'_0 , and ℓ'_D . In the second line of Eq. (C99), we use $|\bar{F}| \leq |F| + c_{\text{bar}}|\partial F|$ for some constant c_{bar} , $|F| \leq c_d R^d$ from Eq. (C95), and the geometrical fact that boundary should grow with volume as

$$|\partial F| \geq c'_d |F|^{1-1/d}, \tag{C100}$$

for some constant c'_d determined by d . We have chosen $c_R = 1/(4c_{\text{DA}})$ in Eq. (C55) so that the last line of Eq. (C99) holds. It remains to further sum over S :

$$\begin{aligned}
\|[D', A]\|_K &\leq \max_i \sum_{S, S'} \sum_{S': S' \cap \bar{F} \neq \emptyset, i \in (S' \cup S)} \|[D'_{S', S'}, A_{S, S}(\bar{F})]\| e^{K|S' \cup S|} \\
&\leq \max_i \sum_{S \ni i, S'} \sum_{S': S' \cap \bar{F} \neq \emptyset, S'} \|[D'_{S', S'}, A_{S, S}(\bar{F})]\| e^{K|S' \cup S|} + \max_i \sum_{S' \ni i, S'} \sum_{S: S' \cap \bar{F} \neq \emptyset, S} \|[D'_{S', S'}, A_{S, S}(\bar{F})]\| e^{K|S' \cup S|} \\
&\leq \frac{1}{4} \max_i \sum_{S \ni i, S} \|V_{S, S}\| e^{K|S|} + \max_i \sum_{S' \ni i, S'} \sum_{j \in S'} \sum_{S: j \in \bar{F}, S} 2\|D'_{S', S'}\| \|A_{S, S}(\bar{F})\| e^{K|S| + c_d(\ell'_D)^d} \\
&\leq \frac{1}{4} \|V\|_K + 2c_d(\ell'_D)^d e^{c_d(\ell'_D)^d} \|D'\|_0 \|A\|_K \leq \left(\frac{1}{4} + 2c_d(\ell'_D)^d e^{c_d(\ell'_D)^d} \frac{d'}{\Delta'} \right) \|V\|_K \\
&\leq \frac{1}{3} \|V\|_K. \tag{C101}
\end{aligned}$$

Here, in the second line, the two terms correspond to the two situations that either S or S' contains i . We use Eq. (C99) for the first term in the third line and invoke Eq. (C93) in the fourth line. The last line follows from choosing a sufficiently small constant c_V in Eq. (C54). ■

6. A stronger notion of metastability

We have proven a stability timescale $t_* \gtrsim e^{(\ell')^{-d}}$ for perturbing H'_0 , as defined in previous sections, using the idea that $H'_0 + D'$ is metastable in volume $\lesssim R^d$ for any D' that is locally block diagonal with respect to H'_0 . Here, we make this intuition rigorous and, thus, connect our proof of slow false vacuum decay to the metastability theory developed in this paper.

Theorem 26. Let $|\psi_0\rangle$ be a ground state of H'_0 that is R' locally nondegenerate. Suppose D' is locally block diagonal with respect to H'_0 , annihilates $|\psi_0\rangle$ locally:

$$D'_{S, S}|\psi_0\rangle = 0, \tag{C102}$$

and has finite range ℓ'_D and local norm (C30). There exists a constant c_R determined by d , ℓ'_0 , and ℓ'_D , so that if $R < R'$

with R defined by Eq. (C55), then $(H_0, |\psi_0\rangle)$ with $H_0 := H'_0 + D'$ obeys

$$\begin{aligned}
H_0[(\mathcal{O} - \langle \mathcal{O} \rangle_{\psi_0})|\psi_0\rangle] &\geq \Delta'/2, \quad \text{for any} \\
\mathcal{O} &\text{ strongly supported in } S \text{ with } |S| \leq c_d R^d. \tag{C103}
\end{aligned}$$

We present several remarks before the proof: Equation (C102) is assumed without loss of generality, because D' automatically stabilizes $|\psi_0\rangle$ due to its locally block diagonality and Eq. (C25), so one can subtract each local term with a c number to achieve Eq. (C102). According to the fattening procedure in Fig. 9(b), the strongly supported conditions in Eq. (C103) can be relaxed to just $\mathcal{O}$ is supported in S , by reducing R by a finite amount.

Equation (C103) immediately implies $(H_0, |\psi_0\rangle)$ is metastable for our original definition for metastability (19) with range R , because any S with $\text{diam} S \leq R$ satisfies $|S| \leq c_d R^d$. However, by counting the volume instead of the diameter, Eq. (C103) is much stronger and constrains, e.g., stripelike regions with $\text{diam} S > R$. Indeed, the stronger criterion (C103) more carefully captures our intuition about metastability for the simplest models, such as the false vacuum decay problem in an Ising ferromagnet.

Using this much sharper notion of metastability is the key idea for the previous proof that obtains the tight d exponent in t_* .

Proof. Let Q be the projector onto the subspace

$$Q = \text{span}((\mathcal{O} - \langle \mathcal{O} \rangle_{\psi_0})|\psi_0\rangle : \mathcal{O} \text{ is strongly supported in } S \text{ with } |S| \leq c_d R^d), \quad (\text{C104})$$

i.e., the subspace of interest that does not include any ground state of H'_0 due to $R < R'$. We aim to show that, for some constant c_R ,

$$QH_0Q \geq \Delta'/2, \quad (\text{C105})$$

which is sufficient to obtain Eq. (C103).

As discussed around Eq. (C11), the total Hilbert space can be separated to subspaces $\tilde{P}_s$ labeled by syndrome s . For example, the ground state satisfies $P_f|\psi_0\rangle = |\psi_0\rangle$ for any f so it has syndrome $s = \emptyset$. For any $|\psi_s\rangle \in Q$ with syndrome s , it is a superposition of states $\mathcal{O}_{S,s}|\psi_0\rangle$ where S contains s and $|S| \leq c_d R^d$. Moreover, since D' is locally block diagonal with $R' > R$, each state in the superposition can be rewritten as $\tilde{\mathcal{O}}_{F,s}|\psi_0\rangle$ where $F = F(s)$ is the (possibly disconnected) subset determined by the shrinking procedure described below Eq. (C85), such that every boundary site in ∂F intersects with the syndrome s , and

$$1 \leq |F| \leq c_d R^d. \quad (\text{C106})$$

Then the different terms in the superposition can be combined, so that $|\psi_s\rangle = \mathcal{O}_{F,s}|\psi_0\rangle$ for a single operator $\mathcal{O}_{F,s}$ supported in $F = F(s)$ determined by s .

We first investigate what happens when acting H_0 on $|\psi_s\rangle$ with fixed syndrome and deal with the general case of superposition of syndromes later. The H'_0 part just stabilizes the state

$$H'_0|\psi_s\rangle = E'_s|\psi_s\rangle, \quad (\text{C107})$$

where

$$E'_s \geq \frac{\Delta'|\partial F|}{c_d \ell_0^d}. \quad (\text{C108})$$

The bound on E'_s is because a constant portion of the boundary checks of F are violated, similar to Eq. (C97). On the other hand, Eq. (C102) implies

$$D'|\psi_s\rangle = \sum_{S: S \cap F \neq \emptyset} D'_S|\psi_s\rangle, \quad (\text{C109})$$

so we immediately have

$$\|D'|\psi_s\rangle\| \leq \mathbb{d}'|F| \leq \frac{\mathbb{d}'|F|^{1/d}|\partial F|}{c'_d} \leq c_d^{1+1/d} \ell_0^d \frac{\mathbb{d}'R}{c'_d \Delta'} E'_s, \quad (\text{C110})$$

where we use the boundary-volume relation (C100) and the energy bound (C107). This proves Eq. (C105) when acting on a state with fixed syndrome.

Expand

$$D'|\psi_s\rangle = \sum_{S: S \cap F \neq \emptyset, s''} D'_{S,s''}|\psi_s\rangle = \sum_{s'} \sum_{S: S \cap F \neq \emptyset} D'_{S,s-s'}|\psi_s\rangle \quad (\text{C111})$$

into syndromes, where the number of syndromes s' that appear is bounded by $c_{\text{syn}}|F|$ with constant c_{syn} determined by d, ℓ_0^d , because due to the locality of D' , s'' and $|s|$ can differ only by $O(1)$ syndromes, and since D' stabilizes $|\psi_0\rangle$, D' must act nontrivially on an already included syndrome in s . Furthermore, each $s' \neq s$ corresponds to a particular $s'' \neq \emptyset$, so is contributed by a finite amount c'_{syn} of local terms $D'_{S,s''}$. Therefore, we have for $s \neq s'$,

$$\begin{aligned} \|\tilde{P}_{s'}QH_0Q\tilde{P}_s\| &\leq \|\tilde{P}_{s'}H_0\tilde{P}_s\| \\ &= \|\tilde{P}_{s'}D'\tilde{P}_s\| \begin{cases} \leq c'_{\text{syn}}\mathbb{d}', & \text{if } s, s' \text{ are connected,} \\ = 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (\text{C112})$$

Here, we say s, s' are connected if $s \neq s'$ and there exists a local term $D'_{S,s-s'}$. We have shown that given s , there are at most $c_{\text{syn}}|F|$ many s' that are connected to it. We use H'_0 preserves syndrome and insert the projector Q freely, because Q does not care about nonempty syndromes and thus commutes with $\tilde{P}_s$.

Equation (C112) bounds the block-off-diagonal elements of matrix H_0 , while Eqs. (C107) and (C110) already bound the block-diagonal elements

$$\begin{aligned} \tilde{P}_sQH_0Q\tilde{P}_s &\geq E'_s - \max_{\psi_s} \|D'|\psi_s\rangle\| \\ &\geq \Delta'|\partial F|/(c_d \ell_0^d) - \mathbb{d}'|F|. \end{aligned} \quad (\text{C113})$$

We then invoke Gershgorin's circle theorem for block matrices (see, e.g., Theorem 1.13.1 in Ref. [147]): For any eigenvalue λ of QH_0Q , there always exists a syndrome s such that either (i) λ is an eigenvalue of $\tilde{P}_sQH_0Q\tilde{P}_s$ or (ii)

$$\begin{aligned} \|(\tilde{P}_sQH_0Q\tilde{P}_s - \lambda)^{-1}\| &\geq \left(\sum_{s' \neq s} \|\tilde{P}_{s'}QH_0Q\tilde{P}_s\| \right)^{-1} \\ &\geq (c_{\text{syn}}|F| \cdot c'_{\text{syn}}\mathbb{d}')^{-1}, \end{aligned} \quad (\text{C114})$$

which implies

$$\begin{aligned}
\lambda &\geq \lambda_{\min}(\tilde{P}_s Q H_0 Q \tilde{P}_s) - c_{\text{syn}} c'_{\text{syn}} \mathfrak{d}' |F| \\
&\geq \frac{\Delta' |\partial F|}{c_d \ell_0^d} - (c_{\text{syn}} c'_{\text{syn}} + 1) \mathfrak{d}' |F| \\
&\geq \left[\frac{\Delta' c'_d |F|^{-1/d}}{c_d \ell_0^d} - (c_{\text{syn}} c'_{\text{syn}} + 1) \mathfrak{d}' \right] |F| \\
&\geq \frac{\Delta' c'_d c_d^{-1-1/d}}{\ell_0^d R} - (c_{\text{syn}} c'_{\text{syn}} + 1) \mathfrak{d}', \tag{C115}
\end{aligned}$$

where $\lambda_{\min}(\cdot)$ denotes the minimal eigenvalue and we use Eqs. (C113) and (C100) together with the geometrical facts (C106). The last expression (C115) is larger than $\Delta'/2$ for R defined by Eq. (C55) with a sufficiently small constant c_R . This proves Eq. (C105) and furthermore Eq. (C103). ■

As a direct implication of this metastability, the following corollary shows that any locally block diagonal D' is “relatively bounded” by H'_0 for states $|\psi\rangle$ obtained by locally exciting the ground state $|\psi_0\rangle$. Intriguingly, this relative boundedness also plays a key role in proving stability of topological order [89], where $|\psi\rangle$ can be any state. Therefore, the following Eq. (C117) serves as a natural generalization of this concept, which is equivalently to

$$Q(D')^2 Q \leq \frac{1}{4} Q(H'_0)^2 Q. \tag{C116}$$

The prefactors $1/2$, $1/4$ are chosen for convenience and can be replaced by any positive constant (which will determine how small c_R should be).

Corollary 27. Given the same conditions as Theorem 26, D' is relatively bounded by H'_0 for sufficiently local excitations above $|\psi_0\rangle$: For all $|\psi\rangle \in \mathcal{Q}$,

$$\|D'|\psi\rangle\| \leq \frac{1}{2} \|H'_0|\psi\rangle\|. \tag{C117}$$

Proof. We have proven Eq. (C105) with Q defined in Eq. (C104). By reducing c_R by a factor of 2, Eq. (C105) also holds for D replaced by $\pm 2D$, so we have

$$QH'_0 Q \pm 2QD'Q \geq 0. \tag{C118}$$

Multiplying the two equations

$$(QH'_0 Q + 2QD'Q)(QH'_0 Q - 2QD'Q) \geq 0, \tag{C119a}$$

$$(QH'_0 Q - 2QD'Q)(QH'_0 Q + 2QD'Q) \geq 0, \tag{C119b}$$

and adding them, we have

$$4(QD'Q)^2 \leq (QH'_0 Q)^2. \tag{C120}$$

In other words,

$$\|QD'|\psi\rangle\| \leq \frac{1}{2} \|QH'_0|\psi\rangle\|, \tag{C121}$$

for any $|\psi\rangle \in \mathcal{Q}$. Since D', H'_0 are finite range, if $|\psi\rangle \in \tilde{\mathcal{Q}}$ where $\tilde{\mathcal{Q}}$ is $\mathcal{Q}$ but with R reduced by a finite constant, the final Q projectors in Eq. (C121) act trivially and can be dropped. Relabeling $\tilde{\mathcal{Q}} \rightarrow \mathcal{Q}$ leads to Eq. (C117). ■

-
- [1] J. S. Langer, *Statistical theory of the decay of metastable states*, *Ann. Phys. (N.Y.)* **54**, 258 (1969).
 - [2] M. Cassandro, A. Galves, E. Olivieri, and M. E. Vares, *Metastable behavior of stochastic dynamics: A pathwise approach*, *J. Stat. Phys.* **35**, 603 (1984).
 - [3] George Gamow, *Zur quantentheorie des atomkernes*, *Z. Phys.* **51**, 204 (1928).
 - [4] G. Nieva, E. Osquiguil, J. Guimpel, M. Maenhoudt, B. Wuyts, Y. Bruynseraede, M. B. Maple, and Ivan K. Schuller, *Photoinduced enhancement of superconductivity*, *Appl. Phys. Lett.* **60**, 2159 (1992).
 - [5] L. Stojchevska, I. Vaskivskiy, T. Mertelj, P. Kusar, D. Svetin, S. Brazovskii, and D. Mihailovic, *Ultrafast switching to a stable hidden quantum state in an electronic crystal*, *Science* **344**, 177 (2014).
 - [6] M. Budden, T. Gebert, M. Buzzi, G. Jotzu, E. Wang, T. Matsuyama, G. Meier, Y. Laplace, D. Pontiroli, M. Riccò et al., *Evidence for metastable photo-induced superconductivity in K_3C_{60}* , *Nat. Phys.* **17**, 611 (2021).
 - [7] Sidney Coleman, *Fate of the false vacuum: Semiclassical theory*, *Phys. Rev. D* **15**, 2929 (1977).
 - [8] Federica Devoto, Simone Devoto, Luca Di Luzio, and Giovanni Ridolfi, *False vacuum decay: An introductory review*, *J. Phys. G* **49**, 103001 (2022).
 - [9] Marc Sher, *Electroweak Higgs potential and vacuum stability*, *Phys. Rep.* **179**, 273 (1989).
 - [10] M. S. Turner and F. Wilczek, *Is our vacuum metastable?*, *Nature (London)* **298**, 633 (1982).
 - [11] Joan Elias-Miró, José R. Espinosa, Gian F. Giudice, Gino Isidori, Antonio Riotto, and Alessandro Strumia, *Higgs mass implications on the stability of the electroweak vacuum*, *Phys. Lett. B* **709**, 222 (2012).
 - [12] S. B. Rutkevich, *Decay of the metastable phase in $d = 1$ and $d = 2$ Ising models*, *Phys. Rev. B* **60**, 14525 (1999).
 - [13] Gianluca Lagnese, Federica Maria Surace, Márton Kormos, and Pasquale Calabrese, *False vacuum decay in quantum spin chains*, *Phys. Rev. B* **104**, L201106 (2021).
 - [14] Anders Andreassen, David Farhi, William Frost, and Matthew D. Schwartz, *Direct approach to quantum tunneling*, *Phys. Rev. Lett.* **117**, 231601 (2016).
 - [15] Chi-Fang Chen, Hsin-Yuan Huang, John Preskill, and Leo Zhou, *Local minima in quantum systems*, in *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, STOC 2024 (Association for Computing Machinery, New York, 2024), pp. 1323–1330.
 - [16] Chao Yin and Andrew Lucas, *Prethermalization and the local robustness of gapped systems*, *Phys. Rev. Lett.* **131**, 050402 (2023).

- [17] A. D. Fokker, *Die mittlere Energie rotierender elektrischer Dipole im Strahlungsfeld*, *Ann. Phys. (Berlin)* **348**, 810 (1914).
- [18] M. Planck, *Über einen Satz der statistischen Dynamik und seine Erweiterung in der Quantentheorie*, Sitzungsberichte der Königlich-Preussischen Akademie der Wissenschaften zu Berlin (Reimer, 1917).
- [19] C. Gardiner, *Stochastic Methods*, 4th ed. (Springer, New York, 2009).
- [20] The reader may instead be familiar with the *even more heuristic* textbook argument whereby one calculates the WKB wave function and estimates the “bouncing time” between the walls of the classically bounding region. This method is difficult to generalize to higher dimensions [21], nor does it give an intrinsically quantum mechanical justification for the prescription.
- [21] Thomas Banks, Carl M. Bender, and Tai Tsun Wu, *Coupled anharmonic oscillators. I. Equal-mass case*, *Phys. Rev. D* **8**, 3346 (1973).
- [22] A. J. F. Siegert, *On the derivation of the dispersion formula for nuclear reactions*, *Phys. Rev.* **56**, 750 (1939).
- [23] Curtis G. Callan and Sidney Coleman, *Fate of the false vacuum. II. First quantum corrections*, *Phys. Rev. D* **16**, 1762 (1977).
- [24] Nicholas Metropolis, Arianna W. Rosenbluth, Marshall N. Rosenbluth, Augusta H. Teller, and Edward Teller, *Equation of state calculations by fast computing machines*, *J. Chem. Phys.* **21**, 1087 (1953).
- [25] Anders Andreassen, David Farhi, William Frost, and Matthew D. Schwartz, *Precision decay rate calculations in quantum field theory*, *Phys. Rev. D* **95**, 085011 (2017).
- [26] P. W. Anderson, *Infrared catastrophe in fermi gases with local scattering potentials*, *Phys. Rev. Lett.* **18**, 1049 (1967).
- [27] Alessio Lerose, Federica M. Surace, Paolo P. Mazza, Gabriele Perfetto, Mario Collura, and Andrea Gambassi, *Quasilocalized dynamics from confinement of quantum excitations*, *Phys. Rev. B* **102**, 041118(R) (2020).
- [28] Aritra Sinha, Titas Chanda, and Jacek Dziarmaga, *Non-adiabatic dynamics across a first-order quantum phase transition: Quantized bubble nucleation*, *Phys. Rev. B* **103**, L220302 (2021).
- [29] O. Pomponio, M. A. Werner, G. Zarand, and G. Takacs, *Bloch oscillations and the lack of the decay of the false vacuum in a one-dimensional quantum spin chain*, *SciPost Phys.* **12**, 061 (2022).
- [30] Ashley Milsted, Junyu Liu, John Preskill, and Guifre Vidal, *Collisions of false-vacuum bubble walls in a quantum spin chain*, *PRX Quantum* **3**, 020316 (2022).
- [31] Gianluca Lagnese, Federica Maria Surace, Sid Morampudi, and Frank Wilczek, *Detecting a long-lived false vacuum with quantum quenches*, *Phys. Rev. Lett.* **133**, 240402 (2024).
- [32] M. Lencsés, G. Mussardo, and G. Takács, *Variations on vacuum decay: The scaling Ising and tricritical Ising field theories*, *Phys. Rev. D* **106**, 105003 (2022).
- [33] D. Szász-Schagrin and G. Takács, *False vacuum decay in the (1 + 1)-dimensional ϕ^4 theory*, *Phys. Rev. D* **106**, 025008 (2022).
- [34] Jeffrey Allan Maki, Anna Berti, Iacopo Carusotto, and Alberto Biella, *Monte Carlo matrix-product-state approach to the false vacuum decay in the monitored quantum Ising chain*, *SciPost Phys.* **15**, 152 (2023).
- [35] Laura Batini, Aleksandr Chatrchyan, and Jürgen Berges, *Real-time dynamics of false vacuum decay*, *Phys. Rev. D* **109**, 023502 (2024).
- [36] A. Zenesini, A. Berti, R. Cominotti, C. Rogora, I. G. Moss, T. P. Billam, I. Carusotto, G. Lamporesi, A. Recati, and G. Ferrari, *False vacuum decay via bubble formation in ferromagnetic superfluids*, *Nat. Phys.* **20**, 558 (2024).
- [37] Siva Darbha, Milan Kornjača, Fangli Liu, Jan Balewski, Mark R. Hirsbrunner, Pedro L. S. Lopes, Sheng-Tao Wang, Roel Van Beeumen, Daan Camps, and Katherine Klymko, *False vacuum decay and nucleation dynamics in neutral atom systems*, *Phys. Rev. B* **110**, 155103 (2024).
- [38] Jaka Vodeb, Jean-Yves Desaulles, Andrew Hallam, Andrea Rava, Gregor Humar, Dennis Willsch, Fengping Jin, Madita Willsch, Kristel Michielsens, and Zlatko Papić, *Stirring the false vacuum via interacting quantized bubbles on a 5,564-qubit quantum annealer*, *Nat. Phys.* **21**, 386 (2025).
- [39] J. M. Deutsch, *Quantum statistical mechanics in a closed system*, *Phys. Rev. A* **43**, 2046 (1991).
- [40] Mark Srednicki, *Chaos and quantum thermalization*, *Phys. Rev. E* **50**, 888 (1994).
- [41] Luca D’Alessio, Yariv Kafri, Anatoli Polkovnikov, and Marcos Rigol, *From quantum chaos and eigenstate thermalization to statistical mechanics and thermodynamics*, *Adv. Phys.* **65**, 239 (2016).
- [42] Takashi Mori, Tatsuhiko N Ikeda, Eriko Kaminishi, and Masahito Ueda, *Thermalization and prethermalization in isolated quantum systems: A theoretical overview*, *J. Phys. B* **51**, 112001 (2018).
- [43] Dmitry A. Abanin, Wojciech De Roeck, and François Huveneers, *Exponentially slow heating in periodically driven many-body systems*, *Phys. Rev. Lett.* **115**, 256803 (2015).
- [44] Dmitry Abanin, Wojciech De Roeck, Wen Wei Ho, and François Huveneers, *A rigorous theory of many-body prethermalization for periodically driven and closed quantum systems*, *Commun. Math. Phys.* **354**, 809 (2017).
- [45] Tomotaka Kuwahara, Takashi Mori, and Keiji Saito, *Floquet-Magnus theory and generic transient dynamics in periodically driven many-body quantum systems*, *Ann. Phys. (Amsterdam)* **367**, 96 (2016).
- [46] Takashi Mori, Tomotaka Kuwahara, and Keiji Saito, *Rigorous bound on energy absorption and generic relaxation in periodically driven quantum systems*, *Phys. Rev. Lett.* **116**, 120401 (2016).
- [47] Dmitry A. Abanin, Wojciech De Roeck, Wen Wei Ho, and François Huveneers, *Effective Hamiltonians, prethermalization, and slow energy absorption in periodically driven many-body systems*, *Phys. Rev. B* **95**, 014112 (2017).
- [48] Krishnanand Mallayya, Marcos Rigol, and Wojciech De Roeck, *Prethermalization and thermalization in isolated quantum systems*, *Phys. Rev. X* **9**, 021027 (2019).
- [49] Wojciech De Roeck and Victor Verreet, *Very slow heating for weakly driven quantum many-body systems*, *arXiv:1911.01998*.

- [50] David J. Luitz, Roderich Moessner, S. L. Sondhi, and Vedika Khemani, *Prethermalization without temperature*, *Phys. Rev. X* **10**, 021046 (2020).
- [51] Dominic V. Else, Wen Wei Ho, and Philipp T. Dumitrescu, *Long-lived interacting phases of matter protected by multiple time-translation symmetries in quasiperiodically driven systems*, *Phys. Rev. X* **10**, 021032 (2020).
- [52] Federica Maria Surace and Olexei Motrunich, *Weak integrability breaking perturbations of integrable models*, *Phys. Rev. Res.* **5**, 043019 (2023).
- [53] Hannes Bernien *et al.*, *Probing many-body dynamics on a 51-atom quantum simulator*, *Nature (London)* **551**, 579 (2017).
- [54] Naoto Shiraishi and Takashi Mori, *Systematic construction of counterexamples to the eigenstate thermalization hypothesis*, *Phys. Rev. Lett.* **119**, 030601 (2017).
- [55] Christopher J. Turner, Alexios A. Michailidis, Dmitry A. Abanin, Maksym Serbyn, and Zlatko Papić, *Weak ergodicity breaking from quantum many-body scars*, *Nat. Phys.* **14**, 745 (2018).
- [56] Sanjay Moudgalya, Stephan Rachel, B. Andrei Bernevig, and Nicolas Regnault, *Exact excited states of nonintegrable models*, *Phys. Rev. B* **98**, 235155 (2018).
- [57] Sanjay Moudgalya, Nicolas Regnault, and B. Andrei Bernevig, *Entanglement of exact excited states of Affleck-Kennedy-Lieb-Tasaki models: Exact results, many-body scars, and violation of the strong eigenstate thermalization hypothesis*, *Phys. Rev. B* **98**, 235156 (2018).
- [58] Cheng-Ju Lin and Olexei I. Motrunich, *Exact quantum many-body scar states in the Rydberg-blockaded atom chain*, *Phys. Rev. Lett.* **122**, 173401 (2019).
- [59] Michael Schechter and Thomas Iadecola, *Weak ergodicity breaking and quantum many-body scars in spin-1xy magnets*, *Phys. Rev. Lett.* **123**, 147201 (2019).
- [60] He-Ran Wang, Dong Yuan, Shun-Yao Zhang, Zhong Wang, Dong-Ling Deng, and L.-M. Duan, *Embedding quantum many-body scars into decoherence-free subspaces*, *Phys. Rev. Lett.* **132**, 150401 (2024).
- [61] Sanjay Moudgalya, B. Andrei Bernevig, and Nicolas Regnault, *Quantum many-body scars and Hilbert space fragmentation: a review of exact results*, *Rep. Prog. Phys.* **85**, 086501 (2022).
- [62] Maksym Serbyn, Dmitry A. Abanin, and Zlatko Papić, *Quantum many-body scars and weak breaking of ergodicity*, *Nat. Phys.* **17**, 675 (2021).
- [63] Anushya Chandran, Thomas Iadecola, Vedika Khemani, and Roderich Moessner, *Quantum many-body scars: A quasiparticle perspective*, *Annu. Rev. Condens. Matter Phys.* **14**, 443 (2023).
- [64] Cheng-Ju Lin, Anushya Chandran, and Olexei I. Motrunich, *Slow thermalization of exact quantum many-body scar states under perturbations*, *Phys. Rev. Res.* **2**, 033044 (2020).
- [65] Federica Maria Surace, Matteo Votto, Eduardo Gonzalez Lazo, Alessandro Silva, Marcello Dalmonte, and Giuliano Giudici, *Exact many-body scars and their stability in constrained quantum chains*, *Phys. Rev. B* **103**, 104302 (2021).
- [66] Shane Dooley, *Robust quantum sensing in strongly interacting systems with many-body scars*, *PRX Quantum* **2**, 020330 (2021).
- [67] Pablo Sala, Tibor Rakovszky, Ruben Verresen, Michael Knap, and Frank Pollmann, *Ergodicity breaking arising from Hilbert space fragmentation in dipole-conserving Hamiltonians*, *Phys. Rev. X* **10**, 011047 (2020).
- [68] Vedika Khemani, Michael Hermele, and Rahul Nandkishore, *Localization from Hilbert space shattering: From theory to physical realizations*, *Phys. Rev. B* **101**, 174204 (2020).
- [69] Sanjay Moudgalya, Abhinav Prem, Rahul Nandkishore, Nicolas Regnault, and B. Andrei Bernevig, *Thermalization and its absence within Krylov subspaces of a constrained Hamiltonian*, in *Memorial Volume for Shoucheng Zhang* (World Scientific, Singapore, 2021), pp. 147–209.
- [70] Zhi-Cheng Yang, Fangli Liu, Alexey V. Gorshkov, and Thomas Iadecola, *Hilbert-space fragmentation from strict confinement*, *Phys. Rev. Lett.* **124**, 207602 (2020).
- [71] David T. Stephen, Oliver Hart, and Rahul M. Nandkishore, *Ergodicity breaking provably robust to arbitrary perturbations*, *Phys. Rev. Lett.* **132**, 040401 (2024).
- [72] Charles Stahl, Rahul Nandkishore, and Oliver Hart, *Topologically stable ergodicity breaking from emergent higher-form symmetries in generalized quantum loop models*, *SciPost Phys.* **16**, 068 (2024).
- [73] Atsuki Yoshinaga, Hideaki Hakoshima, Takashi Imoto, Yuichiro Matsuzaki, and Ryusuke Hamazaki, *Emergence of Hilbert space fragmentation in Ising models with a weak transverse field*, *Phys. Rev. Lett.* **129**, 090602 (2022).
- [74] Oliver Hart and Rahul Nandkishore, *Hilbert space shattering and dynamical freezing in the quantum Ising model*, *Phys. Rev. B* **106**, 214426 (2022).
- [75] This path is defined along the edges of the graph, but since we do not need to reference the edge set we do not bother to give it an explicit label.
- [76] This is common in the literature of mathematical statistical physics [77], where a decomposition like Eq. (15) is sometimes referred to as choosing “potentials.”
- [77] Barry Simon, *The Statistical Mechanics of Lattice Gases*, Volume I (Princeton University Press, Princeton, NJ, 2014), Vol. 260.
- [78] Elliott H. Lieb and Derek W. Robinson, *The finite group velocity of quantum spin systems*, *Commun. Math. Phys.* **28**, 251 (1972).
- [79] Chi-Fang (Anthony) Chen, Andrew Lucas, and Chao Yin, *Speed limits and locality in many-body quantum dynamics*, *Rep. Prog. Phys.* **86**, 116001 (2023).
- [80] If this equation holds, we say $\mathcal{O}$ is a stabilizer of $|\psi_0\rangle$: $\mathcal{O}$ need not be a Pauli string.
- [81] Strictly speaking, in finite volume Λ the two lowest energy states are the cat states $(1/\sqrt{2})(|\psi_+\rangle \pm |\psi_-\rangle)$, with an energy splitting $\sim \exp(-\Lambda)$.
- [82] While our physical intuition is that this statement should also hold for $|\psi_0\rangle$, one has to be somewhat careful: There can be short-time transient dynamics in state $|\psi_0\rangle$, but this transient dynamics should not, e.g., wholly decay a correlation function from 1 to 0. We were unable to prove this intuition in complete generality, while in contrast we could prove very strong statements about the decay of $|\tilde{\psi}_0\rangle$.
- [83] Strictly speaking, the quasiadiabatic unitary may be generated by a Hamiltonian that decays slower than exponential with the support diameter (see, e.g., Ref. [84]).

Then $H = \tilde{H} + \tilde{V}$, where $\tilde{H}$ is H truncated at some support diameter and thus quasilocal in the sense of Eq. (16), while the remaining $\tilde{V}$ has very small local norm but decays slower than exponential. We expect our results also apply to this case, since we treat slower-decaying perturbations in Theorem 9 anyway, but again we have kept the assumptions a bit simpler for the sake of clarity.

- [84] Sven Bachmann, Spyridon Michalakis, Bruno Nachtergaele, and Robert Sims, *Automorphic equivalence within gapped phases of quantum lattice systems*, *Commun. Math. Phys.* **309**, 835 (2012).
- [85] However, the state is not guaranteed to be metastable: We could consider the metastable state to be $\cos\theta|\mathbf{0}\rangle_S + \sin\theta|\max\rangle_S$, where $|\max\rangle_S$ is the highest energy state in the S -restricted Hamiltonian. This state is less metastable than Eq. (24).
- [86] A similar argument shows that $(H, |\mathbf{0}\rangle)$ is $((1-\gamma)\Delta, R')$ metastable for any $0 < \gamma < 1$ if $R' = \min(R, (\gamma\Delta/c_d\epsilon)^{1/d})$.
- [87] Strictly speaking, the proof of this result does not include time-dependent $A(s)$, but it is straightforward to generalize to this case.
- [88] Definition 7 provides the implications (S23) and (S25) from Theorem 3 (references refer to Ref. [16]) which go into the proof of Corollary 4 in Ref. [16]. In this paper, we prove that metastability can also imply these properties, in Theorem 9.
- [89] Sergey Bravyi, Matthew B. Hastings, and Spyridon Michalakis, *Topological quantum order: Stability under local perturbations*, *J. Math. Phys. (N.Y.)* **51**, 093512 (2010).
- [90] Formally speaking, H_0 here is also covered by the assumptions of the generalized prethermal bounds of Ref. [49] since all terms in H_0 commute. However, we could also study modified H_0 where this commutativity assumption is relaxed, or where the energy levels are highly incommensurate, and our metastability bounds continue to hold.
- [91] Jutho Haegeman, J. Ignacio Cirac, Tobias J. Osborne, Iztok Pizorn, Henri Verschelde, and Frank Verstraete, *Time-dependent variational principle for quantum lattices*, *Phys. Rev. Lett.* **107**, 070601 (2011).
- [92] Jutho Haegeman, Christian Lubich, Ivan Oseledets, Bart Vandereycken, and Frank Verstraete, *Unifying time evolution and optimization with matrix product states*, *Phys. Rev. B* **94**, 165116 (2016).
- [93] Johannes Hauschild and Frank Pollmann, *Efficient numerical simulations with tensor networks: Tensor network Python (TeNPy)*, *SciPost Phys. Lect. Notes* **5** (2018), code available from <https://github.com/tenpy/tenpy>.
- [94] C. J. Turner, A. A. Michailidis, D. A. Abanin, M. Serbyn, and Z. Papić, *Quantum scarred eigenstates in a Rydberg atom chain: Entanglement, breakdown of thermalization, and stability to perturbations*, *Phys. Rev. B* **98**, 155134 (2018).
- [95] Vedika Khemani, Chris R. Laumann, and Anushya Chandran, *Signatures of integrability in the dynamics of Rydberg-blockaded chains*, *Phys. Rev. B* **99**, 161101(R) (2019).
- [96] Soonwon Choi, Christopher J. Turner, Hannes Pichler, Wen Wei Ho, Alexios A. Michailidis, Zlatko Papić, Maksym Serbyn, Mikhail D. Lukin, and Dmitry A. Abanin, *Emergent SU(2) dynamics and perfect quantum many-body scars*, *Phys. Rev. Lett.* **122**, 220603 (2019).
- [97] Thomas Iadecola, Michael Schecter, and Shenglong Xu, *Quantum many-body scars from magnon condensation*, *Phys. Rev. B* **100**, 184312 (2019).
- [98] Wen Wei Ho, Soonwon Choi, Hannes Pichler, and Mikhail D. Lukin, *Periodic orbits, entanglement, and quantum many-body scars in constrained models: Matrix product state approach*, *Phys. Rev. Lett.* **122**, 040603 (2019).
- [99] Kieran Bull, Jean-Yves Desaulles, and Zlatko Papić, *Quantum scars as embeddings of weakly broken lie algebra representations*, *Phys. Rev. B* **101**, 165139 (2020).
- [100] C. J. Turner, J.-Y. Desaulles, K. Bull, and Z. Papić, *Correspondence principle for many-body scars in ultra-cold Rydberg atoms*, *Phys. Rev. X* **11**, 021021 (2021).
- [101] Bennet Windt and Hannes Pichler, *Squeezing quantum many-body scars*, *Phys. Rev. Lett.* **128**, 090606 (2022).
- [102] Lei Pan and Hui Zhai, *Composite spin approach to the blockade effect in Rydberg atom arrays*, *Phys. Rev. Res.* **4**, L032037 (2022).
- [103] Keita Omiya and Markus Müller, *Quantum many-body scars in bipartite Rydberg arrays originating from hidden projector embedding*, *Phys. Rev. A* **107**, 023318 (2023).
- [104] Note that R counts the set diameter defined in Sec. II; $R = 0$ for a single-site operator.
- [105] A. A. Ovchinnikov, D. V. Dmitriev, V. Ya. Krivnov, and V. O. Chervanovskii, *Antiferromagnetic Ising chain in a mixed transverse and longitudinal magnetic field*, *Phys. Rev. B* **68**, 214406 (2003).
- [106] Of course, if the value of Δ is quite small at this value, better bounds may arise from treating the model as (Δ', R') metastable with $R' < R$ but $\Delta' > \Delta$.
- [107] Cheng-Ju Lin and Olexei I. Motrunich, *Quasiparticle explanation of the weak-thermalization regime under quench in a nonintegrable quantum spin chain*, *Phys. Rev. A* **95**, 023621 (2017).
- [108] Jacob H. Robertson, Riccardo Senese, and Fabian H. L. Essler, *Decay of long-lived oscillations after quantum quenches in gapped interacting quantum systems*, *Phys. Rev. A* **109**, 032208 (2024).
- [109] Cheng-Ju Lin, Anushya Chandran, and Olexei I. Motrunich, *Slow thermalization of exact quantum many-body scar states under perturbations*, *Phys. Rev. Res.* **2**, 033044 (2020).
- [110] Federica Maria Surace, Matteo Votto, Eduardo Gonzalez Lazo, Alessandro Silva, Marcello Dalmonte, and Giuliano Giudici, *Exact many-body scars and their stability in constrained quantum chains*, *Phys. Rev. B* **103**, 104302 (2021).
- [111] Wladislaw Krinitsin, Niklas Tausendpfund, Matteo Rizzi, and Markus Schmitt, *Domain wall dynamics of a two dimensional quantum Ising model using tree tensor networks*, *Bull. Am. Phys. Soc.* (to be published).
- [112] Federico Balducci, Andrea Gambassi, Alessio Lerose, Antonello Scardicchio, and Carlo Vanoni, *Interface dynamics in the two-dimensional quantum Ising model*, *Phys. Rev. B* **107**, 024306 (2023).
- [113] Federico Balducci, Andrea Gambassi, Alessio Lerose, Antonello Scardicchio, and Carlo Vanoni, *Localization*

- and melting of interfaces in the two-dimensional quantum Ising model, *Phys. Rev. Lett.* **129**, 120601 (2022).
- [114] Luka Pavešić, Daniel Jaschke, and Simone Montangero, *Constrained dynamics and confinement in the two-dimensional quantum Ising model*, [arXiv:2406.11979](#).
- [115] Takahiro Fukui and Norio Kawakami, *Breakdown of the Mott insulator: Exact solution of an asymmetric Hubbard model*, *Phys. Rev. B* **58**, 16051 (1998).
- [116] Biao Lian, *Quantum breakdown model: From many-body localization to chaos with scars*, *Phys. Rev. B* **107**, 115171 (2023).
- [117] Chao Yin, Rahul Nandkishore, and Andrew Lucas, *Eigenstate localization in a many-body quantum system*, *Phys. Rev. Lett.* **133**, 137101 (2024).
- [118] Boris Altshuler, Hari Krovi, and Jérémie Roland, *Anderson localization makes adiabatic quantum optimization fail*, *Proc. Natl. Acad. Sci. U.S.A.* **107**, 12446 (2010).
- [119] C. L. Baldwin, C. R. Laumann, A. Pal, and A. Scardicchio, *Clustering of nonergodic eigenstates in quantum spin glasses*, *Phys. Rev. Lett.* **118**, 127201 (2017).
- [120] Tatsuhiko N. Ikeda, Sho Sugiura, and Anatoli Polkovnikov, *Robust effective ground state in a non-integrable Floquet quantum circuit*, *Phys. Rev. Lett.* **133**, 030401 (2024).
- [121] As a simple example, consider the Dicke state with two particles: $|1100\rangle + |1010\rangle + |0110\rangle + \dots$; here, “1” represents a quasiparticle-occupied site. A two-site operator $|11\rangle(\langle 01| - \langle 10|) + \text{H.c.}$ will, even on average, remove one quasiparticle from this state.
- [122] F. J. Dyson, *Existence of a phase-transition in a one-dimensional Ising ferromagnet*, *Commun. Math. Phys.* **12**, 91 (1969).
- [123] R. B. Israel, *Existence of phase transitions for long-range interactions*, *Commun. Math. Phys.* **43**, 59 (1975).
- [124] R. J. McCraw, *Metastability in a long range one-dimensional Ising model*, *Phys. Lett. A* **75**, 379 (1980).
- [125] Aernout C. D. van Enter, Bruno Kimura, Wioletta Ruszel, and Cristian Spitoni, *Nucleation for one-dimensional long-range Ising models*, *J. Stat. Phys.* **174**, 1327 (2019).
- [126] Mario Collura, Andrea De Luca, Davide Rossini, and Alessio Lerose, *Discrete time-crystalline response stabilized by domain-wall confinement*, *Phys. Rev. X* **12**, 031037 (2022).
- [127] Michael Kastner, *Diverging equilibration times in long-range quantum spin models*, *Phys. Rev. Lett.* **106**, 130601 (2011).
- [128] Alessio Lerose, Bojan Žunkovič, Jamir Marino, Andrea Gambassi, and Alessandro Silva, *Impact of nonequilibrium fluctuations on prethermal dynamical phase transitions in long-range interacting spin chains*, *Phys. Rev. B* **99**, 045128 (2019).
- [129] Fangli Liu, Rex Lundgren, Paraj Titum, Guido Pagano, Jiehang Zhang, Christopher Monroe, and Alexey V. Gorshkov, *Confined quasiparticle dynamics in long-range interacting quantum spin chains*, *Phys. Rev. Lett.* **122**, 150601 (2019).
- [130] Nicolò Defenu, *Metastability and discrete spectrum of long-range systems*, *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2101785118 (2021).
- [131] Guido Giachetti and Nicolò Defenu, *On the conditions for a quantum violent relaxation*, [arXiv:2312.14768](#).
- [132] Chi-Fang Chen and Andrew Lucas, *Finite speed of quantum scrambling with long range interactions*, *Phys. Rev. Lett.* **123**, 250605 (2019).
- [133] Tomotaka Kuwahara and Keiji Saito, *Strictly linear light cones in long-range interacting systems of arbitrary dimensions*, *Phys. Rev. X* **10**, 031010 (2020).
- [134] Minh C. Tran, Andrew Y. Guo, Christopher L. Baldwin, Adam Ehrenberg, Alexey V. Gorshkov, and Andrew Lucas, *Lieb-Robinson light cone for power-law interactions*, *Phys. Rev. Lett.* **127**, 160401 (2021).
- [135] Francisco Machado, Dominic V. Else, Gregory D. Kahanamoku-Meyer, Chetan Nayak, and Norman Y. Yao, *Long-range prethermal phases of nonequilibrium matter*, *Phys. Rev. X* **10**, 011043 (2020).
- [136] Norbert Schuch, Sarah K. Harrison, Tobias J. Osborne, and Jens Eisert, *Information propagation for interacting-particle systems*, *Phys. Rev. A* **84**, 032309 (2011).
- [137] Tomotaka Kuwahara and Keiji Saito, *Lieb-Robinson bound and almost-linear light cone in interacting boson systems*, *Phys. Rev. Lett.* **127**, 070403 (2021).
- [138] Chao Yin and Andrew Lucas, *Finite speed of quantum information in models of interacting bosons at finite density*, *Phys. Rev. X* **12**, 021039 (2022).
- [139] Jérémy Faupin, Marius Lemm, and Israel Michael Sigal, *Maximal speed for macroscopic particle transport in the Bose-Hubbard model*, *Phys. Rev. Lett.* **128**, 150602 (2022).
- [140] J. Eisert and D. Gross, *Supersonic quantum communication*, *Phys. Rev. Lett.* **102**, 240501 (2009).
- [141] Sergey Bravyi and David Gosset, *Gapped and gapless phases of frustration-free spin-1/2 chains*, *J. Math. Phys. (N.Y.)* **56**, 061902 (2015).
- [142] In this subsection, we set $E_0 = 0$ in Eq. (29) without loss of generality.
- [143] Strictly speaking, Eq. (B50) bounds Eq. (B65) only for sets S that are not larger than the metastability range R . Equation (B65) for larger S are bounded in the previous proof straightforwardly by Lieb-Robinson bounds.
- [144] Itai Arad, Tomotaka Kuwahara, and Zeph Landau, *Connecting global and local energy distributions in quantum spin models on a lattice*, *J. Stat. Mech.* (2016), 033301.
- [145] We show H_0 satisfies the metastability condition for completeness in Appendix C 6; however, this is not necessary for the main result Theorem 24 of bounding the lifetime under $H_0 + V_1$.
- [146] Chao Yin and Andrew Lucas, *Low-density parity-check codes as stable phases of quantum matter*, [arXiv:2411.01002](#).
- [147] Christiane Tretter, *Spectral Theory of Block Operator Matrices and Applications* (World Scientific, Singapore, 2008).